

The Delbaen–Schachermayer Closedness Theorem

A Mathematical Account of the Lean Proof

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<https://github.com/selpoG/lean-ftap-closedness>

Abstract

We prove Theorem 4.2 for bounded real-valued semimartingales. Starting from NFLVR for the terminal market of general stochastic integrals, we obtain Fatou closedness and weak-star closedness. The proof constructs auxiliary decompositions for the selected gains, component estimates with common convex weights, and realization of the limit as a general integral against the original price. The assumptions and proof boundaries correspond to those used by the public Lean theorem.

Contents

1	Setting and main theorem	3
1.1	Proof structure	4
1.2	Correspondence with the original proof	5
2	The boundary between the main proof and stochastic analysis	6
2.1	Two data definitions and notation	6
2.2	Seven analytic inputs	7
2.3	Proof locations and the isolation check	10
3	Convex compactness of terminal gains	10
3.1	NFLVR and boundedness in probability	10
3.2	Forward convex limits	11
3.3	Maximal terminal claims	12
4	Maximality, uniform path limits, and a common envelope	12
4.1	Excluding persistent terminal improvements	12
4.2	Chronological pasting	13
4.3	One measure and separate gain decompositions	13

5	Component estimates for the same convexified sequence	14
5.1	Predictable transforms and class maximal estimates	14
5.2	Stopped tails and common Hilbert convexification	15
5.3	Applying the finite Hahn input to the same sequence	17
5.4	Finite-variation Cauchy estimates from maximality	17
6	Realization of maximal closure claims	18
7	Fatou and weak-star closedness	19
7.1	Propagating a terminal lower bound	19
7.2	Fatou closedness	19
7.3	Bounded slices in L^1	19
A	Decomposition, compensation and component norms	21
A.1	Initial values and stopping	21
A.2	The bounded-source decomposition input	21
A.3	A Doléans–Dade–Yen decomposition	22
A.4	Integrable localization of the finite large-jump process	24
A.5	Removing finite large jumps	25
A.6	Gluing predictable compensators and bounding jumps	26
A.7	General quadratic variation and exact j^1	30
A.8	From an arbitrary decomposition to canonical component bounds	43
B	Common localization and component series	46
B.1	Passing to test-uniform limits	46
B.2	Auxiliary topology and common localization	46
C	The Emery topology and realization by one integrand	59
C.1	Good integrators and comparison of topologies	59
C.2	One coefficient for both limiting components	72
D	Proofs of the seven analytic inputs	75
D.1	Regular limits and measurable envelopes	75
D.2	Internal theorems for the integral construction	75
D.3	Deriving decomposition under an envelope	78
D.4	Construction and transfer of the integral calculus	79
D.5	Proof of closedness of the general integral graph	84
D.6	Connecting decomposition and market operations to the inputs	86
D.7	Quantitative auxiliary-gain estimates	86

D.8 Prefix energy and martingale approximation	90
D.9 Finite Hahn trades and closed-stop estimates	91
D.10 Terminal realization from component estimates	94
E Mathematical steps and Lean declarations	95

1 Setting and main theorem

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $(\mathcal{F}_t)_{t \geq 0}$ satisfying the usual conditions. The price S is a bounded real-valued adapted càdlàg semimartingale. Boundedness means that one deterministic constant B satisfies $\sup_{t \geq 0} |S_t| \leq B$ almost surely. The Lean source chooses an everywhere càdlàg representative; the bound is required almost surely. The public theorem also explicitly includes `SigmaFiniteFiltration`, which holds because restrictions of a probability measure to the filtration are finite.

Semimartingale behavior is expressed by the good-integrator property. An elementary predictable strategy J has a pathwise finite-sum integral; its value at each finite time is required to be measurable up to completion. If a sequence J^n satisfies

$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N \ \forall (t, \omega), \quad |J_t^n(\omega)| \leq \varepsilon,$$

then $(J^n \cdot S)_T \rightarrow 0$ in probability for every finite T . Section 2 constructs integrals of general predictable coefficients from these assumptions. We write $H \cdot S$ for such an integral, always with initial value zero.

Definition 1.1. An integrable strategy H is a -admissible if, on one set of probability one,

$$(H \cdot S)_t \geq -a \quad \text{for every } t \geq 0.$$

Let K_0 be the terminal gains of strategies admissible for some finite constant and possessing a finite almost-sure limit as $t \rightarrow \infty$. Let K_1 be the terminal gains of such 1-admissible strategies. Set

$$C_0 = K_0 - L_+^0, \quad C = C_0 \cap L^\infty.$$

The set K_0 is a convex cone and K_1 is convex, with $K_1 \subseteq K_0$. The cone C_0 is downward solid for the almost-sure order. NFLVR means

$$\overline{C}^{\|\cdot\|^\infty} \cap L_+^\infty = \{0\}.$$

A set $D \subseteq L^0$ is Fatou closed if $f_n \in D$, a common constant lower bound on f_n , and $f_n \rightarrow f$ almost surely imply $f \in D$.

For the implementation, distinguish $C_0 \subseteq L^0$ from its raw-function extension

$$\widehat{C}_0 = \{f : \Omega \rightarrow \mathbb{R} : \exists g \in K_0, f \leq g \text{ a.s.}\}.$$

Here K_0 denotes raw representatives of terminal gains. Lean uses $\widehat{C}_0$. If f is measurable up to completion, then so is $g - f$, because terminal gains are measurable up to completion. Thus $f \in \widehat{C}_0$ if and only if $f = g - h$ almost surely for some $g \in K_0$ and an h measurable with respect to completion and nonnegative almost surely. Passing to almost-sure classes recovers exactly $C_0 = K_0 - L_+^0$. The raw Fatou-closedness result therefore implies the stated L^0 result, and its intersection with L^∞ gives exactly the same cone C and NFLVR condition.

Theorem 1.2 (Delbaen–Schachermayer). *Assume NFLVR. Then C_0 is Fatou closed, and C is closed in the topology $\sigma(L^\infty, L^1)$.*

The integral in this statement is the general integral graph of Section 2: bounded coefficients are integrated in a local $M^2 \oplus A^1$ completion, and general gains arise as limits of coefficient truncations uniformly over elementary tests. A terminal claim is the continuous-time almost-sure limit of its own gain process. Lean’s definition of Fatou closedness normalizes the common lower bound to -1 . For a cone stable under positive scaling, divide a sequence bounded below by $-b$ by $\max(1, b)$ and then scale its limit back.

This is the real-valued version of [1, Theorem 4.2]. The essential probabilistic step realizes an almost-sure maximal element of $\overline{K_1}^{L^0}$ as an admissible integral’s terminal value. Weak-star closedness then follows by functional analysis.

1.1 Proof structure

First construct the general integral and its closedness under regular Émery limits. This construction does not assume market closedness. NFLVR gives boundedness in probability and forward convex compactness of K_1 . For a maximal closure claim h , chronological pasting yields integral gains converging uniformly over all time to a process X with $X_\infty = h$. Choose one equivalent measure Q_* and decompose each selected gain separately. Common convex weights make the martingale components Cauchy by tail estimates and Hilbert convexification. Hahn improvements and maximality make the finite-variation components Cauchy. These estimates imply Émery convergence of the same gains. Integral closedness yields $h \in K_1$, from which the two closedness conclusions follow. The market argument ends in Section 7. Its boundary with the appendices consists of the seven analytic propositions (twenty clauses) and two data definitions in Section 2. Each clause corresponds to one input in the Lean isolation check. The main proof uses only these conclusions. A reader wishing to check an input can turn to its indicated appendix proof.

For decomposition, the appendix explicitly cites the bounded Bichteler–Dellacherie characterization and existence of dual predictable projections as standard results, with their hypotheses specified. Lean constructs these too. Integral-range closedness, simultaneous realization by one coefficient, and original-market component estimates are proved within this account. The final appendix relates the mathematical stages to Lean declarations.

1.2 Correspondence with the original proof

The main argument follows Delbaen–Schachermayer[1, Section 4]: selection of a maximal closure claim, pasting, common convexification under an equivalent measure, and a maximality contradiction using Hahn improvements. The table matches the content used in the main proof; it does not restate every original result in its full generality.

Original paper	Corresponding argument here
Lemma A1.1	Proposition 3.2: forward convex compactness in a lower-bounded convex set
Lemma 4.3	Proposition 3.3: existence of a maximal terminal claim
Lemma 4.5	Proposition 4.2: all-time Cauchy estimates from maximality
Lemma 4.7	Proposition 5.1: class bounds for martingale maxima
Lemma 4.8	Proposition 5.3 and its corollary: convex stopped-tail estimates
Lemma 4.9	Proposition 5.5: test-uniform tail estimates
Lemma 4.10	Proposition 5.6: common martingale convexification
Lemma 4.11	Section 5.4: finite-horizon FV variation Cauchy estimates from Hahn improvements
Role of Lemma A3.1	Lemma 4.1: persistent terminal improvement contradicts maximality
Final use of Mémin’s theorem	Proposition 2.8 and Section 6: realizing the limit as an original-price integral
Use of Theorem 2.1	Proposition 7.4: from Fatou to weak-star closedness

We obtain maximal elements by convex compactness and maximization of a functional, replacing the original transfinite induction. Component estimates allow separate gain decompositions, but every contradiction strategy returns to the same original market. For martingale components we first establish Cauchy estimates and identify the limit later; for FV components we prove the needed finite-horizon estimates and combine them with the separately retained infinite-time terminal limit. Proposition 2.5 supplies the finite-scale analytic estimates needed in Lemmas 4.7–4.9. The integral-range closedness cited by the original paper enters the proof of terminal realization, Proposition 2.8. These are reorganizations preserving the roles of the original argument, not a claim to a new proof strategy for the closedness theorem.

2 The boundary between the main proof and stochastic analysis

The main argument receives its analytic results through the seven propositions in this section. Each clause identifier corresponds to one theorem replaced by an assumption in the Lean isolation check. There are twenty clauses, together with two data definitions for the integral graph and the market. These are the complete inputs to the main proof when the analytic implementation is removed. Accepting them suffices to read the entire main argument, including Section 4 onwards, without consulting internal appendix lemmas. Ordinary measure theory, conditional expectation, Doob inequalities, optional sampling and Hilbert-space convex compactness are shared foundations. NFLVR, maximality, the selection of convex averages, boundedness in probability of whole families, and the M/FV Cauchy properties are not inputs here: they are used or proved in the main argument.

2.1 Two data definitions and notation

Fix the original bounded price S and $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mu)$ satisfying the usual conditions. Whenever $Q \sim \mu$ is used below, Q is a probability measure and the same filtration satisfies the usual conditions under Q . Unless stated otherwise, process identities and all-time bounds are understood up to indistinguishability or on one common full-measure set. An elementary predictable test J , with $|J| \leq 1$, consists of finitely many stopping intervals and coefficients measurable at their left endpoints. Define $J \cdot Y$ by the corresponding finite sum of increments and write

$$e_T^J(Y, Z) = 1 \wedge \sup_{t \leq T} |(J \cdot (Y - Z))_t|, \quad d_T^Q(Y, Z) = \sup_J E_Q e_T^J(Y, Z).$$

Emery convergence means $d_T^Q(Y_n, Y) \rightarrow 0$ for every finite T . The Cauchy quantifiers are $\forall T, \varepsilon > 0 \exists N \forall n, k \geq N \forall J$.

Definition 2.1 (The original-price integral graph and terminal market).

1. [G0] Put $H^{(n)} = H 1_{\{|H| \leq n+1\}}$. A local special integral is obtained by applying the *same* predictable coefficient to the energy integral of the M component and the Stieltjes integral of the FV component on stopped coordinates, then gluing. Write $(H, X) \in \mathcal{G}_S$, or $X = H \cdot S$, when the local special integrals of $H^{(n)}$ converge in Emery topology to X , which has a zero-start strongly adapted càdlàg representative. Neither specialness nor existence of a terminal value is required of X itself.
2. [M0] The gain-process model $\mathcal{M}_S$ is obtained from the Emery completion of elementary integrals of the original price. Its strategies carry a realized gain and its own terminal

value; gain and terminal value respect linear operations. Clauses [M1]–[M3] below identify this model with the market defined by $\mathcal{G}_S$.

Set $K_a = \{f : a > 0, \exists(H, X) \in \mathcal{G}_S, X \geq -a, X_t \rightarrow f \text{ a.s.}\}$ and $K_0 = \bigcup_{a>0} K_a$. These set definitions remain concrete in the isolation check. An “original-market gain” means a zero-start strongly adapted càdlàg process Y with a witness $(H, Y) \in \mathcal{G}_S$. A “special gain” additionally carries a specified zero-start decomposition $Y = M + A$ under Q , where M is a càdlàg local martingale and A is predictable, càdlàg and locally of finite variation. A “finite gain” is a special gain constant after some finite time, whose specified A is globally BV. These data types also remain in the isolation check.

2.2 Seven analytic inputs

Proposition 2.2 (Regular limits, common envelopes and equivalent measures).

1. [P1] *If strongly adapted càdlàg processes Y_n are Cauchy in probability for the all-time supremum, there are a strictly increasing f and a strongly adapted càdlàg X such that $Y_{f(n)} \rightarrow X$ uniformly over all time almost surely.*
2. [P2] *If each process in such a Cauchy sequence has its own finite terminal value, there are a strictly increasing f and a finite nonnegative measurable q such that $|Y_{f(n)}(t)| \leq q$ for every n, t , almost surely.*
3. [P3] *For any finite nonnegative measurable q , there is an equivalent probability $Q \sim \mu$ with $q \in L^2(Q)$; the same filtration satisfies the usual conditions under Q .*

Proposition 2.3 (Special decomposition of a specified gain). [D1] *If an original-market gain Y satisfies $|Y_t| \leq |q|$ for $q \in L^2(Q)$, the same Y has a zero-start special decomposition $Y = M + A$. The decomposition is constructed under Q ; no transfer of specialness to the original measure is asserted.*

Proposition 2.4 (Market identification and finite operations).

1. [M1] *For every real a , K_a equals the a -admissible terminal set of $\mathcal{M}_S$.*
2. [M2] *K_0 equals the admissible terminal set of $\mathcal{M}_S$.*
3. [M3] *K_1 equals the unit-admissible terminal set of $\mathcal{M}_S$.*
4. [M4] *Admissible terminal claims of this model are a.e. strongly measurable, and an admissibility bound $-a$ also holds at the terminal value.*

5. [M5] *Model gains are zero-start, strongly adapted and càdlàg, have their own terminal values, and are realized in $\mathcal{G}_S$. An a.e. equal terminal representative can be substituted without changing the gain. For gains Y, Z , a bounded stopping time $\tau \leq T$, and $E \in \mathcal{F}_\tau$, there is a switched strategy with gain $Y_{t \wedge \tau} + Z_t - Z_{t \wedge \tau}$ on E , and Y_t on E^c . Its terminal value is $Z_\infty + Y_\tau - Z_\tau$ on E , and Y_∞ on E^c .*
6. [M6] *For a special gain $Y = M + A$ and finite $T > 0$, $Y^T = M^T + A^T$ is realized as a finite gain.*
7. [M7] *Finite convex combinations of special gains are realized using the same weights for gains, M components and FV components. In particular this realization is available for specified forward convex weights.*

Proposition 2.5 (Finite-scale quantitative estimates). *None of the following clauses assumes NFLVR.*

1. [F1] *Suppose a special gain $Y = M + A \geq -1$ satisfies $|Y| \leq \xi$ and $\|\xi\|_2 \leq B$. Let $0 < \alpha \leq 1/4$, let n be a positive integer, and let $T > 0$, with*

$$k = \lfloor \alpha n / 4 \rfloor > 0, \quad 6B \leq n^2, \quad (B/n)^2 \leq \alpha, \quad m = \alpha^2 - (4B/(\alpha n^2))^2 \geq 0.$$

If $Q(\sup_{t \leq T} |M_t| > n^3) > 8\alpha$, there is an a.e. measurable $f \in K_0 - L_+^0(Q)$ such that

$$f \geq -n^{-1/4} - \frac{2}{nk}, \quad Q\left(f \geq \frac{\alpha m}{4} - n^{-1/4}\right) \geq \frac{m}{2} - \frac{16\sqrt{n}}{k}.$$

2. [F2] *Let finite gains $Y_i = M_i + A_i \geq -1$ have a common envelope $|q| \in L^2(Q)$ and suppose $\{M_i^*\}$ is bounded in probability. Put $\tau_i(c) = \inf\{t : |M_i(t)| > c\}$ and $L^{w,c} = \sum_i w_i (M_i - M_i^{\tau_i(c)})$. Let $B(w, c, \varepsilon)$ be the event that the passage time of $|L^{w,c}|$ across level ε is finite. For every $\varepsilon > 0$ there is $N \geq 0$ such that for every $0 < \delta \leq \varepsilon/4$ there is $c_0 \geq 0$ with the following property. For every finite convex weight w and $c \geq c_0$, if $Q(B(w, c, \varepsilon)) > \varepsilon$, there are an original-market special gain $V = M^V + A^V$ and an envelope g with*

$$V \geq -1, \quad |V| \leq g, \quad \|g\|_2 \leq 1, \quad Q((M^V)^* > \varepsilon/[2(1+N)\delta]) > \varepsilon/2.$$

3. [F3] *Assume the finite-gain, common-envelope and M -maximal boundedness conditions of the preceding clause, but not the lower bound -1 . For every $\eta > 0$ there is $c_0 \geq 0$ such that for every $w, c \geq c_0, T > 0$ and J ,*

$$E_Q e_T^J(L^{w,c}, 0) \leq 4\eta + Q(B(w, c, \eta)).$$

4. [F4] For a special gain $Y = M + A$, $q \in L^2(Q)$, $|Y| \leq q$, $c \geq 0$ and $T > 0$, $P = M^{\tau(c) \wedge T}$ is a true martingale, $P_T \in L^2(Q)$, and $\|P_T\|_2 \leq c + 6\|q\|_2$.
5. [F5] The same stopped component $P = M^{\tau(c) \wedge T}$ is right-continuous and satisfies $P_0 = 0$ almost surely for every real c and $T \geq 0$, without an envelope assumption.

Proposition 2.6 (Martingale approximation uniformly over tests).

1. [A1] Let X_n be strongly progressive. Suppose for every $T, \varepsilon > 0$ there is a strongly progressive sequence P_n with $\sup_n d_T^Q(X_n, P_n) \leq \varepsilon$, Cauchy uniformly over tests at this T . Then X_n is Emery Cauchy.
2. [A2] Let P_n be right-continuous zero-start true martingales and let $U \geq 0$ be finite. If $(P_n(U))$ is Cauchy in $L^2(Q)$, then for every $\varepsilon > 0$ there is N such that $E_Q e_T^J(P_n, P_k) \leq \varepsilon$ for every $n, k \geq N$, every $T \leq U$, and every J .

Proposition 2.7 (Finite Hahn trades for two specified gains). [H1] Let special gains $Y = M_Y + A_Y$ and $Z = M_Z + A_Z$ share an envelope $q \in L^2(Q)$ and both be bounded below by -1 . Fix $T > 0$ and $\delta \geq 0$, and assume $E_Q e_T^J(M_Y, M_Z) \leq b$ and the reverse-order estimate for every J . The following data can be chosen simultaneously for both orientations. For orientation (Y, Z) put $L = M_Y - M_Z$ and $A = A_Y - A_Z$. There are strongly adapted N, C and $V = N + C$, with N right-continuous, $C_0 = 0$, and

$$0 \leq C_c - C_a, \quad A_{c \wedge T} - A_{a \wedge T} \leq C_c - C_a \quad (a \leq c), \quad E_Q[1 \wedge \sup_{t \leq T} |N_t|] \leq b.$$

Put $\tau = \inf\{t : N_t - \max(L_t, 0) < -\delta\}$, $\sigma = \tau \wedge T$, and $E = \{T < \tau\}$. If $0 < \delta \leq 1$, then $Q(\tau \leq T) \leq 8b/\delta$; for every $U \geq T$,

$$Z_t^{\text{paste}} = (Z + V)_{t \wedge \sigma} + 1_E(Z_{t \wedge U} - Z_{t \wedge T}), \quad Z_U^{\text{paste}} \in K_{1+\delta}.$$

Writing C' for the FV component in the reverse orientation, for every real α ,

$$Q(\text{Var}_{[0, T]} A > \alpha) > \alpha \implies Q(C_T > \alpha/2) > \alpha/2 \text{ or } Q(C'_T > \alpha/2) > \alpha/2.$$

This is a finite construction for two gains; it assumes neither a Cauchy sequence, NFLVR nor maximality.

Proposition 2.8 (Realizing the intrinsic terminal value from component estimates). [C1] Let $(H_n, Y_n) \in \mathcal{G}_S$, with Y_n strongly adapted and right-continuous. Suppose Y_n converges almost surely uniformly over all time to a zero-start strongly adapted càdlàg $X \geq -1$, and $X_t \rightarrow h$ almost surely. Under $Q \sim \mu$, let the same gains satisfy $Y_n = M_n + A_n$, where each A_n is

right-continuous and every difference $A_m - A_n$ is locally BV on every path. If for every finite T and $\varepsilon > 0$ there is N such that for every $m, n \geq N$ and every J ,

$$Q(e_T^J(M_m, M_n) > \varepsilon) \leq \varepsilon, \quad Q(\text{Var}_{[0, T]}(A_m - A_n) > \varepsilon) \leq \varepsilon,$$

then $h \in K_1$. No additional martingale assumption on M_n is imposed here.

2.3 Proof locations and the isolation check

Appendix D assembles the proofs of the seven inputs.

Input	Proof location
Proposition 2.2	Section D.1
Propositions 2.3, 2.4	Sections D.2–D.6
Proposition 2.5	Sections D.7, D.8
Proposition 2.6	Section D.8
Proposition 2.7	Section D.9
Proposition 2.8	Section D.10

Appendices A–C divide the internal construction into three stages. Their internal theorems need not be held in mind simultaneously to follow the main proof. The Lean correspondence table gives the fully qualified name for each identifier. The isolation check compares the Japanese and English mappings with the actual set of 22 declarations replaced by assumptions, and rebuilds the unchanged main proof without `Stochastic/`. It also verifies that the resulting main theorem uses all 22 specified inputs and no other analytic axioms. This verifies the formal dependency boundary. Agreement in mathematical meaning between the prose and the Lean types still requires comparing the hypotheses and conclusions stated here with those types.

3 Convex compactness of terminal gains

We henceforth identify the general-graph terminal sets with the gain-model sets using [M1]–[M3]. Clause [M4] supplies a.e. measurability and the admissible lower bound at the terminal value. Linear operations belong to the model structure [M0].

3.1 NFLVR and boundedness in probability

A set $D \subseteq L^0$ is bounded in probability if for every $\varepsilon > 0$ there exists $R > 0$ with

$$\sup_{f \in D} \mathbb{P}(|f| > R) \leq \varepsilon.$$

Proposition 3.1 (Delbaen–Schachermayer, Proposition 3.1). *Under NFLVR, K_1 is bounded in probability.*

Proof. Otherwise, since its elements are at least -1 , choose $1 \leq a_n \uparrow \infty$ and $G_n \in K_1$ with $\mathbb{P}(G_n > a_n) > \varepsilon$ for one $\varepsilon > 0$. Put

$$f_n = \min(G_n/a_n, 1).$$

Positive scaling and subtraction of a nonnegative claim give $f_n \in C_0$. Moreover,

$$-a_n^{-1} \leq f_n \leq 1, \quad \mathbb{E}f_n \geq \mathbb{P}(G_n > a_n) - a_n^{-1} > \varepsilon - a_n^{-1}.$$

Bounded Komlós convexification gives forward convex averages $W_n f$ converging almost surely to some $0 \leq f \leq 1$. Since a_n increases, these averages still lie between $-a_n^{-1}$ and 1, and their expectations are at least $\varepsilon - a_n^{-1}$. Dominated convergence yields $\mathbb{E}f \geq \varepsilon$, hence $\mathbb{P}(f > 0) > 0$. Choose $b > 0$ with $\mathbb{P}(f \geq b) > 0$. Egorov's theorem provides a measurable subset $A \subseteq \{f \geq b\}$ of positive probability on which $W_n f \rightarrow f$ uniformly. Define

$$U_n = 1_A W_n f - a_n^{-1} 1_{A^c}, \quad U = 1_A f.$$

We have $U_n \leq W_n f$, so solidity of C_0 and boundedness give $U_n \in C$. Also $0 \neq U \in L_+^\infty$ and

$$\|U_n - U\|_\infty \leq \max \left\{ \sup_{\omega \in A} |(W_n f)(\omega) - f(\omega)|, a_n^{-1} \right\} \rightarrow 0.$$

This contradicts NFLVR. □

3.2 Forward convex limits

A forward convex combination of (x_n) is

$$y_n = \sum_{k=n}^{N_n} \alpha_{n,k} x_k, \quad \alpha_{n,k} \geq 0, \quad \sum_{k=n}^{N_n} \alpha_{n,k} = 1.$$

The next proposition corresponds to the case of [1, Lemma A1.1] in which boundedness in probability of the convex hull ensures a finite limit.

Proposition 3.2 (Komlós compactness with a lower bound). *Let $D \subseteq L^0$ be convex and bounded in probability. A sequence $G_n \in D$ with a common constant lower bound has forward convex combinations converging almost surely to a finite random variable.*

Proof. Translate by a constant to make the sequence nonnegative. On each tail convex hull choose approximate maximizers Y_n of $\mathbb{E}[1 - e^{-Y}]$. The midpoint identity for the exponential implies that $e^{-Y_n/2}$ is L^2 -Cauchy. Extract indices $n_j \geq j$ such that $e^{-Y_{n_j}/2} \rightarrow q$ almost surely. If $q = 0$ on a set of positive probability, Y_{n_j} tends to infinity there, contradicting boundedness in probability of the convex set. Thus $q > 0$ almost surely. Set $\tilde{G}_j = Y_{n_j}$. Then $\tilde{G}_j \rightarrow -2 \log q$ almost surely, and its convex weights are supported on $k \geq n_j \geq j$, as required for a forward combination of the original sequence. Finally undo the initial constant translation. □

Boundedness in probability of a sequence alone does not imply boundedness of its growing convex hulls. We use convexity of K_1 and boundedness of the entire set K_1 .

3.3 Maximal terminal claims

The next proposition supplies the maximal element of [1, Lemma 4.3]. Here convex compactness and maximization of a functional replace transfinite induction.

Proposition 3.3. *If $D \subseteq L^0$ is convex, bounded in probability, and sequentially closed in probability, each $g \in D$ is dominated by an almost-sure maximal $h \in D$.*

Proof. Maximize $\mathbb{E}[1 - e^{-(f-g)}]$ over $D_g = \{f \in D : g \leq f \text{ a.s.}\}$. The common lower bound g need not be constant. Apply Proposition 3.2 to the nonnegative convex set $D_g - g$, which is bounded in probability, and to an approximate maximizing sequence. Add g back to the convex averages. Their almost-sure limit h belongs to D_g by sequential closedness. Concavity preserves the tail lower bounds on the objective, and bounded convergence shows that h attains its supremum. If $k \in D$, $k \geq h$, and $\mathbb{P}(k > h) > 0$, strict monotonicity of $1 - e^{-x}$ contradicts this maximizing property. $\square$

The probability closure $\overline{K_1}^{L^0}$ has these three properties. In particular, any forward convex limit g of claims from K_1 is dominated by a maximal $h \in \overline{K_1}^{L^0}$.

4 Maximality, uniform path limits, and a common envelope

Fix an almost-sure maximal $h \in \overline{K_1}^{L^0}$. Choose 1-admissible integral gains Y_n with their own terminal values satisfying $Y_n(\infty) \rightarrow h$ almost surely. We first obtain uniform convergence of the paths over all time; test-uniform Cauchy estimates are proved separately in the next section.

4.1 Excluding persistent terminal improvements

The next lemma serves the role of [1, Lemma A3.1] in the original proof of Lemma 4.11: persistent improvement contradicts maximality, here allowing a vanishing lower error.

Lemma 4.1 (Vanishing lower error and maximality). *Suppose f_n are measurable, $f_n \rightarrow h$ almost surely, $g_n \in K_1$, and $(f_n - g_n)^+ \rightarrow 0$ almost surely. There cannot exist fixed $\eta, p > 0$ and measurable events E_n such that $\mathbb{P}(E_n) \geq p$ and $g_n - f_n \geq \eta$ on E_n for every n .*

Proof. Put $e_n = (f_n - g_n)^+$ and $b_n = f_n - e_n = \min(f_n, g_n)$. Then $b_n \leq g_n$ and $b_n \rightarrow h$ almost surely. Apply forward convex compactness to the uncorrected claims $g_n \in K_1$. For the resulting weights W_n , we have $W_n g \rightarrow g \in \overline{K_1}^{L^0}$ and $W_n b \rightarrow h$ almost surely. Thus $g \geq h$, and maximality gives $g = h$ almost surely. The nonnegative gaps $d_n = g_n - b_n$ exceed η on E_n . Their truncations $c_n = \min(d_n, \eta)$ satisfy $\mathbb{E}c_n \geq \eta p$ and

$$0 \leq W_n c \leq \min(W_n d, \eta), \quad \mathbb{E}(W_n c) \geq \eta p.$$

But $W_nd = W_ng - W_nb \rightarrow 0$ almost surely, so bounded convergence forces $\mathbb{E}(W_nc) \rightarrow 0$, a contradiction. No membership assumption on f_n is used. $\square$

4.2 Chronological pasting

The next proposition corresponds to [1, Lemma 4.5]: maximality and time-ordered pasting give Cauchy estimates for the all-time supremum.

Proposition 4.2. *After extraction, the approximating gains satisfy*

$$\sup_{t \geq 0} |Y_n(t) - Y_k(t)| \longrightarrow 0 \quad \text{in probability} \quad (n, k \longrightarrow \infty).$$

Proof. Failure supplies pairs in every tail whose gap exceeds a fixed positive amount with probability bounded below. Right continuity reduces the all-time supremum to a countable time set; approximate its event by a finite subset, order it chronologically, and choose one of the two orientations with positive probability bounded below. Hold the higher candidate until the first grid time detecting the positive gap, then follow the other candidate's increments. The decision is measurable at the switching time, so [M5] supplies this strategy and its specified terminal value. The positive gap preserves the lower bound -1 . The terminal gain equals the other candidate's terminal value plus the detected gap when switching occurs. Both original terminal sequences converge to h ; therefore the pasted claims have vanishing lower error and persistent positive improvement, contradicting the preceding lemma. The countable enumeration only detects the supremum; trades always switch on the chronological grid. $\square$

Proposition 4.3. *A further subsequence converges almost surely, uniformly over all time, to a zero-start strongly adapted càdlàg process X with $X \geq -1$ and $X_t \rightarrow h$ almost surely as $t \rightarrow \infty$.*

Proof. Apply [P1] to obtain a subsequence with a strongly adapted càdlàg uniform limit. The initial value and lower bound pass to the limit. Finally,

$$|X_t - h| \leq \sup_u |X_u - Y_n(u)| + |Y_n(t) - Y_n(\infty)| + |Y_n(\infty) - h|.$$

First choose large n , then large t , to identify the terminal limit of this same X with h . $\square$

4.3 One measure and separate gain decompositions

Use [P2] for a further subsequence and a common finite nonnegative measurable envelope q . Then [P3] supplies one equivalent measure Q_* with $q \in L^2(Q_*)$. Apply [D1] to each Y_i to obtain the zero-start special decomposition $Y_i = M_i + A_i$ of the same gain, with predictable locally finite-variation A_i . Each gain's L^2 envelope feeds the decomposition construction separately;

the auxiliary source may depend on i . The next section estimates arbitrary indexed classes, including the convex and normalized tail families themselves. All selections retain the same measure, gains, path limit X , and terminal h .

5 Component estimates for the same convexified sequence

Fix the selected gains $Y_i \rightarrow X$, the terminal $X_\infty = h$, and the one equivalent measure Q_* from the preceding section. Their common $L^2(Q_*)$ envelope supplies zero-start special decompositions $Y_i = M_i + A_i$. The auxiliary sources may differ with i . We return transformations of each auxiliary gain to the same original market before using NFLVR.

5.1 Predictable transforms and class maximal estimates

The next proposition corresponds to the class estimate of [1, Lemma 4.7]. Auxiliary decompositions may vary with the index; all contradiction strategies return to the same original market.

Proposition 5.1 (An indexed auxiliary-decomposition estimate from NFLVR). *Suppose the original bounded-price general market satisfies NFLVR. Fix $Q \sim P$ and any index set I . Let $Y_i = M_i + A_i$ be gains realized in that market with special decompositions under Q , $M_i(0) = 0$, and locally finite-variation A_i . If*

$$Y_i \geq -1, \quad |Y_i(t)| \leq \xi_i \ (\forall t), \quad \sup_i \|\xi_i\|_{L^2(Q)} \leq B < \infty,$$

then $\{\sup_t |M_i(t)| : i \in I\}$ is bounded in Q -probability. There is no common-auxiliary-source assumption.

Proof. Suppose boundedness fails. The all-time maximum of a càdlàg path is the increasing limit of its integer-horizon maxima. Thus for fixed $0 < \alpha \leq 1/4$ and every sufficiently large n , choose an index i_n and finite $T_n > 0$ with $Q(\sup_{t \leq T_n} |M_{i_n}(t)| > n^3) > 8\alpha$. Apply [F1] to this gain. For $k_n = \lfloor \alpha n/4 \rfloor$, all its finite-scale hypotheses hold for sufficiently large n . The resulting $f_n \in K_0 - L_+^0(Q)$ satisfy, for fixed $\eta, p > 0$,

$$f_n \geq -\delta_n, \quad \delta_n = n^{-1/4} + 2/(nk_n) \rightarrow 0, \quad Q(f_n \geq \eta) > p.$$

Indeed, the stated profit levels tend to $\alpha^3/4$ and the probability lower bounds to $\alpha^2/2$. Admissibility in the same market is invariant under equivalent measures, as is convergence in probability. Transfer NFLVR to Q and apply it to this vanishing-risk sequence of terminal claims, a contradiction. There is no need to lift f_n back to a gain process. $\square$

The path limit X and its own terminal h are retained through every subsequence used to select the envelope and the gains.

Corollary 5.2 (Finite convex classes). *If the gains have one common envelope $q \in L^2(Q)$, then the all-time maximal functions of $\sum_i w_i M_i$, over all finite convex weights w , are uniformly bounded in probability.*

Proof. Use [M7] to realize $Y^w = \sum_i w_i Y_i$ with the same component sums $M^w = \sum_i w_i M_i$ and $A^w = \sum_i w_i A_i$. Convexity preserves the lower bound -1 and common envelope q . Apply the preceding proposition to the class indexed by w itself. This uses no general boundedness assertion for convex hulls in L^0 . $\square$

5.2 Stopped tails and common Hilbert convexification

The next proposition and its corollary correspond to the convex-tail estimate of [1, Lemma 4.8]. We first use a finitely stopped family, then remove the stops.

Proposition 5.3 (Convex tails of finitely stopped gains). *Choose positive finite horizons T_i and set $\bar{Y}_i = Y_i^{T_i}$, stopping both components in the same way. Let $\tau_i(c)$ be the first passage of $|\bar{M}_i|$ at level c and put*

$$L^{w,c} = \sum_i w_i (\bar{M}_i - \bar{M}_i^{\tau_i(c)}).$$

For every $\varepsilon > 0$ there is $c_0 \geq 0$ such that, for $c \geq c_0$ and all finite convex weights, the probability that $|L^{w,c}|$ exceeds ε at a finite time is at most ε .

Proof. Clause [M6] gives a finite-gain sequence with the same stopped components, lower bound and common envelope. Proposition 5.1 bounds its M maxima in probability; pass this to [F2]. If the conclusion fails, some $\varepsilon > 0$ allows a tail with finite-passage probability greater than ε beyond every proposed cutoff. Fix N from [F2] and set

$$\delta_r = \frac{\varepsilon}{4(N+1)(r+1)} \quad (r \geq 0).$$

For each r , first choose the cutoff in [F2], then an offending weight and level. The resulting original-market gains $V_r = M^{V_r} + A^{V_r}$ satisfy $V_r \geq -1$, $|V_r| \leq g_r$, $\|g_r\|_2 \leq 1$, and

$$Q((M^{V_r})^* > 2(r+1)) > \varepsilon/2.$$

Apply Proposition 5.1 to this normalized family itself. It gives $R \geq 0$ such that $Q((M^{V_r})^* > R) \leq \varepsilon/4$ for every r . Choosing $2(r+1) > R$ gives a contradiction. The auxiliary decompositions may vary: all the constructed gains belong to the same original market. $\square$

The cutoff may depend on the prescribed sequence (T_i) .

Corollary 5.4 (Unstopped all-time convex tails). *The same conclusion holds for the original M_i and their passage tails.*

Proof. Encode $(i, r) \in \mathbb{N}^2$ by one natural number and apply the preceding proposition to the entire family Y_i^{r+1} with the same stopped components. The cutoff is now fixed before r is chosen. Embed any original finite convex weights into the indices with second coordinate r . For the original passage $\tau(c)$ and the passage $\tau^{r+1}(c)$ of M^{r+1} ,

$$\tau^{r+1}(c) \wedge (r+1) = \tau(c) \wedge (r+1),$$

so the stopped passage tail equals $(M - M^{\tau(c)})^{r+1}$, including its value at the passage and horizon. The probability estimate therefore holds on $[0, r+1]$ with one common cutoff. Increase r and use continuity of measure from below. $\square$

The next proposition corresponds to the Emery tail estimate of [1, Lemma 4.9], stated uniformly over elementary tests and finite horizons.

Proposition 5.5 (Test-uniform convex-tail estimates). *For every $\varepsilon > 0$ there exists $c_0 \geq 0$ such that for all finite convex weights w , $c \geq c_0$, $T > 0$, and elementary predictable $|J| \leq 1$,*

$$E_Q \left[1 \wedge \sup_{t \leq T} |(J \cdot L^{w,c})_t| \right] \leq \varepsilon.$$

Proof. First use the finite-gain family encoding the index and an integer horizon. Put $\eta = \varepsilon/5$ and take the larger of the cutoff in [F3] and the cutoff in Proposition 5.3 making the level- η passage probability at most η . Then $E_Q e_T^J(L^{w,c}, 0) \leq 4\eta + \eta = \varepsilon$ for all weights, horizons and tests. The cutoff precedes all these choices. For the original unstopped sequence choose an integer r with $T \leq r+1$ and embed the weights at second coordinate r . The stopping identity from the preceding corollary makes the gains coincide on $[0, T]$. Their finite-sum test integrals also coincide, transferring the estimate. $\square$

The next proposition corresponds to the common convexification of [1, Lemma 4.10]. We first obtain Emery Cauchy estimates for the martingale components and identify the limit later.

Proposition 5.6 (Common convexification of the martingale components). *There is one sequence of forward convex weights w_n such that $\widetilde{M}_n = \sum_i w_{n,i} M_i$ is elementary-Émery Cauchy under Q . The gains with the same weights converge uniformly over all time to the original X , whose terminal value remains h .*

Proof. Put $P_{i,j} = M_i^{\tau_i(j+1) \wedge (j+1)}$. Clauses [F4] and [F5] make $P_{i,j}$ a right-continuous zero-start true martingale, with

$$\|P_{i,j}(j+1)\|_2 \leq j+1 + 6\|q\|_2$$

uniformly in i . Normalize these coordinate bounds by geometric weights and embed in a Hilbert direct sum. One convexification makes the terminal sums $\sum_i w_{n,i} P_{i,j}(j+1)$ L^2 -Cauchy

for every j . For fixed T, ε , choose $j + 1$ beyond both T and the tail cutoff. Clause [A2] makes the stopped sums Cauchy uniformly over tests and horizons up to T . On this interval $M_i - P_{i,j} = M_i - M_i^{\tau_i(j+1)}$. The stopped sums are strongly progressive by adaptedness and [F5]. The preceding tail estimate and [A1] give the desired Cauchy property, with cutoff independent of the test. Forward convex weights preserve all-time uniform convergence and the previously retained terminal certificate of X . $\square$

This only proves the martingale estimate. The finite-variation estimate is obtained next for these same weights, gains, and limit.

5.3 Applying the finite Hahn input to the same sequence

Proposition 5.7 (Hahn improvement data with a common pair cutoff). *The same weights from the preceding proposition realize $R_n = \sum_i w_{n,i} Y_i$ with components $\widetilde{M}_n, \widetilde{A}_n$ in the original market. For every $T, b > 0$ there is a common cutoff such that every later pair and every $\delta \geq 0$ admit the two orientations of improvement data in [H1]. The same X and $X_\infty = h$ are retained.*

Proof. Clause [M7] realizes gains and both components with the same weights. Convexity gives $R_n \geq -1$ and $|R_n| \leq q$. The M component's Emery Cauchy property gives a cutoff common to all pairs and tests, with bound b in either order. Apply [H1] to the two specified gains. Forward convex averages preserve the all-time uniform limit, so the same X, h remain. $\square$

5.4 Finite-variation Cauchy estimates from maximality

This subsection corresponds to the maximality argument for the FV components in [1, Lemma 4.11]. We prove variation Cauchy estimates on each finite horizon and retain the infinite-time terminal value through the previously constructed all-time uniform limit.

Proposition 5.8 (Normalized terminal gaps and maximality). *Suppose the variations are not Cauchy in probability on fixed $[0, T]$, $T > 0$. Then some $\alpha > 0$ has, beyond every cutoff, pairs with $Q_*(\text{Var}_{[0,T]}(\widetilde{A}_n - \widetilde{A}_k) > \alpha) > \alpha$. Choose*

$$c = \min(\alpha/8, 1), \quad \delta_r = c2^{-(r+1)}, \quad b_r = \delta_r^2/8, \quad U_r = \max(T, r + 1).$$

Choose pairs beyond both r and the martingale cutoff for error b_r . Choose an orientation satisfying

$$Q_*(C_r(T) > \alpha/2) > \alpha/2.$$

Let $\ell_r \geq r$ be its base index and Z_r its continued gain. The sequences

$$g_r = \frac{Z_r(U_r)}{1 + \delta_r}, \quad f_r = \frac{R_{\ell_r}(U_r)}{1 + \delta_r}$$

contradict maximality.

Proof. Positive scaling gives $g_r \in K_1$. The failure event $D_r = \{\tau_r \leq T\}$ satisfies

$$Q_*(D_r) \leq \frac{8b_r}{\delta_r} = \delta_r.$$

Write L_r for the chosen martingale component difference. On survival,

$$N_r(T) - \max(L_r(T), 0) \geq -\delta_r,$$

hence $N_r(T) \geq -\delta_r$ and

$$g_r - f_r \geq \frac{C_r(T) - \delta_r}{1 + \delta_r}.$$

Thus $(f_r - g_r)^+ \leq \delta_r$. On $E_r = \{C_r(T) > \alpha/2\} \setminus D_r$ we have $g_r - f_r \geq \alpha/8$ and $Q_*(E_r) > \alpha/2 - \delta_r \geq \alpha/4$. No extra martingale-error event is required. Choose one orientation per r , never an orientation depending on the sample point. Since $\sum_r Q_*(D_r) < \infty$, Borel–Cantelli gives eventual survival almost surely, hence vanishing lower error. The all-time uniform convergence to the same X , together with $\ell_r \rightarrow \infty$, $U_r \rightarrow \infty$, and $X_\infty = h$, gives $R_{\ell_r}(U_r) \rightarrow h$ almost surely; therefore $f_r \rightarrow h$. No new terminal limit of each R_n is needed. The same original K_1 , its forward convex candidates, its probability closure, and maximality transfer to Q_* by mutual absolute continuity. The persistent-improvement lemma applies to these measurable bases and claims, the vanishing lower error, and the varying events E_r , giving a contradiction. Neither $f_r \in K_1$ nor a common improvement event is required. This proves the finite-horizon variation Cauchy property; at $T = 0$ variation is zero. $\square$

6 Realization of maximal closure claims

This section corresponds to the application of Mémin’s closedness theorem after component convergence in the proof of [1, Theorem 4.2]. We use [C1] to realize the same limit and its terminal value as an original-price integral.

The preceding construction yields gains $R_n = \widetilde{M}_n + \widetilde{A}_n$ in the original market, converging almost surely uniformly over all time to $X \geq -1$ with $X_\infty = h$. Their martingale components are elementary-Émery Cauchy; the variations of their finite-variation differences are Cauchy in probability on every finite horizon.

Theorem 6.1 (Maximal-claim realization). *Under NFLVR every almost-sure maximal $h \in \overline{K_1}^{L^0}$ belongs to K_1 .*

Proof. Apply the test-uniform M expectation Cauchy estimate for the same convexified sequence with error ε^2 . Markov’s inequality gives $Q_*(e_T^J(\widetilde{M}_m, \widetilde{M}_n) > \varepsilon) \leq \varepsilon$ with one cutoff for all tests. The preceding maximality contradiction supplies the variation probability estimate

for the FV differences. The same R_n are strongly adapted and càdlàg, and differences of the $\tilde{A}_n$ are locally BV. Together with the retained all-time uniform limit $X \geq -1$ and its own terminal h , these verify every hypothesis of [C1]. Hence $h \in K_1$. $\square$

7 Fatou and weak-star closedness

7.1 Propagating a terminal lower bound

Lemma 7.1. *Let Y already be admissible for a finite constant $b > 0$ and have a finite terminal limit. Under NFLVR, $Y_\infty \geq -a$ almost surely, with $a > 0$, implies $Y_t \geq -a$ almost surely for all times on one common full-measure set.*

Proof. Suppose $B = \{Y_t < -a\} \in \mathcal{F}_t$ has positive probability for a deterministic t . On B start trading only after t : set $V_u = 1_B(Y_u - Y_t)$ for $u > t$ and $V_u = 0$ for $u \leq t$. Clause [M5], switching from the zero strategy at time t , gives an integral against the original price. Since $Y_t < 0$ on B , it remains bounded below by $-b$. Its terminal claim $1_B(Y_\infty - Y_t)$ is nonnegative and positive on B . Truncating this claim above by 1 gives a nonzero element of $C \cap L_+^\infty$, contradicting NFLVR. The bound therefore holds at each deterministic time; use a countable dense time set and right continuity to obtain one common null set. $\square$

7.2 Fatou closedness

Theorem 7.2. *C_0 is Fatou closed.*

Proof. Normalize a common lower bound to -1 by positive scaling. Let $f_n \in C_0$, $f_n \geq -1$, and $f_n \rightarrow f$ almost surely. Choose $f_n \leq G_n \in K_0$. Since $G_n \geq -1$, the preceding lemma makes its gain 1-admissible, so $G_n \in K_1$. Forward convex compactness provides averages $\tilde{G}_n \rightarrow g$ almost surely. The same averages of f_n tend to f , hence $f \leq g$. Choose a maximal closure claim $h \geq g$ and use Theorem 6.1 to obtain $h \in K_1 \subseteq K_0$. The inequality $f \leq h$ gives $f \in K_0 - L_+^0 = C_0$. $\square$

7.3 Bounded slices in L^1

Lemma 7.3. *Let D be an almost-surely saturated, downward-solid convex cone which is Fatou closed. For every $R \geq 0$, the slice $D_R = \{f \in D \cap L^\infty : \|f\|_\infty \leq R\}$ is weak-star closed.*

Proof. On a finite measure space use the embedding $L^\infty \hookrightarrow L^1$. Any L^1 -convergent sequence in the image of D_R has an almost-surely convergent subsequence. Choose representatives in $[-R, R]$; the limit stays in this interval and belongs to D by Fatou closedness. The image is therefore norm-closed and convex in L^1 , and Hahn–Banach separation makes it weakly closed. The embedding from weak-star L^∞ to weak L^1 is continuous: an L^1 dual test is an

L^∞ function, which belongs to L^1 on a finite measure space, so the composed integral pairing is weak-star continuous. The inverse image of the image slice is exactly D_R , proving the assertion. $\square$

The passage to the next conclusion corresponds to the use of [1, Theorem 2.1]. Here we explicitly combine closedness of bounded slices with the Krein–Šmulian theorem.

Proposition 7.4. *$C = C_0 \cap L^\infty$ is $\sigma(L^\infty, L^1)$ -closed.*

Proof. The preceding lemma applies to the solid convex cone C_0 , so every norm-bounded slice of C is weak-star closed. Identify $L^\infty = (L^1)^*$ on the finite measure space and apply Krein–Šmulian to obtain closedness of C itself. No separability assumption on L^1 is required. $\square$

This proves both conclusions of Theorem 1.2.

A Decomposition, compensation and component norms

This appendix develops bounded-source decomposition, compensation and estimates for the canonical component norms.

A.1 Initial values and stopping

Elementary integrals are sums of increments on intervals $(0, t]$, hence vanish at zero. Completed martingale integrals also vanish there; completed finite-variation integrals do so as uniform limits of Stieltjes integrals. Evaluate indistinguishability with the localized coordinate gains at zero to see that every represented gain X satisfies $X_0 = 0$ almost surely. A decomposition $X = M + A$ admits constant shifts. Replacing its components by $M - M_0$ and $A + M_0$ preserves the gain and, when $X_0 = 0$, normalizes both components to zero. The costs below use these normalized components. For finite τ , write $X_t^\tau = X_{t \wedge \tau}$ for closed stopping. The strict prefix equals X_t for $t < \tau$ and $X_{\tau-}$ for $t \geq \tau$, with $X_{0-} = X_0$. Thus

$$X^\tau = X^{\tau-} + 1_{\{0 < \tau \leq t\}} \Delta X_\tau.$$

Closed-stop variation and component costs require a boundary-jump term in addition to the strict-prefix bound. Integrals use closed stopping; strict prefixes are auxiliary objects for cost comparison.

A.2 The bounded-source decomposition input

The decomposition input used below is the following bounded form of the Bichteler–Dellacherie theorem. We state explicitly which foundational result is used, separately from integral-range closedness.

Theorem A.1 (Bounded good-integrator decomposition). *Under the usual conditions, let S be adapted and càdlàg with $\sup_t |S_t| \leq B$ almost surely for a deterministic $B < \infty$. Suppose elementary integration is continuous from uniform convergence of bounded predictable elementary coefficients to convergence in probability at every finite horizon. Then*

$$S - S_0 = M^S + A^S,$$

where M^S is a zero-start càdlàg local martingale and A^S is zero-start, predictable, càdlàg and locally of finite variation. The centered components are unique up to indistinguishability. The measure remains the original measure, and no market assumption is required.

Proof. We use the bounded case of the Bichteler–Dellacherie characterization as a standard foundational theorem: an adapted càdlàg good integrator on a finite interval is a local martingale plus an adapted càdlàg FV process. The continuity hypothesis is exactly the elementary-integral criterion in [4, Definition 1.1 and Theorem 1.2]; restriction to a finite interval and

time rescaling give that formulation. Almost-sure and everywhere uniform bounds on elementary coefficients give the same criterion after modifying coefficients on a time-zero null set under the usual conditions. Thus, on a fixed horizon, write $S - S_0 = N + D$, with zero-start components. Here D is initially only adapted. Its predictable replacement needs the following additional argument; it is not supplied by arbitrary-decomposition uniqueness.

Choose localizers making N a true martingale up to this horizon and stop further at the first strict passages of $|N|$ and $\text{Var}(D)$ above r . Let the resulting closed stop be σ_r . Before it both quantities are at most r . At the endpoint,

$$|\Delta D_{\sigma_r}| \leq 2B + |\Delta N_{\sigma_r}| \leq 2B + r + |N_{\sigma_r}|, \quad \text{Var}(D^{\sigma_r}) \leq 2r + 2B + |N_{\sigma_r}|.$$

Optional sampling of the localized true martingale makes the last bound integrable. These stops exhaust the fixed horizon, by pathwise boundedness and finite variation. The compensator theorem, stated with its norm estimate in Section A.8, applies to D^{σ_r} and gives

$$S^{\sigma_r} - S_0 = (N^{\sigma_r} + D^{\sigma_r} - (D^{\sigma_r})^p) + (D^{\sigma_r})^p.$$

This is a special decomposition on each coordinate. On common stops, the difference of two predictable FV components is a zero-start predictable FV local martingale, hence zero. This proves compatibility and centered uniqueness. Glue the stopped martingale coordinates and the predictable FV coordinates, then glue over positive integer horizons. Countably many agreements and right continuity give one indistinguishability event. $\square$

The Lean proof constructs the bounded decomposition directly: the sampled Doob rows have increments $\Delta A_j = E[\Delta S_j \mid \mathcal{F}_{t_j}]$ and $\Delta M_j = \Delta S_j - \Delta A_j$; predictable sign tests control their variation, and common stopping, Hilbert convexification and regularization produce compatible canonical limits. The mathematical account here takes the stated bounded characterization as a foundational theorem. It does not take any theorem about closedness of stochastic-integral ranges as input. In particular, the later decomposition of general good integrators uses only this bounded input, after removing large jumps and stopping the remainder; it does not feed back into this proof.

A.3 A Doléans–Dade–Yen decomposition

A general local martingale need not be locally M^2 . An L^1 bound on its running supremum neither makes the unobserved jump at first passage square integrable nor bounds that jump deterministically. Thus an M^2 localizing sequence cannot be assumed for an arbitrary local martingale.

Let N be the local-martingale component of a given special decomposition. We first need a Doléans–Dade–Yen decomposition

$$N = L + B,$$

where L is a local martingale with deterministically bounded jumps after localization, and B is adapted and locally of finite variation. The construction has four steps:

1. A càdlàg path has finitely many jumps above any positive threshold on a finite interval.
2. Enumerate these jumps chronologically by stopping times, obtaining an adapted càdlàg finite-variation process.
3. Choose an exhaustive localization with integrable variation and take predictable projections.
4. Control the compensated martingale jumps at announced times and prove compatibility with stopping.

The enumeration order of a nonmonotone dense skeleton cannot be used as chronological jump order.

For the deterministic path argument, let x be real-valued and càdlàg, $c > 0$, and $T < \infty$. The set

$$J_{c,T} = \{t \in (0, T] : c < |x_t - x_{t-}|\}$$

is finite. Form the finite step path retaining those jumps in time order:

$$P_t = \sum_{s \in J_{c,T}, s \leq t} (x_s - x_{s-}).$$

It starts at zero, is right-continuous with left limits, and has exactly the retained jumps, with zero jump at every other time. Its cumulative absolute jump sum C is increasing. For $0 \leq u \leq v$, its real variation satisfies the exact extended-real identity

$$\text{eVariationOn}(P, [u, v]) = \text{ofReal}(C(v) - C(u)).$$

The increment $C(v) - C(u)$ is the finite sum of absolute retained jumps in $(u, v]$. This identity holds for each of the finite step paths. Local oscillation events on the factorial skeleton give a measurable, pathwise description of a large left jump occurring before a given time. The infimum of the finite jump set, clamped to the horizon if it is empty, is therefore a stopping time. When the set is nonempty, that time attains its earliest member.

Passing to stochastic processes requires iteration of the chronological debut, not merely measurability at each fixed time. This specialized first debut uses factorial-skeleton oscillations and infima of finite sets; it does not assume a generic optional debut theorem. Restrict to the portion after the preceding stop by deterministic path operations. A countable union over rational times describes the successor before T ; when testing times beyond T , include the jump at T explicitly. Iteration produces stopping times whose active coordinates are strictly

increasing, belong to the exact jump set including its endpoint T , and cover every large left jump. After this finite set is exhausted, the coordinates equal $\top$.

For the chronological sum put

$$J_t^{(n)}(\omega) = 1_{\{\tau_n(\omega) \leq t\}}(N_{\tau_n(\omega)} - N_{\tau_n(\omega)-})(\omega),$$

with value zero when $\tau_n = \top$, and set $J = \sum_n J^{(n)}$. Strong adaptation of stopped values and indicators of stopping-time events makes each coordinate strongly adapted; the countable sum is measurable. Pathwise the enumeration becomes inactive after all large jumps have been listed, so this sum agrees exactly with the finite step path above. It inherits zero start, right continuity, left limits, global and local bounded variation, and the exact cumulative-variation identity. It copies every jump larger than c in $(0, T]$ and has zero left jump at other times. It is constant after T , and its finite partial sums are eventually equal to the whole process on each path. This constructs the finite large-jump process. Neither right-continuous level hitting nor debut on a predictable interval algebra is being used as a substitute for debut of a general optional graph.

A.4 Integrable localization of the finite large-jump process

For a zero-start càdlàg local martingale N , write

$$J_t = \sum_{0 < s \leq t \wedge T, |\Delta N_s| > c} \Delta N_s, \quad V_t = \sum_{0 < s \leq t \wedge T, |\Delta N_s| > c} |\Delta N_s|.$$

A càdlàg path has only finitely many jumps larger than $c > 0$ on a finite interval. Thus J has finite variation and is constant after T . Using the deterministic horizons

$$H_n = T + (n + 1),$$

define the strict variation passages

$$\rho_n = \min\{H_n, \inf\{s : V_s > n + 1\}\}.$$

The horizons are cofinal. Monotonicity of both the variation levels and the horizons makes (ρ_n) an increasing sequence of stopping times. Finite total variation and constancy after T imply, on every path, that

$$\rho_n = H_n$$

for all sufficiently large n . Hence (ρ_n) is an exhaustive localizing sequence.

At first use the strict-prefix source

$$C_t^n = \begin{cases} J_t, & t < \rho_n, \\ J_{\rho_n-}, & t \geq \rho_n. \end{cases}$$

It is adapted and càdlàg, with finite variation at most $n + 1$ up to H_n . Its terminal cumulative variation therefore belongs to L^1 . Each C^n supplies normalized adapted càdlàg finite-variation L^1 data. The value-truncation theorem yields a predictable finite-variation limit and a martingale residual. The source agrees exactly with J on the open prefix $t < \rho_n$.

Refine ρ_n with a martingale localizer λ_n and an absolute-passage localizer κ_n , putting $\tau_n = \rho_n \wedge \lambda_n \wedge \kappa_n$. These finite stops satisfy

$$\text{Var}(J^{\tau_n}; [0, H_n]) \leq n + 1 + |\Delta J_{\tau_n}|.$$

The next subsection proves integrability of the boundary term. Thus every closed source J^{τ_n} also supplies finite-variation L^1 data and a value-truncation predictable projection. Fix all these choices as one family. Each of $(\rho_n), (\lambda_n), (\kappa_n)$ is almost surely increasing and exhaustive, so their minimum has the same properties. For $m \leq n$, stopping absorption gives

$$(J^{\tau_n})^{\tau_m} \sim J^{\tau_m}.$$

Here $\sim$ denotes indistinguishability: everywhere monotonicity of the stops is not required. For $m \leq n$, the difference between the upper predictable projection stopped at τ_m and the lower projection is a local martingale, by this closed-source identity. It is also strongly predictable, right-continuous, and of finite variation. Rigidity of predictable finite-variation local martingales makes it indistinguishable from its initial value. Both projections start at zero almost surely, so their difference vanishes. This establishes compatibility on ordered overlaps.

A.5 Removing finite large jumps

Subtract the finite large-jump process from the original local martingale:

$$R_t = N_t - J_t, \quad J_t = \sum_{0 < s \leq t \wedge T, |\Delta N_s| > c} \Delta N_s.$$

The left limits give the exact identity, at every time,

$$\Delta R_t = \Delta N_t - \Delta J_t.$$

For $0 < t \leq T$, the jump ΔJ_t equals ΔN_t when $|\Delta N_t| > c$, and is zero otherwise. Consequently,

$$0 < t \leq T \implies |\Delta R_t| \leq c.$$

For $t > T$, $\Delta J_t = 0$ and hence $\Delta R_t = \Delta N_t$. This bound concerns the source residual. Subtracting a predictable projection or compensator of J requires an additional jump estimate, using announced times or conditional expectations; that estimate does not follow merely from the displayed identity. The nested overlaps of the closed sources and their predictable projections allow us to glue a global predictable process P . Exhaustion of the stopping sequence

then makes $Q = J - P$ a local martingale. The following subsection constructs this P and proves the required predictable-jump estimate.

Integrability of the jump introduced by closed stopping follows after a further localization. Suppose $\tau \leq H$ for a deterministic $H < \infty$, and that τ lies below both one local-martingale localizing coordinate and the absolute strict passage at level r . The first condition identifies N_τ with a bounded optional sample of a true martingale, so it is integrable. The passage condition and existence of left limits imply

$$|N_{\tau-}| \leq r.$$

Since the jump of J is either the corresponding jump of N or zero,

$$|\Delta J_\tau| \leq |N_\tau| + |N_{\tau-}| \leq |N_\tau| + r.$$

Thus ΔJ_τ is integrable. No announcement of τ is needed, and the boundary correction itself need not be predictable. In the preceding common refinement the bound $\tau_n \leq H_n$ supplies the deterministic horizon. This estimate therefore gives integrable total variation of the closed stopped slices and permits their predictable projections.

A.6 Gluing predictable compensators and bounding jumps

For each n , let P^n be the dual predictable projection (compensator) of J^{τ_n} and $Q^n = J^{\tau_n} - P^n$ its martingale residual. The local construction gives ordered overlaps

$$m \leq n \implies (P^n)^{\tau_m} \sim P^m. \quad (1)$$

The gluing theorem for locally finite-variation processes instead requires

$$(P^n)^{\tau_n \wedge \tau_m} \sim (P^m)^{\tau_n \wedge \tau_m} \quad (n, m \in \mathbb{N}). \quad (2)$$

Taking $m = n$ in (1) shows that each coordinate is unchanged by its own stop. On one full-measure set the sequence (τ_n) is increasing; hence $\tau_n \wedge \tau_m = \tau_m$ when $m \leq n$. Stopping absorption and the total order on the indices prove (2). Thus the ordered overlaps supply the all-pairs compatibility needed for gluing.

The local limit in (2) is eventually constant at each point, and predictability passes to that limit. Compatibility and the almost-sure monotonicity and exhaustion of the schedule give right continuity and local finite variation almost surely. Set the whole path to zero on the exceptional null set, obtaining a predictable version P with these path properties everywhere. No replacement of the original compatibility or schedule by pointwise monotone data is necessary. Initially, however, $P_0^n = 0$ and therefore $P_0 = 0$ hold only almost surely. The exceptional set $\{P_0 \neq 0\}$ is measurable at time zero. Set the whole path to zero there

too. This second normalization preserves predictability, right continuity, local finite variation and stopped indistinguishability with every coordinate. All these properties now hold for the same version P .

The closed sources have the same minimum-stop compatibility as (2); subtracting the projections gives it for (Q^n) too. These coordinates are càdlàg true martingales. Martingale gluing produces one strongly adapted càdlàg local martingale $\tilde{Q}$, whose stop at every τ_n is indistinguishable from Q^n . Thus

$$J^{\tau_n} \sim \tilde{Q}^{\tau_n} + P^{\tau_n}.$$

Intersect the countably many stopped-agreement events with the exhaustion event. For any t , some n satisfies $t \leq \tau_n$, proving $J \sim \tilde{Q} + P$. On the normalized representatives define $Q = J - P$. Then $Q \sim \tilde{Q}$, so Q is a local martingale; it is also adapted, càdlàg, locally of finite variation, and zero-start. This proof uses gluing of true-martingale coordinates and transfer by indistinguishability, rather than expanding the time-zero indicators in the definition of a local property.

Set $L = N - Q = R + P$. Then $N = L + Q$, and L is a càdlàg local martingale. In addition to $|\Delta R| \leq c$, we need the following compensator estimate.

Lemma A.2 (Compensator jump estimate). *For fixed $c > 0$ and $T < \infty$, on one full-measure set,*

$$0 < t \leq T \implies |\Delta P_t| \leq c, \quad |\Delta L_t| \leq 2c.$$

Proof. Stop $R = L + (-P)$ at T . All nonzero jumps of the predictable càdlàg finite-variation process P^T are covered by countably many announced predictable stopping graphs (σ_k) , constructed from rational level crossings of cumulative variation. Each completed time is at most $T + 1$. Apply the localized conditional-expectation estimate to L^T and $-P^T$. Without assuming integrability of all jumps of the original local martingale, this gives

$$|\Delta(P^T)_{\sigma_k}| \leq E[c \mid \mathcal{F}_{\sigma_k-}] = c \quad \text{a.s.}$$

Its input is $|\Delta(R^T)_t| \leq c$; after T the stopped process has zero jumps. Combine the countably many null sets and use the graph coverage. Times with zero jump of P satisfy the estimate automatically. This proves the first bound. The second follows from $\Delta L = \Delta R + \Delta P$ and the triangle inequality. For $T = 0$ the relevant interval is empty. At the dummy enumeration value $T + 1$, the stopped process has zero jump. $\square$

Thus compensation first gives the bound c for ΔP , and then the bound $2c$ for ΔL ; it does not assert a bound c for ΔQ . From the original zero-start càdlàg local martingale, a threshold, a finite horizon, and the usual conditions, the construction supplies $N = L + Q$ with these same components. Both components are zero-start càdlàg local martingales and Q is adapted

and locally of finite variation. Predictability of Q is not asserted. For every pathwise stop $\rho \leq T$, the bound $|\Delta(L^\rho)_t| \leq 2c$ also holds at every time outside the same null set. The decomposition at this stage depends on the fixed horizon.

Lemma A.3 (Compatibility between horizons). *Write $P^{(T)}, Q^{(T)}$ for the components at horizon T and $L^{(T)} = N - Q^{(T)}$. For $U \leq T$ we have pathwise*

$$(J^{c,T})^U = J^{c,U}.$$

On one full-measure set, at every time,

$$(P^{(T)})^U = (P^{(U)})^U, \quad (Q^{(T)})^U = (Q^{(U)})^U, \quad (L^{(T)})^U = (L^{(U)})^U.$$

The localizing sequences used within the individual horizons need not agree.

Proof. The large-jump set in $(0, U]$ is contained in that in $(0, T]$. Any time in the difference exceeds U and does not contribute to the finite sum at $t \wedge U$. This proves the first identity. Next consider

$$A = (P^{(T)})^U - (P^{(U)})^U = (Q^{(U)})^U - (Q^{(T)})^U.$$

The first expression is predictable and has finite variation over the whole time axis, because of the finite horizon stop. The second is a difference of càdlàg local martingales. Predictable finite-variation rigidity makes A indistinguishable from its initial value, which is zero. This proves the identity for P , then for Q , and finally for $L = N - Q$. When $U = 0$, the identities follow directly from zero initial values. $\square$

Construct these decompositions for $T_n = n+1$, always starting from the same original local martingale. A countable intersection gives all ordered overlaps, for all times and all $m \leq n$, on one full-measure set. The jump bounds of the same $P^{(T_n)}$ can also be made simultaneous in n .

Lemma A.4 (Gluing the predictable component over all time). *The coherent horizon family gives a zero-start predictable, right-continuous locally finite-variation process P such that, on one full-measure set, for every n, t , $P_{t \wedge T_n} = P_{t \wedge T_n}^{(T_n)}$. On one full-measure set $|\Delta P_t| \leq c$ for every t .*

Proof. Stop $P^{(T_n)}$ at T_n to obtain coordinates of global finite variation. Apply the appropriate ordered overlap according to the order of the two indices: the coordinates agree after stopping at $T_n \wedge T_m$. The deterministic schedule is increasing and exhaustive, so predictable finite-variation gluing applies. Its initial value is almost surely zero. The usual conditions make the exceptional set measurable at time zero; set the whole path to zero on that set. Take a countable intersection to preserve every horizon identity at once. For a given t , choose n with $t \leq T_n$. The left-jump identity before and at the closed stop gives $|\Delta P_t| = |\Delta P_t^{(T_n)}| \leq c$. $\square$

Lemma A.5 (Common refinement of local-martingale coordinates). *Suppose zero-start càdlàg local martingales X^n satisfy $(X^n)^{\tau_n \wedge \tau_m} \sim (X^m)^{\tau_n \wedge \tau_m}$ for an almost-surely increasing exhaustive stopping sequence (τ_n) . There is a càdlàg local martingale Y such that $Y^{\tau_n} \sim (X^n)^{\tau_n}$ for every n .*

Proof. Let $(\eta_j^n)_j$ localize X^n . Choose $k(n)$ by a countable extraction so that $\sigma_n = \tau_n \wedge \eta_{k(n)}^n$ tends to infinity almost surely. Put $\rho_n = \inf_{j \geq n} \sigma_j$. Right continuity of the filtration makes it a stopping time; the sequence increases to infinity and satisfies $\rho_n \leq \tau_n \wedge \eta_{k(n)}^n$. Zero initial values remove the time-zero indicator in a localizing stop, and optional stopping makes $(X^n)^{\rho_n}$ a true martingale. If W is the pointwise limit of the original compatible family, then $W^{\rho_n} \sim (X^n)^{\rho_n}$. The refined coordinates are therefore compatible, and true-martingale gluing applies. Its result Y agrees with W at all ρ_n , hence $Y \sim W$ by exhaustion. Stopping again at the original τ_n proves the required identities. $\square$

Proposition A.6 (A global Doléans–Dade–Yen decomposition). *For a zero-start càdlàg local martingale N and $c > 0$, there are zero-start càdlàg local martingales L, Q with $N = L + Q$, with Q locally of finite variation, and with $|\Delta L_t| \leq 2c$ for every time on one full-measure set.*

Proof. Apply common refinement and gluing to $Q^{(T_n)}$ from the same horizon family. The horizon identities give local bounded variation and zero initial value almost surely. Correct the paths to zero on the exceptional time-zero measurable set, making both properties hold everywhere. The processes P, Q glued from that family satisfy $(J^{c, T_n})^{T_n} = (Q + P)^{T_n}$ simultaneously over all horizons. Set $L = N - Q$. For $t \leq T_n$, $\Delta L_t = \Delta(N - J^{c, T_n})_t + \Delta P_t$. For any positive t choose a horizon containing it and add the two bounds c . The time-zero jump is zero. The other properties of L follow by subtracting local martingales. $\square$

Apply the quadratic construction below to this same global L . Its application to a finite-horizon L^T is another instance of that construction.

Lemma A.7 (Quadratic localization of a bounded-jump local martingale). *Let M be a zero-start càdlàg local martingale, with deterministic jump bounds $|\Delta M_t| \leq b_n$ on each integer horizon $n + 1$, where $b_n \geq 0$. Let (η_n) be a local-martingale localizing sequence and put*

$$\theta_n = \inf\{t \geq 0 : |M_t| > n + 1\}, \quad \rho_n = \eta_n \wedge \theta_n \wedge (n + 1).$$

This sequence is exhaustive, each M^{ρ_n} is a true martingale, and simultaneously at all times

$$|M_t^{\rho_n}| \leq n + 1 + b_n \quad a.s.$$

In particular, each coordinate's terminal value belongs to L^2 .

Proof. Both the horizon-capped absolute passages and (η_n) are localizing sequences, as is their minimum. At zero, the zero initial value identifies the indicator-corrected localizing stop with the usual closed stop, even on paths with $\eta_n = 0$. Thus M^{η_n} is a true martingale. Bounded optional stopping at $\rho_n \leq \eta_n$ gives the same property for M^{ρ_n} . Before passage the left-limit magnitude is at most $n + 1$, and the boundary jump adds at most b_n . This proves the all-time bound. The deterministic majorant is in L^2 , so the terminal value is square integrable. $\square$

Apply the quadratic construction on each finite horizon and glue at the minimum of the coordinate stops, obtaining one global optional quadratic variation V . The localizer root of this same V equals the coordinate's terminal root almost surely, so their expectation costs agree. Applying Davis's inequality to each coordinate gives

$$E \sup_{t \leq n+1} |M_t^{\rho_n}| \leq 6E \sqrt{V_{\rho_n}}.$$

The supremum is detected on a countable time set by càdlàg regularity.

For the fixed-horizon decomposition take $M = L^T$ and $b_n = 2c$. Thus the quadratic and Davis estimates apply to the same chosen DDY component. This V is the quadratic variation of L^T , not of the original N . The coordinate estimates do not assert global H^1 integrability of unstopped L^T .

A.7 General quadratic variation and exact j^1

For bounded-jump martingales we have finite-horizon M^2 quadratic processes, stopping consistency, gluing, uniqueness, terminal roots, and the one-sided Davis estimate with constant 6. Horizonwise jump bounds suffice for local martingales; the DDY decomposition supplies them for its component L^T .

Lemma A.8 (Absolute convergence of finite-variation jump corrections). *For the same DDY decomposition $N = L + Q$, let $V_T = \text{Var}_{[0,T]} Q$. On one full-measure set, for every $T \geq 0$,*

$$\sum_{0 < t \leq T} |\Delta Q_t| \leq V_T, \quad \sum_{0 < t \leq T} |\Delta L_t \Delta Q_t| \leq 2cV_T, \quad \sum_{0 < t \leq T} (\Delta Q_t)^2 \leq V_T^2.$$

Moreover the absolutely convergent correction is

$$D_T = \sum_{0 < t \leq T} ((\Delta N_t)^2 - (\Delta L_t)^2) = 2 \sum_{0 < t \leq T} \Delta L_t \Delta Q_t + \sum_{0 < t \leq T} (\Delta Q_t)^2, \quad |D_T| \leq 4cV_T + V_T^2.$$

Proof. For a right-continuous BV path, the variation of its signed Stieltjes measure at a singleton is the absolute jump. Every finite sum of absolute jumps in $(0, T]$ is bounded by the measure's variation there, proving summability and the first estimate. Apply this to a locally BV path stopped at T . Multiply by $|\Delta L| \leq 2c$ and $|\Delta Q| \leq V_T$ for the other bounds. Expand $\Delta N = \Delta L + \Delta Q$ and add the absolutely convergent series. The common all-time jump-bound event works for every T . $\square$

Lemma A.9 (Squared increments on the same finite grid). *For a finite ordered grid $u_0 \leq \dots \leq u_m$, define*

$$\begin{aligned} E_G(Y, t) &= \sum_{k < m} (Y_{t \wedge u_{k+1}} - Y_{t \wedge u_k})^2, \\ C_G(Y, Z, t) &= \sum_{k < m} (Y_{t \wedge u_{k+1}} - Y_{t \wedge u_k})(Z_{t \wedge u_{k+1}} - Z_{t \wedge u_k}). \end{aligned}$$

For the same DDY components, pathwise at every time,

$$\begin{aligned} E_G(N, t) &= E_G(L, t) + 2C_G(L, Q, t) + E_G(Q, t), \\ |\sqrt{E_G(N, t)} - \sqrt{E_G(L, t)}| &\leq \text{Var}_{[0, t]}(Q). \end{aligned}$$

The estimate contains no grid-dependent constant.

Proof. Expanding the square of each cell increment gives the first identity. Cauchy–Schwarz gives $C_G(Y, Z, t) \leq \sqrt{E_G(Y, t)}\sqrt{E_G(Z, t)}$. Together with the square expansion this proves the root triangle inequality $\sqrt{E_G(Y + Z, t)} \leq \sqrt{E_G(Y, t)} + \sqrt{E_G(Z, t)}$. The sum of absolute increments over ordered cells is at most path variation, so

$$\sqrt{E_G(Q, t)} \leq \sum_{k < m} |Q_{t \wedge u_{k+1}} - Q_{t \wedge u_k}| \leq \text{Var}_{[0, t]}(Q).$$

Apply the root triangle inequality to both $N = L + Q$ and $L = N - Q$. □

Lemma A.10 (Factorial cross-increment limits). *Let f be a real càdlàg path and Q a right-continuous locally BV path. Let η_T be the signed Stieltjes measure of $Q_t^T = Q_{t \wedge T}$ and G_r the grid $k/(r+1)!$, $0 \leq k \leq (r+1)(r+1)!$. Then*

$$C_{G_r}(f, Q, T) \longrightarrow \int_{(0, T]} \Delta f_s d\eta_T(s).$$

Consequently, for the same DDY decomposition, at every path and finite horizon,

$$E_{G_r}(N, T) - E_{G_r}(L, T) \longrightarrow 2 \int_{(0, T]} \Delta L_s d\eta_T(s) + \int_{(0, T]} \Delta Q_s d\eta_T(s).$$

Proof. For the half-open cell $(a_r(s), b_r(s)]$ containing $s > 0$, put $D_r f(s) = f(b_r(s) \wedge T) - f(a_r(s) \wedge T)$. The endpoints approach s with $a_r(s) < s \leq b_r(s)$, so $D_r f(s) \rightarrow \Delta f_s$ for $s \leq T$. This also holds if s itself is a mesh point. Compact boundedness of càdlàg paths gives $|D_r f(s)| \leq 2 \sup_{u \leq T} |f(u)|$. Dominated convergence under the finite measure $|\eta_T|$ makes the L^1 error over $(0, T]$ tend to zero, hence the signed integrals converge. The integral on each stopped cell equals $(f(b \wedge T) - f(a \wedge T))(Q(b \wedge T) - Q(a \wedge T))$. For sufficiently large r , the cells cover $(0, T]$, so the integral is the cross-increment sum. Apply this with $f = L$ and $f = Q$ and use the finite square expansion to obtain the second limit. □

Lemma A.11 (Jump integrals as series). *For a real càdlàg path f and a finite signed measure η ,*

$$\int_{(0,T]} \Delta f_t d\eta(t) = \sum_{t \in (0,T]} \Delta f_t \eta(\{t\}).$$

In particular, for η_T and the same DDY components, pathwise,

$$E_{G_r}(N, T) - E_{G_r}(L, T) \longrightarrow 2 \sum_{0 < t \leq T} \Delta L_t \Delta Q_t + \sum_{0 < t \leq T} (\Delta Q_t)^2.$$

Proof. For each n , only finitely many times in $(0, T]$ have $|\Delta f_t| > 1/(n+1)$. Thus the support of nonzero jumps is countable. The uniform bound on the measurable cell approximations passes to their limit Δf , making it bounded on $(0, T]$ and integrable for $|\eta|$. Decompose the support into disjoint singletons and sum the integrals to obtain the first identity. For a stopped FV path, $\eta_T(\{t\}) = \Delta Q_t$ whenever $0 < t \leq T$, including the terminal jump. Apply the identity with $f = L, Q$ and rewrite the preceding integral limit to obtain the series formula. $\square$

Lemma A.12 (Nonnegativity of the corrected candidate). *Let V now denote the glued quadratic variation of L . On one full-measure set, $V_t + D_t \geq 0$ for every t .*

Proof. On coordinate n , at times before its localizer and horizon, the original and stopped squared-increment sums agree on the same factorial grids. Use exactly the forward weights and cutoff retained by the construction of the quadratic variation of L^{τ_n} . Full-sequence convergence of the correction is preserved by these weights. Hence the convex squared sums of N^{τ_n} tend to the coordinate variation plus D_t and remain nonnegative. Coordinate agreement with V and exhaustion remove the stops. Take the countable intersection of coordinate events before choosing t . $\square$

Lemma A.13 (Regularity of the correction). *The processes D and $V + D$ are strongly adapted and are right-continuous and locally of finite variation on one full-measure set.*

Proof. On a finite horizon, the coefficients $a_s = 2\Delta L_s \Delta Q_s + (\Delta Q_s)^2$ are absolutely summable. Dominated convergence for the series makes the cumulative sum $\sum_{s \leq t} a_s$ right-continuous. Write $a_s = (a_s + |a_s|) - |a_s|$; both cumulative sums are bounded increasing functions, so their difference has finite variation. These cumulative sums locally agree with D . Adaptation follows from the pathwise limit of the measurable finite-grid square differences at each time. Add the regular process V . $\square$

Lemma A.14 (Monotonicity of the candidate). *On one full-measure set $V + D$ is increasing at all times.*

Proof. Use the same convex convergence as in the nonnegativity argument. For $s \leq t$ choose a coordinate whose localizer exceeds $t + 1$. Approximate s, t from the right by capped factorial

grid points $u_r \leq v_r$ below that localizer. The tail condition on the weights ensures that later convex sums use grids refining this common coarse grid. Squared sums are increasing between grid points, so convex averaging preserves their order. Pass first to the convex limit, identify with the candidate, and then let $r \rightarrow \infty$ by right continuity. Coordinate events were intersected before choosing s, t . $\square$

Lemma A.15 (An everywhere regular representative). *Under the usual conditions there is a strongly adapted $A \sim V + D$ whose paths are right-continuous, increasing, locally BV, have left limits, and satisfy $A_0 = 0$.*

Proof. The set where right continuity or monotonicity fails is null, hence measurable at time zero. Replace entire paths there by zero. This preserves adaptation and indistinguishability. Initial values are zero by the empty jump sum and zero start of V . Increasing paths have local bounded variation and left limits. $\square$

Lemma A.16 (Jump-square identity). *On one full-measure set, $\Delta A_t = (\Delta N_t)^2$ for all times.*

Proof. For an absolutely summable local family a_s , dominated convergence as $u \uparrow t$ shows that the left limit of $\sum_{0 < s \leq u} a_s$ is $\sum_{0 < s < t} a_s$. Thus its jump at $t > 0$ is a_t . Apply this to the correction and use $\Delta V_t = (\Delta L_t)^2$:

$$\Delta(V + D)_t = (\Delta L_t)^2 + 2\Delta L_t \Delta Q_t + (\Delta Q_t)^2 = (\Delta N_t)^2.$$

Both sides are zero at time zero. Transfer the all-time identity to A by indistinguishability. $\square$

Lemma A.17 (Closed stopping of the correction). *For a finite cutoff τ , the correction formed from L^τ, Q^τ equals D^τ . If V^n is the quadratic coordinate at localizer τ_n , then*

$$A^{\tau_n} \sim (V^n)^{\tau_n} + 2 \sum_{0 < s \leq \cdot} \Delta L_s^{\tau_n} \Delta Q_s^{\tau_n} + \sum_{0 < s \leq \cdot} (\Delta Q_s^{\tau_n})^2.$$

Proof. Closed-stop jumps equal original jumps for $s \leq \tau$ and are zero afterward. Summing to t is therefore summing original jumps to $t \wedge \tau$, including the boundary. Combine this pathwise identity with coordinate agreement for V and $A \sim V + D$. $\square$

Lemma A.18 (The finite-variation square residual). *For a càdlàg locally BV path Q and finite T ,*

$$Q_T^2 - Q_0^2 - \sum_{0 < s \leq T} (\Delta Q_s)^2 = 2 \int_{(0, T]} Q_{s-} dQ_s.$$

Proof. Factorial left-step approximations converge to Q_{s-} , since the left cell endpoint approaches every positive time strictly from below. Compact boundedness and finite Stieltjes variation give dominated convergence. On each finite grid the identity

$$2 \sum Q_u(Q_v - Q_u) + \sum (Q_v - Q_u)^2 = Q_T^2 - Q_0^2$$

telescopes, with endpoints capped at T . The preceding cross-increment and atomic integral results identify the square-sum limit. Pass to the limit. $\square$

For the same DDY components, put $I_Q(t) = \int_{(0,t]} Q_{s-} dQ_s$. Since $Q_0 = 0$, on a common full-measure set,

$$N_t^2 - A_t = (L_t^2 - V_t) + 2 \left(L_t Q_t - \sum_{0 < s \leq t} \Delta L_s \Delta Q_s + I_Q(t) \right).$$

We next localize and pass martingale identities through dominated limits.

Lemma A.19 (Common localization with integrable variation). *For the same L, Q , choose finite $\rho_n \uparrow \infty$, $\rho_n \leq n + 1$, such that L^{ρ_n}, Q^{ρ_n} are true martingales,*

$$|L_t^{\rho_n}| \leq n + 1 + 2c, \quad |Q_s| \leq n + 1 \quad (s < \rho_n).$$

The first bound holds at all times on one full-measure set. Moreover, $\text{Var}_{[0, \rho_n]} Q$ is integrable and dominates every horizon variation of Q^{ρ_n} .

Proof. The cumulative variation $A_t = \text{Var}_{[0,t]} Q$ is adapted by finite-grid representation, right-continuous and increasing. Take the minimum of true localizers for both martingales, the level $n + 1$ passages of $|L|$ and A , and the horizon $n + 1$. This is exhaustive and preserves true martingales, using zero start to remove the time-zero indicators. The bounded jumps of L give the closed-stop bound. Before the stop $|Q_s| \leq A_s \leq n + 1$. With $a = n + 1$, the variation jump identity gives

$$A_{\rho_n} \leq a + |\Delta Q_{\rho_n}| \leq 2a + |Q_{\rho_n}|.$$

The last stopped value is integrable by bounded optional sampling. Measurability and this domination give integrability of A_{ρ_n} ; stopped variation equals $A_{t \wedge \rho_n} \leq A_{\rho_n}$. $\square$

On this localization the finite-grid sums $I_G(Q^\rho, Q^\rho)$, $I_G(Q^\rho, L^\rho)$, and $I_G(L^\rho, Q^\rho)$ are true martingales. For the first two replace coefficients by $Q_s 1_{\{s < \rho\}}$, which are bounded and adapted; the sums agree because integrator increments after stopping vanish. For the third use the closed-stop bound on L . Neither boundedness nor square integrability of Q_ρ is needed.

Lemma A.20 (Martingale limits under integrable domination). *If martingales M^k converge almost surely at each time to strongly adapted Y , and $|M_t^k| \leq B_t$ for one integrable B_t at each time, then Y is a martingale.*

Proof. The limit is integrable by domination. For $s \leq t$ and $E \in \mathcal{F}_s$, dominated convergence in the identities $\int_E M_s^k = \int_E M_t^k$ yields $\int_E Y_s = \int_E Y_t$. Use the characterization of conditional expectation. $\square$

Lemma A.21 (Self-integral and cross-term local martingales). *The processes*

$$I_t = \int_{(0,t]} Q_{s-} dQ_s, \quad K_t = L_t Q_t - \sum_{0 < s \leq t} \Delta L_s \Delta Q_s$$

are local martingales.

Proof. Fix a common localization and write $\bar{L} = L^{\rho_n}$, $\bar{Q} = Q^{\rho_n}$, $a = n + 1$, $b = n + 1 + 2c$, and $A = \text{Var}_{[0, \rho_n]} Q$. Strict-prefix coefficients give the full self-integral bound

$$|I_G(\bar{Q}, \bar{Q})_t| \leq a \text{Var}_{[0,t]}(\bar{Q}) \leq aA.$$

Factorial sums converge to the stopped Stieltjes integral. Compare the square identities before and after stopping and use the stopped jump-square series to identify the limit with $I_{t \wedge \rho_n}$. For cross terms, the finite-grid product formula with last grid point u_G is

$$I_G(\bar{L}, \bar{Q})_t + I_G(\bar{Q}, \bar{L})_t = \bar{L}_{t \wedge u_G} \bar{Q}_{t \wedge u_G} - C_G(\bar{L}, \bar{Q}, t).$$

The product is bounded by bA and the cross sum by $2bA$, so the total is bounded by $3bA$. Its factorial limit is $K_{t \wedge \rho_n}$ by the cross-increment limit, the jump-series identity, and stopping compatibility. The pathwise limits retain strong adaptation. Integrability of A and the preceding lemma make both closed stops true martingales. Zero start agrees with the localizing indicator convention. We need not separately construct a completed integral $\int Q_- dL$. $\square$

Theorem A.22 (Quadratic variation of a local martingale). *Every zero-start strongly adapted càdlàg local martingale N on a filtered probability space satisfying the usual conditions has an adapted càdlàg increasing process V with*

$$V_0 = 0, \quad \Delta V = (\Delta N)^2, \quad N^2 - V \text{ a local martingale.}$$

The jump identity holds simultaneously at all times almost surely, and V is unique up to indistinguishability.

Proof. Use the regularized corrected candidate. The square formula gives $N^2 - V = (L^2 - [L]) + 2(K + I)$. Transfer the second summand to the indistinguishable regular process $N^2 - V - (L^2 - [L])$, align localizers, and add the local martingales. For uniqueness, allow even adapted càdlàg locally BV candidates V, W without monotonicity, provided their initial values and jumps agree and both square residuals are local martingales. Then $D = V - W$ is a locally BV local martingale with zero jumps, hence has continuous paths almost surely. Stop at

integer horizons. The left-limit process is predictable and agrees with this continuous process; correct the common null set and use predictable FV local-martingale rigidity to make the difference zero. Countably many horizons give all-time equality. $\square$

Thus

$$[N]_T = [L]_T + D_T.$$

Different DDY thresholds and schedules give indistinguishable certificates for the same source. Nonconvex full-sequence square-sum convergence is not inferred from this existence result alone.

Lemma A.23 (Stopping consistency). *For any stopping time τ , possibly infinite,*

$$[N^\tau] \sim [N]^\tau.$$

The left-hand side may be independently constructed.

Proof. A zero-start right-continuous local martingale remains one after arbitrary closed stopping: stop each true localized martingale again, commute stops, and remove the indicator at zero. Thus existence applies to N^τ . The candidate V^τ retains adaptation, càdlàg regularity, monotonicity, and zero start. Before and at τ the jumps agree with the original ones and afterward are zero. Also

$$(N^\tau)^2 - V^\tau = (N^2 - V)^\tau$$

is a local martingale. Uniqueness for the stopped source proves the claim. $\square$

Lemma A.24 (The root triangle inequality). *For zero-start strongly adapted càdlàg local martingales X, Y ,*

$$\sqrt{[X + Y]_t} \leq \sqrt{[X]_t} + \sqrt{[Y]_t}$$

at all times almost surely.

Proof. Write $V = [X]$, $W = [Y]$, $S = [X + Y]$. For real a put

$$A^a = aS + (1 - a)V + (a^2 - a)W.$$

This is adapted càdlàg locally BV with zero start, the correct squared jumps for $X + aY$, and square residual

$$(X + aY)^2 - A^a = a((X + Y)^2 - S) + (1 - a)(X^2 - V) + (a^2 - a)(Y^2 - W).$$

The uniqueness comparison, which does not require monotonicity, gives $A^a \sim [X + aY]$. Intersect the events for rational a . At every time on this common event,

$$0 \leq aS_t + (1-a)V_t + (a^2-a)W_t = W_t a^2 + (S_t - V_t - W_t)a + V_t$$

for all rational a , hence all real a by continuity. Its discriminant gives $(S_t - V_t - W_t)^2 \leq 4V_t W_t$. Since $V_t, W_t \geq 0$, the root bound follows. $\square$

Lemma A.25 (Finite-grid L^1 Davis estimates). *For a real discrete martingale Y , set*

$$Y_n^* = \max_{0 \leq k \leq n} |Y_k|, \quad R_n = \sqrt{Y_0^2 + \sum_{k < n} (Y_{k+1} - Y_k)^2}.$$

Both are integrable, and

$$EY_n^* \leq 6ER_n, \quad ER_n \leq 7EY_n^*.$$

No square-integrability hypothesis is needed.

Proof. They are bounded respectively by $\sum_{k \leq n} |Y_k|$ and $|Y_0| + \sum_{k < n} |Y_{k+1} - Y_k|$, proving integrability. A bounded adapted coefficient times a martingale increment is integrable with zero conditional expectation; hence the finite left-endpoint integral has mean zero. Integrate the pathwise Davis inequality, whose coefficients are bounded by one, to obtain the first estimate. For the reverse inequality use, with $m \geq |x|$ and $m, q \geq 0$,

$$F(x, m, q) = -2m + \sqrt{m^2 + q} + \frac{m^2 - x^2}{2\sqrt{m^2 + q}},$$

interpreting division by zero as zero. With $M = \max(m, |x + d|)$,

$$F(x + d, M, q + d^2) - F(x, m, q) \leq -\frac{xd}{\sqrt{m^2 + q}} + 5(M - m).$$

To check this put $r = \sqrt{m^2 + q}$ and $u = \sqrt{M^2 + q + d^2}$. For $r > 0$ and $M \leq 2m$, the square-root tangent estimate and $u \geq r$ bound the left side plus xd/r by $(M^2 - m^2)/r - 2(M - m) \leq M - m$. For $M > 2m$, $M = |x + d|$, $|d| \leq M + m \leq 3(M - m)$, and $u - r \leq 4(M - m)$. Also $xd/r \leq |d|$. The new potential numerator $M^2 - (x + d)^2$ is zero and the old numerator is nonnegative, giving the stated bound. If $r = 0$ then $x = m = q = 0$ and direct calculation suffices. Sum with $x = Y_k, m = Y_k^*, q = R_k^2, d = Y_{k+1} - Y_k$. Since the initial potential is nonpositive and the final one at least $R_n - 2Y_n^*$,

$$R_n \leq 7Y_n^* - \sum_{k < n} \frac{Y_k}{\sqrt{R_k^2 + (Y_k^*)^2}} (Y_{k+1} - Y_k).$$

The coefficients are adapted and bounded by one; take expectations. $\square$

Apply these estimates on the same DDY common localizers to the original $N^{\rho_j} = L^{\rho_j} + Q^{\rho_j}$. Concavity prevents transferring the reverse estimate directly to roots of convexified energies; we first need convergence of the original square sums.

Lemma A.26 (Unconvexified square sums of a bounded martingale). *Under the usual conditions, let M be a strongly adapted càdlàg martingale satisfying $|M_t| \leq C$ on every path up to a finite horizon T . On factorial grids π_i , put*

$$I_i = 2 \sum_{[u,v] \in \pi_i} M_u(M_v - M_u), \quad E_i = \sum_{[u,v] \in \pi_i} (M_v - M_u)^2.$$

If Y, V are the martingale and increasing components retained by the quadratic construction, then $I_i \rightarrow Y_T$ and $E_i \rightarrow V_T$ in L^2 . If the bound holds almost surely at all times on one common set, then $E_i \rightarrow V_T$ in probability as well.

Proof. Let ν be the predictable energy measure of the same V . The grid left-endpoint coefficients H_i are bounded predictable elementary processes converging pointwise to M_- on $(0, T]$. The measure ν is finite and gives no mass to time zero or to times after T . Dominated convergence yields $H_i \rightarrow M_-$ in $L^2(\nu)$. The grid-independent isometry

$$\|((H_i - H_j) \cdot M)_T\|_{L^2(P)} = \|H_i - H_j\|_{L^2(\nu)}$$

makes I_i Cauchy in $L^2(P)$; denote its limit by I . The retained forward convex weights preserve this limit. With those same weights, however, the integral sums converge almost surely to Y_T . Uniqueness of the probability limit gives $I = Y_T$. Use $I_i + E_i = M_T^2 - M_0^2$ and the retained square decomposition to obtain $E_i \rightarrow V_T$.

If the bound fails only on one common null set, the usual conditions put that set in $\mathcal{F}_0$. Replacing the process by zero there produces an everywhere bounded indistinguishable càdlàg martingale. The same weights and components supply quadratic data for this source too. Original square sums of indistinguishable sources agree almost surely, so their probability convergence transfers back. $\square$

Theorem A.27 (Reverse Davis for general local martingales). *Under the usual conditions let N be a zero-start strongly adapted càdlàg local martingale. For any intrinsic quadratic variation and finite T ,*

$$E\sqrt{[N]_T} \leq 7EN_T^*,$$

allowing infinite expectations.

Proof. At a common DDY stop ρ_j , the L component is uniformly bounded almost surely, so its original square sums converge in probability. Their difference from the original N^{ρ_j} sums

converges pathwise to the FV correction. Stopping consistency identifies the sum limit with $[N]_{T \wedge \rho_j}$. Extract an almost-surely convergent subsequence and take roots. The grid maxima are bounded by $(N^{\rho_j})_T^*$, so finite-grid reverse Davis and Fatou give

$$E\sqrt{[N]_{T \wedge \rho_j}} \leq 7E(N^{\rho_j})_T^* \leq 7EN_T^*.$$

Exhaustion makes the root eventually equal to $\sqrt{[N]_T}$ almost surely; apply Fatou again and use uniqueness to cover any certificate. $\square$

Lemma A.28 (Uniform integrability of convex square roots). *Fix a common stop and finite horizon. Let E_i, B_i be the original and L -component factorial square sums. For any forward weights W , $\sqrt{W(E)_k}$ is uniformly integrable.*

Proof. Let $A = \text{Var}(Q^{\rho_j})_\infty$, which is integrable. The grid root bound gives $\sqrt{E_i} \leq \sqrt{B_i} + A$, hence $E_i \leq 2B_i + 2A^2$. Convex weights imply

$$W(E)_k \leq 2W(B)_k + 2A^2, \quad \sqrt{W(E)_k} \leq \sqrt{2W(B)_k + 2A^2} \leq \sqrt{2W(B)_k} + \sqrt{2}A \leq 2\sqrt{W(B)_k} + 2(1+A).$$

The stopped L is bounded, and its quadratic approximation estimates make B_i uniformly integrable, as are their forward convex averages. The remaining majorant is one integrable function. Integrability of A^2 is not needed. $\square$

Lemma A.29 (Terminal root convergence after closed stopping). *For a common stop ρ and horizon T , let V^L and its weights be from the finite-horizon quadratic construction for L^ρ , and E_k the same weighted energies of N^ρ . With DDY correction C ,*

$$\sqrt{E_k(T)} \longrightarrow \sqrt{V_T^L + C_{T \wedge \rho}} \quad \text{in } L^1,$$

and the limit is integrable.

Proof. Stopped grid increments are original increments evaluated at stopped times, so the correction limit holds even when $t > \rho$. The same weights give convergence of the L energy and the correction, hence almost-sure energy convergence. Continuity of square root and the preceding uniform integrability permit Vitali's theorem. The L^2 terminal value of L^ρ supplies the finite-horizon quadratic data. $\square$

Lemma A.30 (Identification with the original stopped quadratic variation). *If $V = [N]$, then*

$$V_T^L + C_{T \wedge \rho} = V_{T \wedge \rho}$$

almost surely, and the preceding L^1 limit is $\sqrt{V_{T \wedge \rho}}$.

Proof. Both V^L and $[L]$ stopped at ρ and T are quadratic certificates for $(L^\rho)^T$: their residuals are martingales and jumps are the source squares. Uniqueness gives $V_T^L = [L]_{T \wedge \rho}$. The indistinguishable identity $V = [L] + C$ also holds at the random time $T \wedge \rho$. Substitute into the root limit. $\square$

Lemma A.31 (Davis for a general local martingale). *Under the usual conditions let N be a zero-start strongly adapted càdlàg local martingale. For finite T ,*

$$E \sup_{t \leq T} |N_t| \leq 6E \sqrt{[N]_T}$$

as extended nonnegative expectations. A finite right side implies integrability of the maximum.

Proof. Fix a common stop ρ . A coarse factorial-grid maximum is below every later refined-grid maximum. Average with the tail weights, use finite-grid Davis, and then root concavity to bound its expectation by $6E \sqrt{E_k(T)}$. The preceding L^1 convergence replaces this by $6E \sqrt{[N]_{T \wedge \rho}}$ in the limit. Refine the coarse grid; right continuity and monotone convergence give

$$E \sup_{t \leq T} |N_t^\rho| \leq 6E \sqrt{[N]_{T \wedge \rho}}.$$

The stopped maximum is integrable at this stage. Now exhaust with ρ_n . Eventually the stopped maximum equals the original one almost surely; the right side is at most $6E \sqrt{[N]_T}$ by monotonicity. Fatou removes localization, and uniqueness removes the choice of certificate. $\square$

Lemma A.32 (Infinite horizons and all-decomposition costs). *Put $R_N = \sup_{t \geq 0} \sqrt{[N]_t}$. Then*

$$E \sup_t |N_t| \leq 6ER_N.$$

For zero-start Y , consider every indistinguishable decomposition $Y = N + A$ with zero-start strongly adapted càdlàg local martingale N and strongly adapted càdlàg locally FV A . Define

$$\mathcal{J}(Y) = \inf_{Y=N+A} E[R_N + \text{Var}(A)_\infty].$$

Then $E \sup_t |Y_t| \leq 6\mathcal{J}(Y)$, with no finite-cost assumption and infimum of the empty family equal to infinity.

Proof. Increase integer horizons and use monotone convergence for the first bound. The root supremum is measurable by integer-time detection and monotonicity. For each decomposition, zero start gives

$$\sup_t |Y_t| \leq \sup_t |N_t| + \text{Var}(A)_\infty.$$

Take expectations, apply Davis, and take the infimum over all decompositions. Existence and uniqueness of quadratic variation make its choices immaterial. $\square$

Lemma A.33 (Closed stops and strict-prefix extension costs). *For any stopping time τ , possibly infinite, $\mathcal{J}(Y^\tau) \leq \mathcal{J}(Y)$. Set*

$$\mathcal{J}_{\tau-}(Y) = \inf\{\mathcal{J}(L) : L_t = Y_t \text{ for all } t < \tau \text{ a.s.}\}.$$

Agreement uses one common null set. Then

$$\mathcal{J}_{\tau-}(Y) \leq \mathcal{J}(Y^\tau) \leq \mathcal{J}(Y), \quad E \sup_{t < \tau} |Y_t| \leq 6\mathcal{J}_{\tau-}(Y).$$

The empty supremum at $\tau = 0$ is zero.

Proof. Stop both components of each decomposition. Stopping consistency bounds the root by R_N ; variation does not increase. Take the infimum. The stopped process itself extends the strict prefix. Every extension L satisfies $\sup_{t < \tau} |Y_t| \leq \sup_t |L_t|$ almost surely; apply the all-time estimate and take the extension infimum. $\square$

Lemma A.34 (Triangle inequalities for both costs). *Under the same assumptions,*

$$\mathcal{J}(X + Y) \leq \mathcal{J}(X) + \mathcal{J}(Y), \quad \mathcal{J}_{\tau-}(X + Y) \leq \mathcal{J}_{\tau-}(X) + \mathcal{J}_{\tau-}(Y).$$

The second inequality uses the same cutoff τ , which need not be a stopping time.

Proof. Add decompositions $X = N + A$ and $Y = M + B$ to obtain the admissible decomposition $X + Y = (N + M) + (A + B)$. The quadratic-root triangle inequality gives $R_{N+M} \leq R_N + R_M$ almost surely, while variation subadditivity gives $\text{Var}(A + B)_\infty \leq \text{Var}(A)_\infty + \text{Var}(B)_\infty$. All-time variation is the supremum of integer-horizon variations and is measurable for right-continuous adapted paths. We may therefore add the expectations, bounding the sum decomposition's cost by the two original costs. Taking independent infima over the two decomposition families proves the first inequality. Adding two global extensions of the same strict prefix gives an extension of the sum target. Apply the first inequality and take independent infima over extensions for the second. Infinite costs and empty families are included. $\square$

Lemma A.35 (Negation and successive differences). *Under the same assumptions, $\mathcal{J}(-X) = \mathcal{J}(X)$ and $\mathcal{J}_{\tau-}(-X) = \mathcal{J}_{\tau-}(X)$. For one fixed τ , therefore,*

$$\mathcal{J}_{\tau-}(Z_k - X) \rightarrow 0 \implies \mathcal{J}_{\tau-}(Z_{k+1} - Z_k) \rightarrow 0.$$

Proof. A quadratic certificate for N also satisfies the conditions for $-N$: the square residual is unchanged and squared jumps ignore the sign. Variation is invariant under negation too, so $(N, A) \mapsto (-N, -A)$ preserves costs. Taking infima and applying negation twice proves the identity for $\mathcal{J}$. Negating extensions of the same prefix gives the corresponding identity for the outer infimum. The triangle inequality and symmetry give

$$\mathcal{J}_{\tau-}(Z_{k+1} - Z_k) \leq \mathcal{J}_{\tau-}(Z_{k+1} - X) + \mathcal{J}_{\tau-}(Z_k - X).$$

Combine this with nonnegativity for the limit. Infinite costs are allowed, but the lemma does not itself construct a common stop. $\square$

Lemma A.36 (Components from finite strict-prefix cost). *If $\tau \leq T$ is a finite stop and $\mathcal{J}_{\tau-}(X) < r$, there are $X' \sim X$, a local martingale N , and globally BV A , both adapted and càdlàg, such that $X'_t = N_t + A_t$ pointwise for $t < \tau$, and*

$$E \sup_{t \leq T} |N_{t \wedge \tau}| + E \text{Var}_{[0, T]}(A^{\tau-}) < 6r.$$

Proof. Choose an extension Y and decomposition $Y = N + B$ with cost less than r . Put $A = B^T$. Its strict-prefix variation is bounded by the original total variation, and all-time Davis bounds the stopped martingale maximum. Define X' as $N + A$ on the strict prefix and as X elsewhere. On the common event of extension and decomposition agreement, $X' = X$ at all times. This gives both the exact prefix identity and indistinguishability. $\square$

Lemma A.37 (Component sequences from summable costs). *Under the usual conditions let $\tau \leq T$ be one finite stopping time. If $\sum_k \mathcal{J}_{\tau-}(X_k) < \infty$, one can choose the preceding exact representatives X'_k and components (N_k, A_k) with*

$$\sum_k \left(E \sup_{t \leq T} |N_{k, t \wedge \tau}| + E \text{Var}_{[0, T]}(A_k^{\tau-}) \right) \leq 6 \left(\sum_k \mathcal{J}_{\tau-}(X_k) + 1 \right) < \infty.$$

On one full-measure set $X'_{k, t} = X_{k, t}$ for every k, t , and

$$\sum_k \left(\sup_{t \leq T} |N_{k, t \wedge \tau}| + \text{Var}_{[0, T]}(A_k^{\tau-}) \right) < \infty.$$

Proof. Put $c_k = \mathcal{J}_{\tau-}(X_k)$ and $\varepsilon_k = 2^{-k-1}$. Each c_k is finite, so $c_k < c_k + \varepsilon_k$. The preceding lemma supplies representatives and components with cost less than $6(c_k + \varepsilon_k)$. Sum and use $\sum_k \varepsilon_k = 1$ to obtain the expectation bound. The martingale maxima and strict-prefix variations are measurable; Tonelli gives pathwise finiteness of their sum almost surely. Intersect with the countably many indistinguishability events to retain agreement with all original targets, at every time, on that same set. $\square$

The common stop and summability are inputs here. These arbitrary decomposition components still require a separate bridge to actual original-price integrals.

Lemma A.38 (Strict-prefix comparison and boundary jumps). *For finite $\tau \leq T$,*

$$\mathcal{J}(X^{\tau-}) \leq 2\mathcal{J}(X).$$

If $X_0 = 0$, then

$$\mathcal{J}_{\tau-}(X) \leq \mathcal{J}(X^{\tau-}) \leq 2\mathcal{J}_{\tau-}(X).$$

For zero-start adapted càdlàg X ,

$$\mathcal{J}(X^\tau) \leq 2\mathcal{J}_{\tau-}(X) + E|\Delta X_\tau|.$$

Infinite costs are allowed.

Proof. For $X = N + A$, stopping A at T leaves the prefix unchanged. Use the decomposition $X^{\tau-} = N^\tau + (A^{\tau-} - B^\tau(N))$, where B^τ is the boundary jump process. The jump-square identity and monotonicity give $|\Delta N_\tau| \leq \sqrt{[N]_\tau}$. The stopped root does not increase; variation increases by at most this jump, so the cost is at most twice the original. Take infima. An extension agreeing before τ has the same left limit there; at $\tau = 0$ use zero start. Apply the same estimate to extension decompositions and take infima. The reverse bound follows because the strict prefix is itself an extension. Finally $B^\tau(X)$ is adapted zero-start FV with cost at most $E|\Delta X_\tau|$. Add it to $X^{\tau-}$ and use the triangle inequality. $\square$

These results concern the zero-start core and its extension costs. The next subsection constructs common prelocal control and supplies the algebraic inputs for the main sequence. Nonzero initial values are separate coordinates. A finite M^2 quadratic root is not itself exact j^1 , and the stopping-jump seminorm is distinct from the root-plus-variation cost. Quadratic variation does not detect constant martingales, which is why components are centered.

A.8 From an arbitrary decomposition to canonical component bounds

We use the compensator theorem in the following precise form. For an adapted zero-start càdlàg process D of integrable total variation, constant after a finite deterministic time, there is a predictable zero-start càdlàg process D^p of integrable variation such that $D - D^p$ is a uniformly integrable martingale. It is the *dual predictable projection*, characterized by

$$E \int K dD^p = E \int K dD \quad \text{for every bounded predictable } K.$$

This is the compensator theorem, not the assertion that $D_t^p = E[D_t \mid \mathcal{F}_{t-}]$. Existence is the standard dual-projection theorem; the characterization and jump formula are recalled in [5, Section 4.3]. Its variation bound is

$$E \operatorname{Var}(D^p) \leq E \operatorname{Var}(D). \tag{3}$$

Indeed, apply the defining identity to a predictable Hahn sign for dD^p ; the left side is its expected total variation and the right side is bounded by $E \operatorname{Var}(D)$. Equivalently compensate the two increasing Jordan components separately and use positivity. For $K = 1_F 1_{(s,t]}$, $F \in \mathcal{F}_s$,

the same identity proves the martingale property of $D - D^p$. The Lean construction obtains this compensator by truncation and limits of predictable finite-variation approximants. In this account the existence theorem is used explicitly as a standard foundational result.

Lemma A.39 (Canonical cost of a constant-after-horizon gain). *Let X be zero-start and càdlàg, constant after $T < \infty$, with a canonical decomposition $X = M^{\text{can}} + A^{\text{can}}$. Under the usual conditions,*

$$\begin{aligned} E \operatorname{Var}_{[0,T]}(A^{\text{can}}) &\leq 12\mathcal{J}(X), \\ E \sup_{t \leq T} |M_t^{\text{can}} - M_0^{\text{can}}| + E \operatorname{Var}_{[0,T]}(A^{\text{can}}) &\leq 30\mathcal{J}(X). \end{aligned}$$

These are extended-value inequalities. The infimum defining $\mathcal{J}$ allows adapted, rather than necessarily predictable, FV components.

Proof. Fix a finite-cost decomposition in that infimum, of cost j . The strict-prefix witness construction gives, at the deterministic stop $\sigma = T + 1$, components N, B representing X before σ , with

$$c_N = E \sup_t |N_t^\sigma|, \quad c_B = E \operatorname{Var}(B^{\sigma-}), \quad c_N + c_B \leq 6j.$$

Keep their initial values. With $J_t^N = \Delta N_\sigma 1_{[\sigma, \infty)}(t)$, put

$$D = B^{\sigma-} - B_0 - J^N.$$

This is adapted and of integrable variation, since

$$E \operatorname{Var}(D) \leq c_B + E|\Delta N_\sigma| \leq c_B + 2c_N.$$

Compensate this specific remainder. The processes

$$\widehat{M} = N^\sigma + D - D^p, \quad \widehat{A} = B_0 + D^p$$

form a special decomposition of $(N + B)^{\sigma-} = X$. The last equality uses constancy of X after T , not just agreement on an open interval. By (3),

$$E \operatorname{Var}(\widehat{A}) \leq c_B + 2c_N \leq 2(c_N + c_B) \leq 12j.$$

Now, and only now, canonical uniqueness identifies the centered components of this constructed decomposition with those of X . Take the infimum over j to obtain the first bound. Zero start gives

$$\sup_{t \leq T} |M_t^{\text{can}} - M_0^{\text{can}}| \leq \sup_t |X_t| + \operatorname{Var}_{[0,T]}(A^{\text{can}}).$$

The already proved maximal estimate $E \sup_t |X_t| \leq 6\mathcal{J}(X)$ therefore gives $6 + 12 + 12 = 30$. Infinite costs cause no difficulty. $\square$

For a prelocal witness for a regular zero-start X before $\tau \leq T$, let c be its sum of expected martingale-maximum and variation costs. The witness comparison in the r^1 localization subsection below gives $\mathcal{J}_{\tau-}(X) \leq 16c$. The boundary-jump correction in the quadratic-cost subsection gives

$$\mathcal{J}(X^\tau) \leq 2\mathcal{J}_{\tau-}(X) + E|\Delta X_\tau| \leq 32c + E|\Delta X_\tau|.$$

If an independently constructed actual integral has gain X^τ , apply Lemma A.39 to that same constant-after- T gain. Thus

$$E \sup_{t \leq T} |M_t^{\text{actual}} - M_0^{\text{actual}}| + E \text{Var}_{[0,T]}(A^{\text{actual}}) \leq 30(32c + E|\Delta X_\tau|). \quad (4)$$

Existence of the actual integral is supplied separately. This argument transfers a quantitative bound to its components; uniqueness alone does not provide that bound.

B Common localization and component series

This appendix combines finite-horizon estimates along common localizers and constructs limits of component series.

B.1 Passing to test-uniform limits

Let X^n be the elementary integral representatives of a realized strategy over a strongly progressive right-continuous source, with designated gain X . Write e_T for the capped absolute envelope at horizon T and put

$$\Phi_{K,T}(Y) = E[e_T(K \cdot Y)], \quad g_{n,T} = \sup(\{0\} \cup \{\Phi_{K,T}(X^n - X) : |K| \leq 1\}).$$

These are finite-sum elementary test integrals; no general predictable integral is used in their definition.

Lemma B.1 (Cauchy estimates and fixed-test limits). *For these representatives, $g_{n,T} \rightarrow 0$ for every finite T , and*

$$\sum_{r \geq 0} 2^{-r-1} g_{n,r+1} \rightarrow 0.$$

Proof. Fix $\varepsilon, T > 0$. The representatives' Émery Cauchy property supplies a cutoff N common to every K , with $\Phi_{K,T}(X^m - X^n) < \varepsilon$ for $m, n \geq N$. Only now fix $n \geq N$ and K . Fixed-test errors tend to zero in probability; extract an auxiliary subsequence in m whose capped errors tend to zero almost surely. They are bounded by one, so their expectations vanish. The triangle inequality gives

$$\Phi_{K,T}(X^n - X) \leq \Phi_{K,T}(X^m - X^n) + \Phi_{K,T}(X^m - X) \leq \varepsilon + \Phi_{K,T}(X^m - X).$$

Pass to the auxiliary limit, then take the supremum over K using the common N . Since $0 \leq g_{n,T} \leq 1$, dominated convergence for the geometric series gives the second assertion. $\square$

This is precisely convergence against the elementary tests defining the topology in this account; no extension to general predictable tests is needed.

B.2 Auxiliary topology and common localization

Lemma B.2 (Pathwise finiteness from capped series). *For concrete zero-start decompositions $X_k = N_k + A_k$, set*

$$B_k(t) = \sqrt{[N_k]_t} + \text{Var}_{[0,t]} A_k.$$

If $\sum_k E[1 \wedge B_k(n)] < \infty$ for every integer $n \geq 0$, then $\sum_k B_k(t) < \infty$ for all times on one full-measure set.

Proof. The clocks are finite, nonnegative, increasing and measurable at each time. At integer n , Tonelli gives almost-sure finiteness of the capped sum. Its terms tend to zero, hence eventually $B_k(n) < 1$ and the cap disappears. The finitely many initial terms are finite too. Intersect over integers; for any t use monotonicity and an integer above t . $\square$

This does not imply integrability of the uncapped sum.

Lemma B.3 (Monotonicity and right continuity of the sum). *On one full-measure set, $U_t = \sum_k B_k(t)$ is finite, increasing and right-continuous.*

Proof. Cumulative variation of a right-continuous locally BV path is right-continuous: its right jump is the absolute right jump of the path, hence zero. Each clock is therefore right-continuous. On $t \leq s < t + 1$ it is dominated by the summable $B_k(t + 1)$. Dominated convergence for series gives right continuity of the sum; monotonicity is termwise. $\square$

Lemma B.4 (A common passage for concrete decompositions). *Under the usual conditions choose a strongly adapted càdlàg version of U . Its strict absolute passage above $r + 1$, capped at horizon $r + 1$, defines one localizing sequence $\tau_r \leq r + 1$. On one full-measure set,*

$$\sum_k B_k(t) \leq r + 1$$

for every r and $t < \tau_r$.

Proof. Cumulative variation at t is recovered from a countable supremum of finite grid variations before t , so is $\mathcal{F}_t$ -measurable. The quadratic clocks and their sum are adapted. Finite increasing paths have left limits; right continuity gives càdlàg regularity. Correct the exceptional null set at time zero. The càdlàg absolute-passage construction is increasing and exhaustive. Before passage its magnitude is at most $r + 1$; indistinguishability returns the bound to the sum. $\square$

The capped estimates are inputs here. The closed-graph comparison below will supply them from Émery convergence.

Lemma B.5 (Boundary correction for a stopped quadratic root). *On one full-measure set,*

$$\sqrt{[N]_t} \leq \sqrt{[N]_{t-}} + |\Delta N_t|$$

for every t . For finite-valued stopping time τ ,

$$E \sup_{t \geq 0} \sqrt{[N^\tau]_t} \leq E[\sqrt{[N]_{\tau-}} + |\Delta N_\tau|].$$

A deterministic bound on τ and finite expectations are unnecessary.

Proof. Use $[N]_t = [N]_{t-} + (\Delta N_t)^2$ with nonnegative left limit. The stopped increasing process attains its all-time supremum at τ . Evaluate the pathwise inequality there and integrate. At zero use the left-limit convention and zero initial variation. $\square$

Lemma B.6 (Prelocal cost from prefix components and jumps). *Under the usual conditions, for a zero-start decomposition $X = N + A$ and finite stopping time $\tau \leq T$,*

$$\mathcal{J}_{\tau-}(X) \leq E[\sqrt{[N]_{\tau-}} + |\Delta N_{\tau}|] + E[\text{Var}(A)_{\tau-}].$$

Local finite variation of A suffices; expectations may be infinite.

Proof. First stop A at the deterministic horizon T . This gives global bounded variation without changing A on $[0, \tau)$ or its strict prefix. Monotonicity of cumulative variation gives

$$\text{Var}(A)_t \leq \text{Var}(A)_{\tau-} \quad (t < \tau).$$

The strict-prefix variation estimate therefore yields

$$\text{Var}(A^{\tau-})_{\infty} \leq \text{Var}(A)_{\tau-}.$$

The process $N^{\tau} + A^{\tau-}$ extends X before τ , with these two components as an admissible decomposition. Bound the martingale cost by the preceding quadratic-root boundary estimate, and integrate the global variation bound for the other component. Finally take the defining infima over extensions and decompositions. $\square$

Lemma B.7 (Left-limit sums at the common passage). *For the same common passages constructed from the capped-series assumption, on one full-measure set, for every r ,*

$$\sum_k \left(\sqrt{[N_k]_{\tau_r-}} + \text{Var}(A_k)_{\tau_r-} \right) \leq r + 1.$$

Proof. The left limit of each clock is the sum of the square root of the left limit of quadratic variation and the left limit of cumulative variation. Fix a finite partial sum. For $t < \tau_r$ it is at most the full clock sum, hence at most $r + 1$. If $\tau_r > 0$, take the left limit of this finite sum. Then take the supremum over finite subsets to obtain the nonnegative-series bound. If $\tau_r = 0$, every left limit equals the zero initial value, so the same conclusion holds. $\square$

Lemma B.8 (Summable costs and witnesses at common stops). *The same τ_r satisfies*

$$\sum_k \mathcal{J}_{\tau_r-}(X_k) \leq r + 1 + \sum_k E|\Delta N_{k, \tau_r}|.$$

Define

$$s(N) = \sup_{\tau \text{ bounded stopping time}} E|\Delta N_{\tau}|.$$

If $\sum_k s(N_k) < \infty$, exact prefix-representation witnesses can be chosen at each r with total cost at most

$$6(r + 1 + \sum_k s(N_k) + 1).$$

Proof. Stopped jumps are measurable. Sum the preceding single-decomposition cost bounds over finite subsets; integrate the left-limit clock bound to get $r+1$, then take the supremum over subsets. Each bounded τ_r makes the jump term at most $s(N_k)$. Approximate the summable prelocal infima with positive errors totaling one and apply the component selection bound with Davis constant 6. $\square$

These are witnesses for arbitrary decompositions, not yet original-price integral certificates.

Definition B.9 (The zero-start auxiliary cost). For one decomposition $D = (N, A)$ and its quadratic variation, put

$$c_{r1}(D) = s(N) + \sum_{n=0}^{\infty} 2^{-n-1} E[1 \wedge (\sqrt{[N]_n} + \text{Var}(A)_n)], \quad R(X) = \inf_{X \sim N+A} c_{r1}(N, A).$$

The two terms use the same decomposition before taking their joint infimum. Infinite values are permitted.

Lemma B.10 (Finiteness on all decomposable zero-start processes). *If X has an admissible zero-start decomposition, $R(X) \leq 3$. Conversely $R(X) < \infty$ implies such a decomposition exists. Finite global cost $\mathcal{J}(X)$ is not required.*

Proof. Apply all-time DDY with threshold 1 to the martingale component: $N = L + Q$, with $|\Delta L| \leq 2$. Then $X \sim L + (Q + A)$ is admissible. Its bounded-stop jump cost is at most 2; each capped expectation is at most one and the weights total one. The joint cost is at most 3. Conversely the infimum over no decompositions is infinity. $\square$

Regular zero-start good integrators enter this space by the decomposition construction and elementary completeness proved later.

Lemma B.11 (Comparison with global cost). *$R(X) \leq 2\mathcal{J}(X)$ as extended values. Thus vanishing global costs imply vanishing auxiliary costs.*

Proof. Fix a decomposition $D = (N, A)$ and denote its global cost by

$$j(D) = E \left[\sup_t \sqrt{[N]_t} + \text{Var}(A)_\infty \right].$$

The jump-square identity and monotonicity give, on one event for all times,

$$|\Delta N_\sigma| \leq \sqrt{[N]_\sigma} \leq \sup_t \sqrt{[N]_t}.$$

Take expectations and the supremum over bounded stopping times to get $s(N) \leq j(D)$. Every clock is bounded by the all-time root plus total variation, so each capped expectation is also at most $j(D)$. The geometric weights total one, hence $c_{r,1}(D) \leq 2j(D)$. Take the infimum over decompositions. All these inequalities hold for extended values; no conversion of an infinite cost to a real number is needed. Finally $0 \leq R(X_k) \leq 2\mathcal{J}(X_k)$ gives the convergence assertion. $\square$

Lemma B.12 (Auxiliary triangle inequalities and differences). *Under the usual conditions,*

$$R(X + Y) \leq R(X) + R(Y), \quad R(-X) = R(X), \quad R(X - Y) \leq R(X) + R(Y).$$

In particular, $R(Z_k - X) \rightarrow 0$ implies $R(Z_{k+1} - Z_k) \rightarrow 0$.

Proof. Fix two admissible decompositions and add their components. Construct quadratic variation of the martingale sum. The root triangle inequality and subadditivity of variation bound its clock by the sum of the two clocks. Integrate $1 \wedge (a+b) \leq (1 \wedge a) + (1 \wedge b)$, multiply by the positive weights, and sum to obtain subadditivity of the capped term. For the jump term, integrate the jump triangle inequality at each bounded stopping time and then take the supremum. Measurability of stopped jumps permits separation of the nonnegative integrals. Take infima over decompositions and quadratic variations, keeping both cost coordinates attached to the same decomposition throughout. Negation preserves quadratic variation, variation, and absolute jumps, so it preserves R . The difference estimate follows by addition and negation. Apply it to $Z_{k+1} - Z_k = (Z_{k+1} - X) - (Z_k - X)$ for the convergence assertion. $\square$

Lemma B.13 (Auxiliary cost under closed stopping). *For any stopping time τ , $R(X^\tau) \leq R(X)$. If $\sigma \leq \tau$, then $R(X^\sigma) \leq R(X^\tau)$. Thus arbitrary termwise stopped versions of a vanishing-cost sequence also have vanishing cost.*

Proof. Stop both components. At any bounded stopping time the stopped jump equals the original jump before or at τ and zero afterward, so its jump cost cannot increase. Stopped quadratic variation and variation bound the clocks. Take the joint infimum. Apply absorption for the second assertion and squeeze for convergence. $\square$

Lemma B.14 (Common localization from vanishing auxiliary costs). *If $R(X_k) \rightarrow 0$, choose one strictly increasing subsequence f and one localizing sequence $\tau_r \leq r + 1$. At every r , exact prefix witnesses for $X_{f(k)}$ have total cost at most $6(r + 4)$, and their costs tend to zero.*

Proof. First suppose $\sum_k R(X_k) < \infty$. Approximate the infima with errors totaling one, giving decompositions whose joint cost sum is at most $\sum_k R(X_k) + 1$. Their jump costs are summable. For each fixed integer horizon the positive time weight implies summability of capped expectations. Apply the common-localization theorem to these same decompositions. The witness

sum is at most $6(r+1+\sum_k R(X_k)+2)$. For a general null sequence choose $R(X_{f(k)}) < 2^{-k-1}$, giving the bound $6(r+4)$. Finite nonnegative sums imply termwise convergence to zero. $\square$

The later topology comparison supplies this hypothesis from Émery convergence. The global-cost comparison already supplies it when $\mathcal{J}(X_k) \rightarrow 0$.

Lemma B.15 (Separation). *Indistinguishable processes have equal auxiliary costs. Under the usual conditions,*

$$R(X) = 0 \iff X \sim 0, \quad R(X - Y) = 0 \iff X \sim Y.$$

Proof. Transfer each decomposition and its quadratic variation to an indistinguishable target without changing its cost; the infima agree. If $R(X) = 0$, the constant sequence $X_k = X$ has cost sum zero. Choose near-minimizing decompositions and apply common localization. For every r , $\sum_k \mathcal{J}_{\tau_r-}(X) < \infty$. This is a constant nonnegative series, so $\mathcal{J}_{\tau_r-}(X) = 0$. For adapted right-continuous X and fixed t , the random variable $|X_t|1_{\{t < \tau_r\}}$ is measurable and has expectation zero by the prefix maximal estimate. Intersect over r and use exhaustion to obtain $X_t = 0$. Intersect next over a countable right-dense skeleton and extend the equality to every time by right continuity. For an arbitrary raw target, $R(X) < 1$ supplies an admissible decomposition; its component sum is a regular representative to which the argument applies. The zero decomposition and zero quadratic variation prove the converse, together with invariance under indistinguishability. Applying this equivalence to differences proves the second assertion. $\square$

Lemma B.16 (Summable costs imply pathwise uniform summability). *If $\sum_k R(X_k) < \infty$, then almost surely, for every finite T ,*

$$\sum_k \sup_{t \leq T} |X_k(t)| < \infty.$$

A vanishing-cost sequence has one subsequence with this property.

Proof. For regular targets define $G_{k,r}$ as the supremum of $|X_k|$ over the countable right-dense skeleton strictly before τ_r . It is measurable and $EG_{k,r} \leq 6\mathcal{J}_{\tau_r-}(X_k)$. Tonelli gives summability almost surely for all countably many r . For any T choose r with $T < \tau_r$; right-continuous approximation by skeleton points bounds the whole horizon supremum by $G_{k,r}$. For raw targets use regular sums of near-minimizing decompositions and countable indistinguishability. Extract a summable-cost subsequence for the last assertion. $\square$

Lemma B.17 (Auxiliary convergence implies ucp convergence). *If $R(X_k - X) \rightarrow 0$, then for each finite T and $\varepsilon > 0$,*

$$P\{\exists t \leq T : |X_k(t) - X(t)| > \varepsilon\} \rightarrow 0.$$

No extra measurability or path regularity of the raw targets is assumed.

Proof. For summable costs use regular decompositions and factorial envelopes to obtain maxima that are measurable up to completion. The preceding lemma makes them finite and tending to zero almost surely. Their real-valued versions converge in probability; finiteness identifies their tails with the extended maximal envelopes. From any subsequence of a vanishing-cost sequence extract a summable one. The subsequence criterion gives convergence of the original tail probabilities. Apply to differences; a strict exceedance at some time is contained in the corresponding maximal tail event. $\square$

Cost contraction also gives ucp convergence after arbitrarily selected termwise stopping times.

Proposition B.18 (Metric quotient and Cauchy sequences). *Let $\mathcal{D}_0$ be the zero-start processes with admissible decompositions. The finite pseudometric $d(X, Y) = R(X - Y)$ has separated quotient equal to the indistinguishability quotient. Metric convergence is equivalent to vanishing R -error and implies ucp convergence. Every Cauchy sequence has one strictly increasing f with*

$$\sum_k R(X_{f(k+1)} - X_{f(k)}) \leq 1;$$

these same differences have common localizers and exact witnesses with cost sum at most $6(r + 4)$ at stage r .

Proof. Differences remain decomposable, so DDY gives finiteness. The auxiliary algebra gives symmetry and triangle; separation identifies the quotient. Its distance is precisely the representative cost. For a Cauchy sequence choose tail bounds $b_n \rightarrow 0$ and increasing f with $b_{f(k)} < 2^{-k-1}$. Sum the adjacent-distance bounds, then apply the summable-cost common-localization theorem to exactly these differences. $\square$

Lemma B.19 (A locally uniform limit for the same subsequence). *If $\sum_k R(Y_{k+1} - Y_k) < \infty$, there is one process Y such that, on one full-measure set, $Y_k \rightarrow Y$ uniformly on every finite interval. This applies to the same Cauchy subsequence carrying the common-localization witnesses in the preceding proposition.*

Proof. Summability of auxiliary costs gives, simultaneously for all finite T ,

$$\sum_k q_k(T) < \infty, \quad q_k(T) = \sup_{t \leq T} |Y_{k+1}(t) - Y_k(t)|.$$

Each term is finite. At any fixed time, the series of distances between successive values is bounded by this series. Completeness of the real numbers gives a pointwise limit. Define Y_t by that limit. The summable uniform step bounds $q_k(T)$ upgrade pointwise convergence to uniform convergence on $[0, T]$. The definition of Y_t does not depend on the horizon. The previously chosen subsequence has summable difference costs, so this construction applies without a further extraction. $\square$

This is initially a raw limit. We next retain global component series to construct its decomposition and prove convergence of all cost coordinates.

Lemma B.20 (Martingale component sums at a fixed stop). *If true martingales M_k satisfy $|M_k(t)| \leq g_k$ for all times with measurable $g_k \geq 0$ and $\sum_k E g_k < \infty$, then $M_t = \sum_k M_k(t)$ is a strongly adapted true martingale. This applies to the centered stopped witness components*

$$\sum_k (N_k - N_k(0))_{t \wedge \tau_r}$$

at each fixed stage.

Proof. Tonelli makes $G = \sum_k g_k$ integrable and finite almost surely. The component series is absolutely convergent at all times on that event, and its sum is measurable at every time. Finite sums are martingales dominated by G , so the dominated martingale-limit lemma applies. For witnesses the stopped centered components are true martingales, and the finite-horizon envelope dominates them at all times because the stop is bounded by that horizon. Centering increases cost by at most a factor of two, preserving envelope summability. $\square$

Lemma B.21 (Centered finite-variation series). *Zero-start paths a_k with $\sum_k \text{Var}(a_k) < \infty$ have a uniformly convergent all-time sum a of finite variation, and*

$$\text{Var} \left(a - \sum_{k < n} a_k \right) \rightarrow 0.$$

At a fixed witness stage this applies almost surely to $a_k = A_k^{\tau-} - A_k^{\tau-}(0)$.

Proof. Zero start bounds $|a_k(t)|$ by variation, giving uniform convergence of the series. The finite-variation limit and variation-distance convergence follow from summable variation of consecutive partial-sum differences. For witnesses, centering does not change variation and the strict prefix is constant after the stop, so its all-time variation is bounded by the finite-horizon witness variation. Tonelli gives summability almost surely. Centering removes initial constants that variation alone cannot control. $\square$

We now keep one global decomposition per difference rather than choose components independently at successive localization stages.

Lemma B.22 (Common estimates retaining global decompositions). *For a Cauchy sequence in the quotient, choose one strictly increasing f and global decompositions of its actual neighboring differences, $X_{f(k+1)} - X_{f(k)} \sim N_k + A_k$. Their auxiliary costs sum to at most 2, and one localizing sequence $\tau_r \leq r + 1$ satisfies*

$$\sum_k \left(E\sqrt{[N_k]_{\tau_r}} + E \operatorname{Var}(A_k^{\tau_r-}) \right) \leq r + 3.$$

The variation is over the entire time axis.

Proof. Choose the subsequence with adjacent auxiliary distances summing to at most one. Approximating the infima by positive errors totaling one gives one sequence of decompositions with total cost at most two. Stop using the common clock of these same decompositions. The expected sum of the pre-stop roots and cumulative variations is at most $r + 1$. For each term,

$$\sqrt{[N_k]_{\tau_r}} \leq \sqrt{[N_k]_{\tau_r-}} + |\Delta N_{k,\tau_r}|, \quad \operatorname{Var}(A_k^{\tau_r-}) \leq \operatorname{Var}_{[0,\tau_r)}(A_k).$$

The boundary-jump expectations are bounded by the same decompositions' costs of jumps at bounded stopping times, whose sum is at most two. Pass from finite sums of nonnegative integrals to their supremum to obtain the claim. Since stopped quadratic variation is increasing and constant after the stop, its root at the stop is its all-time root supremum. $\square$

Lemma B.23 (Sum of the same global martingale components). *Set $M_t = \sum_k N_k(t)$. Each M^{τ_r} is a true martingale, and M is a zero-start local martingale.*

Proof. For fixed r , Davis gives

$$\sum_k E \sup_t |N_k(t \wedge \tau_r)| \leq 6 \sum_k E \sqrt{[N_k]_{\tau_r}} < \infty.$$

A measurable finite-horizon envelope dominates the stopped component for all time. Its integrability upgrades the stopped component to a true martingale; the dominated series lemma then applies. Evaluation at a stopped time commutes with the pointwise sum, giving exactly M^{τ_r} . Zero initial values remove the localizing indicator, proving the local property. $\square$

Lemma B.24 (A regular representative of the martingale sum). *The same M is indistinguishable from an everywhere càdlàg, strongly adapted, zero-start local martingale.*

Proof. At every common stop the envelope expectations are summable; Tonelli gives pathwise uniform convergence of the stopped finite sums. Uniform càdlàg limits are càdlàg. Intersect over the stops. For any time t , exhaustion places $[0, t + 1)$ before one stop, transferring that regularity to the global sum. Each time value is measurable. Replace nonregular paths by zero on the time-zero measurable exceptional set and transfer the local-martingale property by indistinguishability. $\square$

Lemma B.25 (Sum of the same global finite-variation components). *For the same decomposition sequence set $A_t = \sum_k A_k(t)$. On one full-measure set A is locally of finite variation and, for every finite T ,*

$$\sum_{k < n} A_k \longrightarrow A \quad \text{uniformly on } [0, T], \quad \text{Var}_{[0, T]} \left(A - \sum_{k < n} A_k \right) \longrightarrow 0.$$

Proof. The capped-clock estimates give $\sum_k \text{Var}_{[0, T]}(A_k) < \infty$ for every T on one full-measure set. Fix T and put $a_k(t) = A_k(t \wedge T)$. Its all-time variation is at most $\text{Var}_{[0, T]}(A_k)$ and $a_k(0) = 0$. The zero-start variation-series theorem shows that $\sum_k a_k$ has global finite variation and that its partial sums converge both uniformly and in all-time variation distance. On $[0, T]$ we have $a_k = A_k$, so both convergences transfer back to the original components. Every compact interval is contained in some $[0, T]$, proving local finite variation of A . $\square$

Lemma B.26 (A regular representative of the variation sum). *The preceding sum has an indistinguishable zero-start strongly adapted representative with everywhere càdlàg locally BV paths.*

Proof. Measurability of each series value gives adaptation. Local uniform convergence of right-continuous finite sums gives right continuity of the limit. Correct the common exceptional set for right continuity and local BV using the usual conditions at time zero. Zero start is preserved, and local BV gives left limits. $\square$

The global components were fixed once: equality of sums would not by itself identify separate components in decompositions with merely adapted FV parts.

Lemma B.27 (A limit inside the decomposable-process space). *Every auxiliary-Cauchy sequence has a subsequence $X_{f(n)}$ converging almost surely locally uniformly to an admissibly decomposable process Y . Its pointwise limit is indistinguishable from this Y and also decomposable.*

Proof. Use the same regular sums M, A and set $Y = X_{f(0)} + M + A$. Add the base term's decomposition to those of the sums. On the common event of all component identities and convergence, choose $\tau_r > T$ to transfer stopped martingale uniform convergence to $[0, T]$. The FV sums also converge uniformly there. Telescoping gives

$$X_{f(0)} + \sum_{k < n} (N_k + A_k) = X_{f(n)},$$

hence convergence to Y . Uniqueness of each time limit on that event gives indistinguishability and transfers the decomposition. $\square$

Lemma B.28 (Stopped-jump lower semicontinuity). *If martingale components U_n tend almost surely locally uniformly to a process U with left limits, then $s(U) \leq \liminf_n s(U_n)$. In particular the same regular sum satisfies*

$$s(M) \leq \sum_k s(N_k).$$

Proof. Fix bounded $\sigma \leq T$. Uniform convergence after deterministic stopping at T gives convergence of left limits and hence of jumps at σ . Fatou yields $E|\Delta U_\sigma| \leq \liminf_n s(U_n)$, with bound independent of σ ; take the supremum. Apply to finite sums, whose jump cost is at most the sum of their costs, using the already proved local uniform convergence. $\square$

Lemma B.29 (Component-series tails). *For the same regular sums M, A and original decompositions, for every n ,*

$$s\left(M - \sum_{k < n} N_k\right) \leq \sum_{k=0}^{\infty} s(N_{k+n}).$$

The left side tends to zero. At every finite horizon T we also have

$$E\left[\min\left\{\text{Var}_{[0,T]}\left(A - \sum_{k < n} A_k\right), 1\right\}\right] \rightarrow 0.$$

Proof. Fix n and shift the original decomposition sequence by n . Its martingale partial sums converge locally uniformly to $M - \sum_{k < n} N_k$ on the same full-measure set. Apply stopped-jump lower semicontinuity to this sequence for the first inequality. Its right side tends to zero since $\sum_k s(N_k) < \infty$. For finite variation, use the pathwise variation convergence already proved. The regular sum and every partial sum are strongly adapted and right-continuous, so the variation of their difference on a finite interval is measurable. Transfer to the same representatives by indistinguishability and apply dominated convergence with the constant bound one. $\square$

Lemma B.30 (Quadratic-root tails of the same martingale sum). *Let $U_n = M - \sum_{k < n} N_k$. At a common bounded stop τ with summable original root expectations,*

$$E\sqrt{[U_n]_\tau} \leq 42 \sum_{k \geq n} E\sqrt{[N_k]_\tau} \rightarrow 0.$$

The intrinsic quadratic variation of each tail is independent of this stop.

Proof. If partial sums converge uniformly on $[0, T]$ to local martingale U , then

$$U_T^* \leq \sum_k (N_k)_T^*,$$

allowing an infinite envelope sum. Reverse Davis, Tonelli, and forward Davis give

$$E\sqrt{[U]_T} \leq 7EU_T^* \leq 7 \sum_k E(N_k)_T^* \leq 42 \sum_k E\sqrt{[N_k]_T}.$$

Apply this to the stopped components and their tail series. Their summable envelope expectations give all-time uniform convergence almost surely to U_n^τ . Choose deterministic $T \geq \tau$ and use stopping consistency. The right side is the tail of a summable nonnegative series. $\square$

Lemma B.31 (Removing common stops from capped roots). *Suppose $\tau_r \rightarrow \infty$ almost surely and $E\sqrt{[U_n]_{\tau_r}} \rightarrow 0$ for every r . Then for each finite T ,*

$$E[\min\{\sqrt{[U_n]_T}, 1\}] \rightarrow 0.$$

Proof. Monotonicity of quadratic variation gives

$$\min\{\sqrt{[U_n]_T}, 1\} \leq \sqrt{[U_n]_{\tau_r}} + 1_{\{\tau_r \leq T\}}.$$

The second term's expectation tends to zero as $r \rightarrow \infty$ by exhaustion and dominated convergence. First fix r making this error small, then send $n \rightarrow \infty$ in the first term. $\square$

Theorem B.32 (Completeness of the auxiliary metric). *Under the usual conditions, the indistinguishability quotient of all zero-start processes admitting a decomposition is complete for $d(X, Y) = R(X - Y)$.*

Proof. Choose a fast subsequence $X_{f(k)}$ of a Cauchy sequence in the quotient, retaining decompositions (N_k, A_k) of its actual neighboring differences. Let M, A be the regular component sums constructed above and put $X = X_{f(0)} + M + A$. This process admits a decomposition. Subtracting the first n terms from these same sums decomposes $X - X_{f(n)}$ into

$$U_n = M - \sum_{k < n} N_k, \quad B_n = A - \sum_{k < n} A_k.$$

The identification of the target is the telescoping identity for the original adjacent differences. The jump-tail estimate gives $s(U_n) \rightarrow 0$, and at each finite T ,

$$E \min\{\sqrt{[U_n]_T}, 1\} \rightarrow 0, \quad E \min\{\text{Var}_{[0, T]}(B_n), 1\} \rightarrow 0.$$

The inequality $\min\{a + b, 1\} \leq \min\{a, 1\} + \min\{b, 1\}$ therefore makes the capped clock expectation of this same decomposition tend to zero at every integer horizon. These expectations are at most one, so dominated convergence also applies to their geometric weighted sum. The decomposition's r^1 cost tends to zero, and hence so does its infimum $R(X - X_{f(n)})$. Negation invariance gives convergence of the subsequence to X in the quotient. A Cauchy sequence with a convergent subsequence converges entirely to the same limit. Any quotient sequence can be lifted to representatives for this argument, proving completeness. $\square$

Lemma B.33 (Continuous scalar multiplication). *For every decomposable zero-start process X ,*

$$R(cX) \leq \max(1, |c|)R(X).$$

If $a_k \rightarrow a$ and $R(X_k - X) \rightarrow 0$, then $R(a_k X_k - aX) \rightarrow 0$.

Proof. Scale one decomposition. The jump and square-residual characterizations give $[cN] = c^2[N]$, so clocks and jump costs scale by $|c|$. Use $1 \wedge (|c|b) \leq \max(1, |c|)(1 \wedge b)$ and take joint infima. For a fixed X and $a_k \rightarrow 0$, that inequality alone is insufficient. Choose a DDY decomposition with finite jump cost. Its scaled jump cost vanishes; each finite clock is finite pathwise, so its scaled capped expectation vanishes by bounded convergence. Apply dominated convergence also to the geometric series. Hence $R(a_k X) \rightarrow 0$. Now split

$$a_k X_k - aX = a_k(X_k - X) + (a_k - a)X$$

and use boundedness of (a_k) , the triangle inequality, and the fixed-process conclusion. Sequential continuity in these metric spaces gives continuity, descending to the quotient. $\square$

C The Emery topology and realization by one integrand

This appendix connects good-integrator behavior with completeness of the Emery topology and represents both component integrals by one predictable coefficient.

C.1 Good integrators and comparison of topologies

Lemma C.1 (Locally FV processes are good integrators). *Let A be a strongly adapted right-continuous real process whose paths all have local finite variation. Under a finite measure it is an elementary good integrator.*

Proof. Let predictable elementary integrands H_n tend uniformly to zero over all times and sample points. Fix T and stop A there, giving paths of global finite variation without changing the elementary gain at T . Pathwise the gain has the Stieltjes representation

$$(H_n \cdot A)_T = \int 1_{(0,T]}(t) H_n(t) dA_t^T.$$

The integrands tend pointwise to zero and eventually have absolute value at most one. The pathwise total-variation measure is finite, so dominated convergence for the vector measure makes the gain tend to zero on each sample path. Strong adaptation and right continuity imply progressive measurability, hence measurability of the elementary gains. Under finite measure their almost-sure convergence implies convergence in probability. No bound on the number of strategy blocks or on expected variation is used. $\square$

Linearity of elementary gains reduces the decomposable case to martingales, and indistinguishability transfers the result even to nonregular raw targets.

Lemma C.2 (Martingale integral estimate). *For a right-continuous true martingale M with $M_T \in L^2$, and elementary predictable H with everywhere bound $|H| \leq B$, $B \geq 0$,*

$$\|(H \cdot M)_T\|_2 \leq B \sqrt{E(M_T - M_0)^2}.$$

A right-continuous martingale square integrable at each time is consequently a good integrator.

Proof. Stop at T and take chronological factorial-grid left-endpoint integrals. Discrete isometry bounds their squared expectation by $B^2 E(M_T - M_0)^2$. The grids cover the stopped terminal horizon, and right continuity gives pathwise convergence to the elementary gain. Fatou for the L^2 norm passes the bound to the limit. For uniformly vanishing H_n , write $C = \sqrt{E(M_T - M_0)^2}$ and eventually use $|H_n| \leq \varepsilon/(C + 1)$ to get norm at most ε . Measurability and L^2 convergence imply convergence in probability. No block count or coefficient absolute-sum bound occurs. $\square$

Lemma C.3 (Removing localization from good-integrator behavior). *If strongly adapted right-continuous S under a finite measure has $\tau_r \rightarrow \infty$ almost surely and every S^{τ_r} is a good integrator, then so is S . Monotonicity of the stops is unnecessary.*

Proof. For fixed T , bounded convergence gives $P(\tau_r \leq T) \rightarrow 0$. On $\{\tau_r > T\}$ all elementary evaluation times lie before the stop, so the gains agree. Therefore

$$P(|(H_n \cdot S)_T| \geq \varepsilon) \leq P(\tau_r \leq T) + P(|(H_n \cdot S^{\tau_r})_T| \geq \varepsilon).$$

First choose r , then send n to infinity. □

Proposition C.4 (Decomposable processes are good integrators). *Under the usual conditions a zero-start process with admissible decomposition $X = N + A$ is an elementary good integrator.*

Proof. Apply DDY with threshold one: $N = L + Q$, $|\Delta L| \leq 2$ and Q locally FV. Common localization makes L^{τ_r} a deterministically bounded true martingale. The L^2 estimate and removal of localization give good-integrator behavior for L . Add the locally FV $Q + A$ and transfer to the raw target by indistinguishability. Local square integrability of N itself is not assumed. □

Lemma C.5 (Finite stopping preserves good integrators). *If S is a good integrator and τ finite-valued, so is S^τ .*

Proof. Stop an elementary strategy by subtracting its predictable post-stop tail. Its integrand is $1_{\{t \leq \tau\}} H_t$, preserving uniform convergence to zero. Commuting minima in each finite summand gives $(H \cdot S^\tau)_T = ((1_{\{t \leq \tau\}} H) \cdot S)_T$. Transfer measurability and convergence by this identity. □

Proposition C.6 (Stopped decomposition of a bounded-jump good integrator). *Let X be zero-start, strongly adapted, càdlàg and a good integrator, with $|\Delta X| \leq c$ outside one common null set, $c \geq 0$. The exhaustive stops*

$$\sigma_n = \inf\{t : |X_t| > n + 1\} \wedge (n + 1)$$

make X^{σ_n} bounded sources with bound $n + 1 + c$. Theorem A.1 therefore supplies a global special decomposition for every coordinate.

Proof. Compact boundedness gives exhaustion. The pre-passage bound and stopping jump give the closed bound. Adaptation and càdlàg regularity survive stopping, as does good-integrator behavior by the preceding lemma. These supply all inputs to Theorem A.1. □

Proposition C.7 (Removing large jumps from a general good integrator). *For zero-start strongly adapted càdlàg good integrator X and $c > 0$, set*

$$J_t = \sum_{0 < s \leq t} \Delta X_s 1_{\{|\Delta X_s| > c\}}, \quad Y = X - J.$$

Then J is zero-start strongly adapted càdlàg locally FV, and Y is such a good integrator with $|\Delta Y_t| \leq c$ at every path and time. Thus the preceding stopped-decomposition proposition applies to Y .

Proof. The finite-horizon sums J^T agree before smaller horizons. Define $J_t = J_t^t$; fixed-time adaptation follows from that coordinate, right continuity from agreement on a larger horizon, and left limits and local BV from compact-interval agreement. At $t > 0$ the residual agrees, including its left limit, with the finite-horizon residual at horizon t , so inherits the jump bound; at zero use the convention of zero jump. Subtract the locally FV good integrator J from X , using linearity and stability of convergence in probability. No compensation of J is needed, since the final FV component need only be adapted. $\square$

Theorem C.8 (Decomposition of regular zero-start good integrators). *Under the usual conditions, a zero-start strongly adapted càdlàg X is an elementary good integrator if and only if it has a decomposition $X = N + B$ with zero-start càdlàg local martingale N and zero-start strongly adapted càdlàg locally FV B .*

Proof. One direction was proved above. For the other remove jumps larger than one, writing $X = Y + J$. Decompose the bounded stopped sources $Y^{\sigma_n} = M^n + A^n$ and normalize

$$\widehat{M}^n = (M^n - M_0^n)^{\sigma_n}, \quad \widehat{A}^n = (A^n - A_0^n)^{\sigma_n}.$$

Local martingales can be centered without global integrability of the initial value. Constant extension of a predictable initial value and finite stopping preserve predictability. Each stopped FV path has global bounded variation. Zero start of the source and idempotence of stopping preserve the decomposition. Stop two coordinates at $\rho = \sigma_n \wedge \sigma_m$. Their FV difference is indistinguishable from a local-martingale difference, and is zero-start predictable right-continuous globally BV. Rigidity makes it zero, giving compatibility of both components. Glue the same martingale and predictable FV families. Exhaustion gives $Y = M + A$ on a common full-measure set. The martingale component is strongly adapted and càdlàg; the other component is predictable, right-continuous and locally of finite variation. Local finite variation gives its left limits. Recenter after gluing and set

$$N = M - M_0, \quad B = A - A_0 + J.$$

Since $M_0 + A_0 = 0$ almost surely, $X = N + B$. The final B is adapted locally FV; predictability is not asserted. $\square$

Proposition C.9 (Passing uniform Cauchy estimates to a ucp limit). *Let X_n be strongly progressively measurable and right-continuous, and let $X_n \rightarrow Y$ ucp. For a unit-bounded predictable elementary test J , put*

$$\bar{e}_T^J(U, V) = E[1 \wedge \sup_{t \leq T} |(J \cdot U)_t - (J \cdot V)_t|].$$

Suppose that, for every $T, \varepsilon > 0$, there is N such that $\bar{e}_T^J(X_m, X_n) \leq \varepsilon$ for all $m, n \geq N$ and all J . Then $\sup_J \bar{e}_T^J(X_n, Y) \rightarrow 0$. Conversely, this uniform convergence implies the Cauchy condition.

Proof. For fixed J , the finite-sum estimate bounds the capped maximal test error by the capped maximal process error multiplied by

$$\max\{8 \text{length}(J), 1\}.$$

This constant is used only for a fixed test. The ucp convergence and boundedness of capped errors in $[0, 1]$ give $\bar{e}_T^J(X_m, Y) \rightarrow 0$.

For T, ε , choose the Cauchy cutoff N first, then fix $n \geq N$ and J . For all $m \geq N$, the triangle inequality gives

$$\bar{e}_T^J(X_n, Y) \leq \bar{e}_T^J(X_m, X_n) + \bar{e}_T^J(X_m, Y) \leq \varepsilon + \bar{e}_T^J(X_m, Y).$$

Send $m \rightarrow \infty$ to get $\bar{e}_T^J(X_n, Y) \leq \varepsilon$. Only now take the supremum over J , since N was chosen independently of the test. The converse uses the triangle inequality through Y and an error bound $\varepsilon/2$ for each of its two terms. $\square$

No closedness of a fixed source's integral range is used. We construct the regular limit and its good-integrator behavior next.

Proposition C.10 (A fast subsequence on increasing horizons). *Suppose strongly progressively measurable right-continuous processes X_n satisfy the uniform elementary-test Cauchy condition and $X_n(0) = X_0(0)$ almost surely for every n . There is a strictly increasing f such that, almost surely, for all sufficiently large k ,*

$$\sup_{t \leq k+1} |X_{f(k+1)}(t) - X_{f(k)}(t)| \leq 2^{-k-1}.$$

Thus on each finite interval the sufficiently late adjacent suprema are summable pathwise.

Proof. The unit test $1_{(0, T]}$ integrates to $X(t) - X(0)$ for $t \leq T$. The common initial value therefore turns the uniform test estimate into

$$E[1 \wedge \sup_{t \leq T} |X_m(t) - X_n(t)|] \leq \varepsilon$$

for all sufficiently large m, n . Put $q_k = 2^{-k-1}$ and let N_k be the cutoff at $T = k + 1$ and error q_k^2 . Set $f(0) = N_0$ and $f(k + 1) = \max(f(k) + 1, N_{k+1})$. For the consecutive capped maximum E_k , we have $EE_k \leq q_k^2$. Markov gives $P(E_k \geq q_k) \leq q_k$. Since $\sum_k q_k < \infty$, Borel–Cantelli gives $E_k < q_k$ eventually, outside one null set. As $q_k < 1$, the cap can be removed. Right continuity makes the stopped factorial grid detect the entire interval supremum, so the estimate holds simultaneously for all $t \leq k + 1$. $\square$

Proposition C.11 (Regular limits of test-uniform Cauchy sequences). *Assume the usual conditions. If strongly progressively measurable càdlàg processes X_n are Cauchy uniformly over elementary tests and $X_n(0) = X_0(0)$ almost surely for every n , there is a strongly progressively measurable càdlàg process Y such that*

$$Y(0) = X_0(0) \quad a.s., \quad \sup_J \bar{e}_T^J(X_n, Y) \longrightarrow 0 \quad (T < \infty).$$

The entire sequence also converges to Y ucp.

Proof. Take the preceding fast subsequence. On each finite interval, outside one common null set, sufficiently late adjacent suprema are bounded by a summable numerical sequence. The triangle inequality and series-tail bound give locally uniform Cauchy paths. Completeness of the reals gives a limit $Z(t)$ at every time, with locally uniform convergence. Use a measurable limit selection defined even on nonconvergent sample points. Then $Z(t)$ is measurable with respect to the information at t , so Z is strongly adapted.

Stopping paths at a finite horizon reduces this to all-time uniform convergence. Uniform limits preserve right continuity and left limits, so Z is almost surely càdlàg. The usual conditions put the exceptional set in the time-zero sigma algebra. Replacing its paths by zero gives strongly adapted, everywhere càdlàg Y . Adapted right continuity implies strong progressive measurability. Locally uniform convergence survives outside the common null set, and convergence at time zero gives the asserted initial value.

The subsequence’s local uniform convergence gives convergence in probability of every capped maximal error. In the Cauchy-to-limit proof let the auxiliary index m run through $f(m)$. Since $f(m) \rightarrow \infty$, the same Cauchy cutoff applies. Take the fixed-test limit first to obtain uniform error bounds for the original entire sequence. Thus that sequence converges against elementary tests. The common initial value and unit test give full-sequence ucp convergence as well. $\square$

This construction does not assume the approximants are good integrators or use integral-range closedness.

Proposition C.12 (Good-integrator stability under test-uniform limits). *Suppose strongly progressively measurable right-continuous X_n, Y satisfy $\sup_J \bar{e}_T^J(X_n, Y) \rightarrow 0$ for each finite T . If every X_n is a good integrator, so is Y .*

Proof. Fix uniformly vanishing predictable elementary integrands H_k and a horizon T . For arbitrary $\varepsilon, \delta > 0$, put $a = \min(\varepsilon/2, 1) > 0$. Choose one m such that $\bar{e}_T^J(X_m, Y) \leq a\delta/4$ for every unit test. With this m fixed, good-integrator behavior of X_m gives, eventually,

$$P(|(H_k \cdot X_m)_T| \geq \varepsilon/2) < \delta/2.$$

On the same tail $|H_k| \leq 1$. The capped maximal integral difference E_k therefore has $EE_k \leq a\delta/4$, so Markov yields $P(E_k \geq a) \leq \delta/4$. Right continuity bounds the capped difference at T by E_k . The triangle inequality gives the event inclusion

$$\{|(H_k \cdot Y)_T| \geq \varepsilon\} \subseteq \{|(H_k \cdot X_m)_T| \geq \varepsilon/2\} \cup \{E_k \geq a\}.$$

Its probability is less than δ . Since δ was arbitrary, this proves convergence in probability. Measurability of the gain follows from strong progressive measurability of Y and its finite-sum formula. $\square$

Corollary C.13 (The Cauchy limit remains decomposable). *Under the usual and filtration sigma-finiteness conditions, a test-uniform Cauchy sequence of regular zero-start good integrators has a regular zero-start decomposable limit, with full-sequence elementary-test convergence.*

Proof. Construct Y as above and set $Z_t = Y_t - Y_0$. Centering preserves all elementary increments, regularity, and adaptation. Stability makes Z a good integrator; apply its zero-start decomposition theorem. The FV component need only be adapted. $\square$

Proposition C.14 (Elementary gauges on the same quotient). *On the indistinguishability quotient $\mathcal{S}$ of decomposable zero-start processes,*

$$e_T([X], [Y]) = \sup_J \bar{e}_T^J(X, Y)$$

is well-defined. Every sequence Cauchy for all finite-horizon gauges has a unique gauge limit in $\mathcal{S}$.

Proof. Indistinguishable sources have identical finite-sum test integrals at every time off one null set. Envelopes, integrals, and test suprema are unchanged. Raw representatives need not be regular: choose each decomposition's component sum, which is zero-start strongly progressive and càdlàg. This preserves the Cauchy condition. Decomposability gives good-integrator behavior, so the preceding corollary constructs a limit in the same domain. Return to the original representatives and take the quotient. For uniqueness use regular zero-start versions, fixed-test uniqueness and the unit test to identify increments, then initial values to identify paths. $\square$

This construction uses no auxiliary-metric completeness or continuity, only identification of its zero-distance relation with indistinguishability.

Proposition C.15 (A time-weighted elementary metric). *On the same quotient,*

$$d_{\mathcal{E}}(x, y) = \sum_{n \geq 0} 2^{-n-1} e_{n+1}(x, y)$$

is a metric bounded by one.

Proof. Regular representatives make test errors integrable and in $[0, 1]$. Symmetry and triangle pass through integration and the test supremum; horizon monotonicity follows from the all-time supremum on each interval. The weights sum to one, so the series is finite and has the metric algebraic properties. If its value is zero, each positive-weight gauge is zero. Horizon monotonicity gives zero at every finite horizon; uniqueness of the gauge limit of a constant sequence gives equality of the two quotient points. $\square$

Proposition C.16 (Independent completeness of the elementary metric). *$(\mathcal{S}, d_{\mathcal{E}})$ is complete.*

Proof. For $T \leq k + 1$,

$$2^{-k-1} e_T(x_m, x_n) \leq d_{\mathcal{E}}(x_m, x_n).$$

A metric-Cauchy sequence is therefore Cauchy for every gauge. Take its already constructed gauge limit and use dominated convergence with majorant 2^{-k-1} to obtain metric convergence. Neither auxiliary completeness nor closedness of an integral range is used. $\square$

Proposition C.17 (The elementary metric and addition). *The natural real-linear structure on $\mathcal{S}$ has translation-invariant $d_{\mathcal{E}}$, and subtraction is 2-Lipschitz for the maximum product metric.*

Proof. Linearity in the source makes each test difference invariant under adding the same process. Integrate, take suprema, and sum the time weights. Symmetry also gives invariance under simultaneous negation. Thus

$$d_{\mathcal{E}}(x - y, x' - y') \leq d_{\mathcal{E}}(x, x') + d_{\mathcal{E}}(y, y') \leq 2 \max(d_{\mathcal{E}}(x, x'), d_{\mathcal{E}}(y, y')).$$

This gives the compatible additive uniform structure and continuous addition and negation. $\square$

Proposition C.18 (Fixed scalar multiplication).

$$d_{\mathcal{E}}(ax, ay) \leq \max(|a|, 1) d_{\mathcal{E}}(x, y)$$

for every real a .

Proof. For positive a use finite-sum linearity and $1 \wedge (au) \leq \max(a, 1)(1 \wedge u)$, then integrate, take suprema, and sum. Zero is immediate; negative a reduces to positive by negation isometry. $\square$

The factor does not tend to zero with a ; the next estimate handles that case.

Lemma C.19 (Uniform boundedness of unit-integral maxima). *For a strongly adapted right-continuous good integrator S , right-continuous filtration, and finite T , the maxima $\sup_{t \leq T} |(H \cdot S)_t|$ over all unit-bounded elementary predictable H are bounded in probability.*

Proof. The good-integrator property gives boundedness of unit integrals at the fixed time T . Choose a tail bound $r \geq 0$ and put $R = r + 1$. Let τ be the first strict absolute passage of $X = H \cdot S$ above R . It is a stopping time, and if finite then $|X_\tau| \geq R$ by right continuity. Stopping H at $\tau \wedge T$ gives another unit strategy K . If the original maximum exceeds R , then $\tau \leq T$ and $|(K \cdot S)_T| \geq R > r$. Its fixed-time tail controls the original maximal tail. $\square$

Proposition C.20 (Joint scalar continuity for the elementary metric). *$(\mathcal{S}, d_{\mathcal{E}})$ is a complete real topological vector space.*

Proof. Fix x and a regular representative S , which is a good integrator. At horizon T choose $R > 0$ making all unit-integral maximal tails at most $\varepsilon/2$. If $|a| \leq \varepsilon/(4R)$, the capped scaled maximum f is at most $\varepsilon/4$ on the event where the unscaled maximum is at most R . Hence $P(f \geq \varepsilon/2) \leq \varepsilon/2$, and $f \leq \varepsilon/2 + 1_{\{f \geq \varepsilon/2\}}$ gives $Ef \leq \varepsilon$. Here the random variable being bounded is

$$f = 1 \wedge \sup_{t \leq T} |a(H \cdot S)_t|.$$

The scalar cutoff is test-independent, so $e_T(ax, 0) \rightarrow 0$, and therefore the weighted metric tends to zero. For $a_n \rightarrow a, x_n \rightarrow x$, use

$$d_{\mathcal{E}}(a_n x_n, ax) \leq \max(|a_n|, 1) d_{\mathcal{E}}(x_n, x) + d_{\mathcal{E}}((a_n - a)x, 0) \rightarrow 0.$$

Sequential continuity is continuity in metric spaces. Addition and completeness were already established. $\square$

Proposition C.21 (The auxiliary complete topological vector space). *The same quotient with auxiliary metric R and its natural real-linear operations is a complete metrizable real topological vector space.*

Proof. Admissible decompositions are stable under addition, negation, and scaling. The triangle inequality gives continuity of addition and

$$R((X - Y) - (X' - Y')) \leq R(X - X') + R(Y - Y') \leq 2 \max(R(X - X'), R(Y - Y')),$$

so subtraction is uniformly continuous. Use the earlier scalar-continuity lemma. The operations descend to the separated quotient and its quotient map is continuous linear. Its equality is indistinguishability, and the already proved completeness is for this same metric. $\square$

Proposition C.22 (Closed graph between the two topologies). *The identity $(\mathcal{S}, d_{\mathcal{E}}) \rightarrow (\mathcal{S}, R)$ has closed graph.*

Proof. Suppose $x_n \rightarrow y$ in the elementary metric and $x_n \rightarrow z$ in the auxiliary metric. Choose regular zero-start representatives from decompositions. Positive time weights and horizon monotonicity turn elementary-metric convergence into every finite-horizon gauge convergence. The unit test and common initial values give ucp convergence to y . The auxiliary maximal-tail estimate gives ucp convergence to z . Uniqueness of right-continuous ucp limits makes y, z indistinguishable. The product is metrizable, so sequential closedness proves closedness of the graph. $\square$

Lemma C.23 (Open mapping and closed graph for complete invariant metrics). *Let E, G be real topological vector spaces with translation-invariant metrics. If E is complete and G is Baire, every continuous surjective linear map $f : E \rightarrow G$ is open. If both are complete, a linear map with closed graph is continuous.*

Proof. A zero neighborhood V absorbs E , giving $G = \bigcup_{n \geq 0} (n+1)\overline{f(V)}$. Baire and the homeomorphism given by nonzero scaling make $\overline{f(V)}$ have nonempty interior. For any zero neighborhood U choose symmetric V with $V + V \subset U$; subtract an interior point to see that $\overline{f(U)}$ is a zero neighborhood. For $\varepsilon > 0$ choose

$$0 < \delta_n \leq 2^{-n}, \quad B_G(0, \delta_n) \subset \overline{f(B_E(0, (\varepsilon/4)2^{-n}))}.$$

For $d_G(y, 0) < \delta_0$, construct successive corrections from $x_0 = 0$ with $d_G(f(x_n), y) < \delta_n$ and $d_E(x_n, x_{n+1}) \leq (\varepsilon/4)2^{-n}$. Completeness gives $x_n \rightarrow x$ with $d_E(x, 0) \leq \varepsilon/2$; continuity and vanishing residual give $f(x) = y$. Thus the image of the ε -ball contains a zero neighborhood. Translate to obtain openness. A closed graph is a complete closed linear subspace of the product. Apply openness to its first projection; its continuous inverse followed by the second projection is the original map. $\square$

Proposition C.24 (From elementary convergence to auxiliary control). *For decomposable processes, elementary-test convergence $X_n \rightarrow X$ implies $R(X_n - X) \rightarrow 0$. For elementary representatives G_n of a realized gain G over a strongly progressive càdlàg good-integrator source, one subsequence n_k and one exhaustive $\tau_r \leq r + 1$ give prefix witnesses for $G_{n_k} - G$ with costs $c_{r,k}$ satisfying*

$$\sum_k c_{r,k} \leq 6(r+4) < \infty, \quad c_{r,k} \longrightarrow 0.$$

A simultaneous selection for consecutive differences has bound $6(r + 5)$. Assume the usual conditions and sigma-finite filtration.

Proof. The closed-graph theorem makes the identity from elementary to auxiliary topology continuous. Every elementary gain has a regular zero-start good-integrator decomposition. Independent elementary completeness constructs a decomposable limit, identified with the designated gain by ucp uniqueness. Apply the auxiliary subsequence and common-localization theorem. For consecutive differences first select $\sum_k R(G_{n_k} - G) \leq 1$. Triangle gives a difference-cost sum at most 2; apply summable-cost localization without another extraction. Realization of the limit as an integral has not been assumed. $\square$

This comparison is independent of closedness of a fixed source's integral range.

Proposition C.25 (One measure for prefix costs and boundary jumps). *For an everywhere bounded source, one can choose simultaneously an equivalent probability Q , a subsequence n_k , and exhaustive $\tau_r \leq r + 1$ so that for $Z_k = G_{n_{k+1}} - G_{n_k}$,*

$$\sum_k c_{r,k} \leq 6(r + 5), \quad \sum_k E_Q |\Delta Z_k(\tau_r)| < \infty.$$

There is also a measurable $\eta \geq 0$ in $L^2(Q)$, defining Q by an exponential tilt with bounded density, which bounds $|G_{n_k} - G|$ on $[0, k + 1]$ simultaneously almost surely. Thus $\xi \in L^2(\mu)$ implies $\xi + \eta \in L^2(Q)$.

Proof. Fix Q first from growing-horizon error envelopes. Both auxiliary costs and these envelopes' $L^1(Q)$ norms vanish; choose a subsequence whose sum is bounded termwise by a geometric sequence. Increasing indices allow shrinking the measured horizon to $k + 1$, retaining the envelope and sum bound one. Consecutive auxiliary costs sum to at most two, giving common stops. For fixed r and $k + 1 \geq r + 1$, if a_k is the error envelope then

$$|\Delta Z_k(\tau_r)| \leq 2(a_{k+1} + a_k),$$

with summable expectations. The finitely many earlier terms are integrable because the bounded source and each row's finite coefficient bound bound its entire gain. No further extraction changes the common stops. $\square$

Comparing prefix witnesses with actual component costs. For the same witnesses define $\hat{N}_k = (N_k - N_k(0))^{\tau_r}$. Intrinsic quadratic variation and reverse Davis give

$$E_Q \sqrt{[\hat{N}_k]_{r+1}} \leq 7E_Q \sup_{t \leq r+1} |\hat{N}_k(t)| \leq 14c_{r,k}.$$

Centering costs at most twice the original maximum; neither bounded jumps nor a separately supplied quadratic schedule is needed. For a prefix witness N, A , set

$$M = ((N - N_0)^\tau)^T, \quad B = A^{\tau-} - A_0 - \Delta N_\tau 1_{[\tau, \infty)}.$$

Use the zero-jump convention at $\tau = 0$. The second component is adapted, zero-start, and globally FV. Their sum agrees with the zero-start target before τ ; when $\tau = 0$ that condition is empty. Prefix variation and the boundary jump give $E \text{Var}(B) \leq 2c$, while the martingale root expectation is at most $14c$. Thus the actual J1 decomposition bounds the strict-prefix infimum, here denoted $J_\tau(X)$, as follows:

$$J_\tau(X) \leq 16c.$$

Indistinguishability of the exact witness transfers initial values and prefix agreement on the same null set. Consequently the same rows have summable $J_{\tau_r}(Z_k)$. If an integral representation is supplied, Lemma A.39 and equation (4) give

$$E \sup_{t \leq T} |M_t^{\text{actual}} - M_0^{\text{actual}}| + E \text{Var}_{[0, T]} A^{\text{actual}} \leq 30(32c + E|\Delta X_\tau|).$$

This estimate does not itself construct the integral representation and does not transfer special certificates unconditionally between measures.

C.1.1 Constructing actual rows from the same selection.

Choose the preceding Q, f, τ_r from the bounded source and elementary representatives. An almost-sure bound uniform over time suffices: the finitely many initial jump terms can be bounded off that common null set, and the bound transfers by $Q \ll \mu$. After fixing Q , construct the original price's decomposition and completed schedule. Each Z_k is the gain of the difference of two elementary strategies, whose coefficient absolute-sum bound is the sum of their row bounds. Construct its stopped graph directly at τ_r , and stop again at $T_r = r + 1$. The gain is still $Z_k^{\tau_r}$ and the coefficient is

$$1_{(0, \tau_r]}(H^{f(k+1)} - H^{f(k)}).$$

Both components are constant after T_r and the FV component is globally BV. These are integral certificates for the original S . Indistinguishability of the stopped and represented gains allows the comparison estimate to apply to these independently constructed certificates. For their components $M^{r,k}, A^{r,k}$ it yields

$$\sum_k \left(E_Q \sup_{t \leq T_r} |M_t^{r,k} - M_0^{r,k}| + E_Q \text{Var}_{[0, T_r]} A^{r,k} \right) < \infty.$$

The tilt, density bound, and $L^2(Q)$ error envelope remain part of the same selection.

All-time component series. Fix r and center $m_k = M^{r,k} - M_0^{r,k}$, $a_k = A^{r,k} - A_0^{r,k}$. The actual rows are restopped elementary realizations retaining martingale left limits. Each m_k is càdlàg and constant after T_r , so its measurable horizon envelope g_k dominates it for all time, with $\sum_k E_Q g_k < \infty$. This upgrades each local martingale to a true martingale; the dominated series theorem makes

$$m(t) = \sum_k m_k(t)$$

a true martingale. Partial sums converge uniformly over all time almost surely. Centering does not change FV, and constancy after T_r gives

$$\text{Var}_{[0,\infty)} a_k \leq \text{Var}_{[0,T_r]} A^{r,k}.$$

Tonelli makes the variation series finite almost surely, giving a globally BV sum $a(t) = \sum_k a_k(t)$ with all-time uniform and variation-distance convergence. It is predictable as a pointwise series of predictable components. All conclusions concern the actually generated rows of the same selection.

Actual approximants including the first term. Write $G_k = H^{f(k)} \cdot S$. At stage r let $P_{r,0}$ be the directly constructed certificate of $G_r^{\tau_r}$ and recursively, in the raw layer,

$$P_{r,n+1} = P_{r,n} + R_{r,r+n}$$

using the same actual increment rows. Telescoping gives coefficient $1_{(0,\tau_r]} H^{f(r+n)}$ and gain $G_{r+n}^{\tau_r}$. Transfer realization from its direct elementary certificate using coefficient equality and gain indistinguishability; a general actual addition calculus is unnecessary. Initial and increment components are zero-start. If c_k bounds the k th elementary gain, the simultaneous error envelope gives

$$|G_{r+n}^{\tau_r}(t)| \leq |G_r^{\tau_r}(t)| + |(G_{r+n} - X)^{\tau_r}(t)| + |(G_r - X)^{\tau_r}(t)| \leq c_r + 2\eta =: \zeta_r.$$

Here $\tau_r \leq r + 1 \leq r + n + 1$. Thus $\zeta_r \in L^2(Q)$ is a common all-time gain envelope for these actual approximants. Neither a new measure nor an additional L^2 assumption on the designated X is needed.

Regular component limits and the Mémín input. Right continuity and left limits pass through all-time uniform convergence. Correct bad paths on time-zero measurable null sets, intersecting the two component agreement events. Preserve the martingale property, predictability, uniform convergence, and variation-distance convergence. Add the initial components to obtain M_r^∞, A_r^∞ ; the former is an everywhere càdlàg local martingale and the latter an everywhere right-continuous predictable globally BV process. Global BV is asserted at this

fixed stopped stage only. Adjacent raw finite sums differ by exactly the stored increment components. Because the FV paths are constant after T_r , their canonical Stieltjes measure's total variation equals the path variation on $[0, T_r]$. This uses the stopped-path theorem, not an auxiliary measure field in a record. Every horizon martingale-difference envelope is likewise bounded by the T_r envelope. The expected-cost sum and Tonelli therefore make both adjacent-difference norm series finite almost surely. Retain the stronger selected error bound

$$\sum_k E_Q \sup_{t \leq k+1} |G_k(t) - X(t)| \leq 1.$$

Tonelli makes the envelopes summable pathwise and tending to zero. On one full-measure set $G_k(t) \rightarrow X(t)$ at all times, hence also at $t \wedge \tau_r$. Gain identification and component convergence give

$$M_r^\infty + A_r^\infty \sim X^{\tau_r}.$$

Together with ζ_r these supply every Mémín component-data field and the stored stopped-gain identification from the source and representatives. No new subsequence or measure is selected.

Component identification by stopping and finite addition. For a right-continuous local martingale M with localizers ρ_n , put $B_n = \{\rho_n > 0\}$. The process $1_{B_n} M$ stopped at ρ_n is a martingale. Stop further at σ and commute the indicator and the two stops. The same localizers establish the local property of M^σ ; this argument does not require $M_0 = 0$. Finite stopping preserves predictability of a predictable FV component and turns local BV into global BV. Thus stop raw gains and both components directly, keeping initial values. On a graph coordinate the raw stop and the completed strategy have the same gain; uniqueness of centered canonical components identifies both parts. Repeat on minimum stops of two graph schedules, identifying energy kernels, variation bridges, and completed integrals. This supplies finite addition on a common schedule. Together with the finite-horizon Stieltjes identifications this constructs the needed FV calculus without assuming a general predictable-restriction calculus. The intrinsic special terminal model using this addition does not require a stopping calculus as input; identifying it with the general market is a separate step.

C.1.2 Direct joint passages and realization of the stopped limit.

At fixed outer stage r combine the source completed schedule with all martingale passages of $P_{r,k}$ to obtain exhaustive finite-horizon stops $\rho_{r,j}$. Construct source prefixes by the direct stopping rule. The pre-passage bound and the Corollary 2.4 jump envelope include overshoot and give

$$\|M_{r,k}^{\rho_{r,j}}(T_j)\|_{L^2(Q)} \leq c_j + 6\|\zeta_r\|_{L^2(Q)}.$$

The stopped local martingale has an integrable all-time envelope, so is a true martingale by dominated convergence. The fixed graph's completed coordinate and the directly stopped raw components have the same gain. Centered uniqueness identifies the integral representation of each approximating component. Apply Lemma C.26 with the summable adjacent FV variations and the common-coordinate L^2 bounds on martingale terminals proved above. That lemma first obtains convergence of the full coefficient sequence under variation control, then makes forward averages of the same coefficients converge in the energy coordinates. One predictable coefficient consequently represents both limiting integrals. Agreement with the variation-control limit also determines the FV integral on energy-null regions. The exhaustive coordinate graph realizes the component-sum limit as an actual strategy. Applying this to the same Q, τ_r component data gives an original-price actual strategy V_r with

$$(V_r \cdot S) \sim_Q X^{\tau_r}, \quad \tau_r \leq r + 1, \quad \tau_r \rightarrow \infty \quad Q\text{-a.s.}$$

The left side denotes its stored gain. No new stopping calculus, measure, or approximating sequence is an input. These are special certificates at each stopped stage under Q . Return to the original measure's general graph, pasting of stopping intervals, and terminal identification are the separate steps already given in Section D.4.

C.2 One coefficient for both limiting components

We spell out the coefficient selection used in the preceding realization. All decompositions in this subsection are under the fixed auxiliary probability Q , on the original price S . Write $S - S_0 = M^S + A^S$, with zero-start canonical components. Let H_n be the actual approximating coefficients and write

$$M_n = H_n \cdot M^S, \quad A_n = H_n \cdot A^S.$$

The preceding component-series construction supplies regular limits M, A , locally uniform convergence of M_n to M , variation convergence of A_n to A , and, almost surely on each finite horizon T ,

$$\sum_n \text{Var}_{[0,T]}(A_{n+1} - A_n) < \infty.$$

It also supplies common joint stops on each of which the martingale terminals are bounded in $L^2(Q)$, uniformly in the approximation index.

Lemma C.26 (Simultaneous coefficient selection). *Under these conditions there is one predictable real coefficient H whose completed martingale and finite-variation integrals on every joint coordinate are the corresponding stops of M and A . No absolute continuity between the energy measure and the variation measure is required.*

Proof. Fix first a deterministic $T > 0$. Write $v_T(\omega) = |d(A^S)^T|(\omega)$ for the source's pathwise variation measure. Choose a finite measurable random variable $R_T \geq 0$ dominating

$$v_T(\omega)([0, \infty)) + \text{Var}_{[0, T]} A_0 + \sum_n \text{Var}_{[0, T]}(A_{n+1} - A_n),$$

Such a variable exists by the pathwise summability already proved. The construction may add the summable martingale envelopes to R_T ; this makes the same weight usable for all component estimates. Set $a_T = (1 + R_T)^{-2}$, replacing it by 1 on the null exceptional set. It is strictly positive almost surely. Define a finite measure on the predictable sigma-field by

$$\nu_T(B) = E_Q \left[a_T \int 1_B(t, \omega) d\nu_T(\omega)(t) \right].$$

This changes only the variation control; martingale and energy estimates remain under Q .

For every approximating integral the pathwise density identity gives

$$\begin{aligned} \int |H_0| d\nu_T &= E_Q[a_T \text{Var}_{[0, T]} A_0] < \infty, \\ \sum_n \int |H_{n+1} - H_n| d\nu_T &= E_Q \left[a_T \sum_n \text{Var}_{[0, T]}(A_{n+1} - A_n) \right] < \infty. \end{aligned}$$

Indeed $d(A_{n+1} - A_n) = (H_{n+1} - H_n)dA^S$ and the variation of a signed measure with density f is $|f|$ times its variation measure. Tonelli and completeness of L^1 imply that the full coefficient sequence has a limit h_T , both ν_T -almost everywhere and in $L^1(\nu_T)$. Equivalently, outside a Q -null set it converges $v_T(\omega)$ -almost everywhere and in pathwise $L^1(v_T(\omega))$: its successive pathwise L^1 distances are summable and its initial term is integrable.

For the countably many joint terminal coordinates choose $b_r > 0$ with $\sup_n \|M_n(\rho_r)\|_2 \leq b_r$. The vectors $(2^{-(r+1)/2} M_n(\rho_r)/b_r)_r$ are bounded in the Hilbert direct sum of the terminal $L^2(Q)$ spaces. Weak subsequential compactness and Mazur's lemma give forward convex averages converging strongly in this direct sum. Transfer the subsequence weights back to the original indices; their support still tends forward. Thus one family $w_{n,k}$ makes every terminal coordinate Cauchy. Define $\bar{H}_n = \sum_k w_{n,k} H_k$, and use the same weights for $\bar{M}_n$. Let q_r be the predictable energy measure on the r th coordinate. The completed isometry gives

$$\|\bar{H}_n - \bar{H}_m\|_{L^2(q_r)} = \|\bar{M}_n(\rho_r) - \bar{M}_m(\rho_r)\|_{L^2(Q)}.$$

Here ρ_r is the finite coordinate stop, including its deterministic horizon, and $\bar{M}_n$ uses precisely the same weights. Thus these coefficients are Cauchy in each of the countably many $L^2(q_r)$ spaces. Choose one strictly increasing f with

$$\|\bar{H}_{f(j+1)} - \bar{H}_{f(j)}\|_{L^2(q_r)} \leq 2^{-j} \quad (r \leq j).$$

Each q_r is finite, so Cauchy-Schwarz and Tonelli give pointwise convergence q_r -almost everywhere. Define H to be the pointwise limit where this sequence converges in $\mathbb{R}$, and zero

elsewhere. Its convergence set is predictable, so H is predictable; L^2 completeness gives $\bar{H}_{f(j)} \rightarrow H$ in $L^2(q_r)$ for every r .

The same definition determines the variation component, including regions where q_r vanishes. At each point where $H_n \rightarrow h_T$, forward convexity gives

$$|\bar{H}_n - h_T| \leq \sup_{k \geq n} |H_k - h_T| \rightarrow 0.$$

Hence $H = h_T$ ν_T -almost everywhere, and also $v_T(\omega)$ -almost everywhere for almost every ω . In particular $H \in L^1(\nu_T)$ and is pathwise integrable against v_T . A joint source prefix is a further stop before its deterministic horizon, so its variation measure is a restriction of v_T . The same integrability therefore holds on that coordinate. This explicitly supplies both $L^2(q_r)$ and L^1 membership for one coefficient.

The completed martingale isometry and Doob's inequality show that the martingale integrals of $\bar{H}_{f(j)}$ converge to that of H on coordinate r . Forward averages preserve the already existing limit M , and uniqueness of limits in probability identifies this integral with M^{ρ_r} . On the variation side, for almost every path,

$$\text{Var}_{[0,T]}(H_n \cdot A^S - H \cdot A^S) = \int |H_n - H| dv_T \rightarrow 0.$$

The left-hand integrals also converge in variation to A^T . Zero initial values and uniqueness of limits identify $H \cdot (A^S)^T = A^T$. Restricting these signed-measure identities to $(0, \rho_r]$ proves the finite-variation identification on the joint coordinate, with the endpoint jump included. Countably many coordinates and right continuity give one null set for all process equalities. Thus the same H realizes both parts. $\square$

The coefficients in different outer stopped stages need not agree. Section D.4 pastes them on disjoint stopping intervals after return to the original measure. Lemma C.26 supplies the stopped realizations required there; it does not assume integral-range closedness in selecting H .

D Proofs of the seven analytic inputs

The boundary with the main argument consists of the seven propositions and two data definitions in Section 2. This appendix assembles their proofs. No internal lemma from the preceding three appendices is needed directly in the main argument.

D.1 Regular limits and measurable envelopes

Proof of Proposition 2.2. For [P1], use the all-time Cauchy condition to choose a strictly increasing f with

$$\mu\{\sup_t |Y_{f(n+1)}(t) - Y_{f(n)}(t)| > 2^{-n}\} \leq 2^{-n}.$$

Right continuity reduces the supremum to nonnegative rational times, so these events are measurable. By Borel–Cantelli, on almost every path the consecutive all-time suprema are at most 2^{-n} for all sufficiently large indices. The resulting tail series is finite, so the sequence is uniformly Cauchy. Completeness of the reals gives an all-time uniform limit. Uniform limits preserve right continuity and left limits at every positive time. The usual conditions put the common exceptional null set in $\mathcal{F}_0$; define the process to be zero there at every time. At each fixed time it is a limit of adapted real-valued random variables, and the null-set correction preserves adaptedness. This gives a strongly adapted càdlàg representative.

For [P2], use the same fast subsequence. Each Y_n has bounded paths over all time, by càdlàg boundedness on compact intervals and its finite limit at infinity. On the common full-measure set of uniform Cauchy convergence,

$$\sup_{n,t} |Y_{f(n)}(t)| < \infty.$$

Choose a countable right-dense set D . The extended-real random variable

$$q_0 = \sup_{n,t \in D} |Y_{f(n)}(t)|$$

is measurable. Set it to zero wherever it is infinite. The resulting finite nonnegative measurable q dominates all the gains on one full-measure set.

For [P3], put $Z = E_\mu e^{-q}$ and $dQ/d\mu = e^{-q}/Z$. Since q is finite, $0 < Z \leq 1$ and the density is positive, giving $Q \sim \mu$. Boundedness of $x^2 e^{-x}$ on $[0, \infty)$ gives $q \in L^2(Q)$. Equivalent measures have the same null sets, so completeness is preserved; right continuity of the filtration is measure-independent. Thus the usual conditions are preserved as well. $\square$

D.2 Internal theorems for the integral construction

D.2.1 Decomposition and the integral domain

Theorem D.1 (Decomposition under an envelope). *Under the usual conditions, let Y be an adapted càdlàg good integrator with $\sup_t |Y_t| \leq q$, $q \geq 0$, and $q \in L^2(Q)$. There is a*

decomposition $Y - Y_0 = M + A$, where M is a zero-start càdlàg local martingale and A is zero-start, predictable, càdlàg and locally of finite variation. Both components are unique up to indistinguishability. The martingale component has an exhaustive sequence of finite stopped coordinates on which it is a true L^2 martingale. For bounded S take a constant envelope. The good-integrator property is preserved by equivalent measures, but the decomposition is constructed anew under the chosen measure.

Location of the proof. Section D.3 of the appendix derives this from decomposition and compensation. Neither NFLVR nor integral-range closedness is an assumption of this input. $\square$

An elementary predictable test J consists of finitely many stopping intervals and bounded coefficients measurable at their left endpoints. Its integral against any càdlàg process is a finite sum of increments. Put

$$d_T^Q(Y, Z) = \sup_{J: |J| \leq 1} E_Q \left[1 \wedge \sup_{t \leq T} |(J \cdot (Y - Z))_t| \right].$$

Emery convergence [2] means $d_T^Q(Y_n, Y) \rightarrow 0$ for every finite T . A countable right-dense time set ensures measurability of the supremum. The Cauchy quantifiers are $\forall T, \varepsilon > 0 \exists N \forall n, k \geq N$, with the same N for every unit-bounded test.

A *local special integral* against a specially decomposed process is obtained by applying the completed L^2 integral for the martingale energy measure and the L^1 integral for the FV variation measure to the same predictable coefficient, then gluing on stopped intervals. The FV integral is the pathwise Stieltjes integral. The required localization, completion and uniqueness are part of the next calculus theorem.

Definition D.2 (The general integral graph). Write $H^{(n)} = H1_{\{|H| \leq n+1\}}$. For predictable H and zero-start strongly adapted càdlàg X , put (H, X) in the general integral graph of S if the local special integrals of $H^{(n)}$ converge to X in Emery topology, and write $X = H \cdot S$. Processes are identified up to indistinguishability. Neither a special decomposition nor an infinite-time terminal value is required of X itself.

Theorem D.3 (Integral calculus and estimates). *For bounded S and the general integral graph above, the following hold.*

1. *Bounded predictable coefficients have local special integrals, unique up to indistinguishability for a fixed coefficient. Local special integrals belong to the general graph with the same coefficient and gain. Every general gain is an Emery limit of elementary integrals against the original price and is a good integrator.*
2. *Local special integrals obey linearity, closed stopping and predictable restriction. For $Y = M + A$ and bounded predictable k , $k \cdot Y = k \cdot M + k \cdot A$ and $\Delta(k \cdot Y)_t = k_t \Delta Y_t$. Finite*

sums use the same weights for both components. Stopping at a finite time T turns local FV into global BV.

3. *Emery convergence and the general graph transfer between equivalent probability measures with the same coefficients and gains. The same admissibility bound and any existing intrinsic terminal value are preserved.*
4. *On a fixed finite horizon, a unit-bounded predictable coefficient has unit-bounded elementary approximations simultaneously for its M and FV integrals: terminal L^2 convergence on square-integrable stopped coordinates on the M side and variation convergence on the FV side. For a zero-start L^2 martingale N and $|J| \leq 1$,*

$$E_Q[1 \wedge \sup_{t \leq T} |(J \cdot N)_t|] \leq 2\|N_T\|_2.$$

For a local martingale M exhausted by such L^2 stopped coordinates, if all unit elementary tests have capped expectation at most b on that horizon, the same is true of $k \cdot M$ for every predictable $|k| \leq 1$.

5. *A predictable FV process A has a predictable positive Hahn set B ; $C = 1_B \cdot A$ is increasing and $C_c - C_a \geq A_c - A_a$. For $|J| \leq 1$, also $\sup_{t \leq T} |(J \cdot A)_t| \leq \text{Var}_{[0,T]} A$. The intervals include their right endpoints, retaining jumps at closed stops.*

Location of the proof. Section D.4 of the appendix proves these assertions from bounded completed integrals, truncation and uniqueness, independently of realization of limits in the general integral domain. $\square$

D.2.2 Limit realization and original-market operations

The next theorem supplies the Mémin closedness result [3] used in the proof of [1, Theorem 4.2].

Theorem D.4 (Closedness of the general integral graph). *If general gains Y_n of the same bounded price S converge in Emery topology to zero-start strongly adapted càdlàg X , some predictable H satisfies (H, X) in the same general integral graph.*

Location of the proof. Section D.5 of the appendix connects topology comparison, common localization and realization of both components by one coefficient. The subsequent argument uses only this conclusion, not the auxiliary measure or stopping sequence selected in its proof. $\square$

Corollary D.5 (Operations in the original market). *Finite linear combinations, closed stops, restrictions to stopping intervals, and finite predictable pastings of general gains are again*

general gains of the original price. Moreover, if a realized zero-start gain $Y = M + A$ has an $L^2(Q)$ envelope, every realized local special integral $V = k \cdot Y$ is a general gain of the original price. Its components remain the same $V = k \cdot M + k \cdot A$. If Q is equivalent to the original measure, the same gain can be returned to that measure. If $V \geq -a$ and V is constant after T , its intrinsic terminal value is V_T , in the terminal set at the same admissibility level.

Proof. Apply finite sums, stops and restrictions to the elementary approximations of Theorem D.3. Finite-sum associativity and the triangle inequality give Emery convergence to the specified transformed gain. Pasting coefficients multiply events known at a stopping time by indicators of subsequent stopping intervals, so are predictable. Theorem D.4 realizes these regular zero-start limits.

For bounded k , approximate the M and FV integrals with the same elementary coefficients. Each elementary transform is a finite sum of stopped increments of Y ; substitute its original-price approximations. First make the error small in each finite sum, then diagonalize over integer horizons. This gives original-price elementary integrals converging to the same V in Emery topology. Apply closedness again. For unbounded realized k , truncation convergence from Theorem D.3 reduces to the bounded case; apply the same closedness theorem. Transfer the general graph between measures. The bound, constancy after T , and terminal value are properties of the unchanged gain and are preserved. $\square$

D.3 Deriving decomposition under an envelope

Proof of Theorem D.1. Apply Theorem C.8 to $Y - Y_0$, obtaining $Y - Y_0 = N + D$ with zero-start local martingale N and adapted locally FV D . That theorem follows from bounded-source decomposition, large-jump removal and stopped pasting, independently of integral-range closedness. We have $|Y - Y_0| \leq 2q$ and $q \in L^2(Q) \subseteq L^1(Q)$.

Fix a finite horizon and first localize so that N is a true martingale. Stop further when $|N|$ or $\text{Var}(D)$ first exceeds r . For the closed stop σ_r , the pre-stop bounds and $\Delta D = \Delta Y - \Delta N$ imply

$$\text{Var}(D^{\sigma_r}) \leq 2r + 2q + |N_{\sigma_r}|.$$

Optional sampling and $q \in L^1$ make the right side integrable. Apply the compensation theorem in Section A.8 to this process, constant after the fixed horizon. Then

$$M^{(r)} = N^{\sigma_r} + D^{\sigma_r} - (D^{\sigma_r})^p, \quad A^{(r)} = (D^{\sigma_r})^p$$

give a special decomposition. A zero-start predictable FV local martingale vanishes, so the components agree on overlapping stopped intervals. Pathwise boundedness and local finite variation make the stops exhaust each horizon. Remove localization and paste over integer horizons to obtain the claimed decomposition and uniqueness. Apply Lemma D.17 to the now

existing decomposition with envelope $2q$. On each fixed T , the jumps of M are bounded by an L^2 variable ξ_T . Stop when $|M|$ first exceeds n and at T . The closed-stop maximum is at most $n + \xi_T$, so this dominated local martingale is a true L^2 martingale. Taking $T = n$ and increasing the levels yields an exhaustive sequence by local boundedness of the paths. The jump lemma is proved for an already supplied special decomposition using conditional expectation and Doob's inequality; its proof does not invoke existence from the present theorem. Finally, the good-integrator property is a condition on convergence in probability at finite times, preserved by equivalent measures. It is this condition, not the special components, that is transferred. $\square$

D.4 Construction and transfer of the integral calculus

We prove Theorem D.3. Local special integrals and the general graph are those of Definition D.2.

Lemma D.6 (Measure invariance without a decomposition). *For strongly progressively measurable X_n, Y , convergence of the elementary test errors uniformly over all unit-bounded tests on every finite horizon is invariant under equivalent probability measures. Explicitly, for every finite T and $\varepsilon > 0$, one N must satisfy*

$$E_\mu \left[1 \wedge \sup_{t \leq T} |(J \cdot X_n)_t - (J \cdot Y)_t| \right] \leq \varepsilon$$

for every $n \geq N$ and every unit-bounded elementary predictable test J . Without right continuity, use the countable horizon skeleton for the supremum.

Proof. Every capped test envelope is measurable and lies in $[0, 1]$. If uniform convergence failed under the target measure, choose $n_k \geq k$ and tests J_k whose target expectations stay above one positive constant. Test-uniform convergence under the source measure makes the expectations of this chosen envelope sequence tend to zero. Pass to convergence in probability, transfer it by absolute continuity, then use boundedness to recover convergence of expectations under the target measure, a contradiction. Reverse the measures for the converse. No special decomposition is involved. $\square$

Associativity of finite sums turns uniform elementary approximation estimates into this process convergence. Tests only see increments, so initial values must be fixed separately. Convergence alone does not produce an integrand.

Lemma D.7 (Uniqueness with fixed initial value). *If a strongly progressively measurable sequence converges in this sense to right-continuous Y, Z with $Y_0 = Z_0$ almost surely, then Y, Z are indistinguishable.*

Proof. For a fixed test, convergence of capped expectations gives convergence in probability uniformly on compact intervals, whose limit is unique. Use the unit test on $(0, T]$ and evaluate at T to obtain $Y_T - Y_0 = Z_T - Z_0$ almost surely. Equality of initial values and right continuity extend equality from a countable right-dense time set to all times. $\square$

Proposition D.8 (Uniqueness and reduction of measure transfer). *For a fixed source and coefficient the graph gain is unique up to indistinguishability. Transfer of bounded-coefficient special graphs between equivalent measures implies transfer of the truncation graph.*

Proof. Two certificate sequences have identical coefficients $H^{(n)}$ row by row. Fixed-source uniqueness identifies their gains. Apply the preceding uniqueness lemma to this common sequence and its zero-start regular limits. For measure transfer apply bounded graph transfer separately to each row, then transfer test-uniform convergence and replace rows by their indistinguishable representatives. The regular limit and designated gain are preserved; no special decomposition of the limit is transferred. $\square$

For $|H| \leq 1$, every truncation equals H . A constant certificate sequence therefore gives graph membership once zero start and left limits are verified.

Proposition D.9 (Unbounded coefficients and coordinate tails). *Let H have a global integral representation. Every $H^{(n)}$ has a local representation over the same source. If h_k is the coefficient on completed coordinate k , with controls q_k, v_k , then for $r_{k,n} = h_k 1_{\{|h_k| > n+1\}}$,*

$$\|r_{k,n}\|_{L^1(v_k)} \rightarrow 0, \quad \|r_{k,n}\|_{L^2(q_k)} \rightarrow 0.$$

The terminal L^2 norm of the corresponding completed martingale tail integral also tends to zero.

Proof. Restrict all coordinates to the same predictable set and glue the compatible completed coordinates. Each h_k already belongs to both control spaces, so the large-value tails of its integrable powers vanish. The controls need not have finite total mass. The terminal martingale norm is the energy norm restricted to the horizon, bounded by the full coefficient tail norm. $\square$

Proposition D.10 (Uniform truncation convergence and inclusion). *If the source martingale component has left limits, every global or local special integral representation, including unbounded coefficients, belongs to the all-time truncation graph with the same coefficient and gain.*

Proof. Fix a completed coordinate. For a zero-start square-integrable martingale M and any elementary predictable test with represented integrand $|J| \leq 1$, discrete L^2 contraction,

chronological grid limits, and Doob's estimate give

$$E \left[1 \wedge \sup_{t \leq T} |(J \cdot M)_t| \right] \leq 2 \|M_T\|_2.$$

This bound does not depend on the number of test intervals or the sum of coefficient magnitudes. A completed martingale integral is constant after its horizon, so the estimate applies at any test horizon. For a variation-integrable coefficient r , identify the completed finite-variation integral B with its pathwise Stieltjes integral. The signed measure density formula yields

$$|dB|(\mathbb{R}_+) \leq \int |r_t| d|A|_t, \quad \sup_t |(J \cdot B)_t| \leq |dB|(\mathbb{R}_+).$$

The canonical variation norm bounds the L^1 norm of the first right-hand side. Transfer convergence in probability from the reference measure to the original probability and use the cap by one to obtain convergence of expectations, uniformly in the test. Apply both estimates to complementary coefficient tails. Additivity identifies the sum of the bounded cut and the tail with the original gain, so each coordinate has test-uniform truncation convergence. Choose a localization coordinate first so that $\mathbb{P}(\tau_k \leq T)$ is small, then the row cutoff on that coordinate. Before the stop all tests agree; after it the capped error is at most one. This removes localization.

For a local representation, stopping at each positive integer horizon gives a global one. Integrate all bounded truncations on one source schedule and identify their stops with the global certificates by uniqueness. Remove localization again. Transfer regular representatives on a countable common full-measure set and correct its complement using the usual conditions. This yields an everywhere regular zero-start representative. No common schedule for the unbounded local coefficient itself is needed. $\square$

Proposition D.11 (Bounded global certificates). *With left limits for the source martingale, a global integral representation whose coefficient satisfies $|H| \leq b$ belongs to the truncation graph. The same holds for any bounded-coefficient local representation stopped at a positive finite deterministic time T .*

Proof. Restrict each completed coefficient to $\{|H| \leq n+1\}$ and glue the compatible coordinates. Completed stopping identities realize every finite stop on the same schedule. For $n+1 \geq b$ the coefficients coincide with H , hence their gains are indistinguishable by uniqueness. Refine the source schedule against itself to obtain martingale coordinates with left limits. Glue the càdlàg martingale and locally finite-variation coordinates into a regular representative of the original gain; the completed graph gives zero initial value. All test errors on the eventual constant sequence vanish. For a local representation stop at T : its coefficient $1_{(0,T]}H$ is bounded by $\max(b, 0)$ and its gain is $X_{t \wedge T}$. Apply the global result. Global finite variation of the unstopped gain is not required. $\square$

Proposition D.12 (Constructing bounded predictable integrals). *The bounded source and any bounded predictable H supply a local integral graph and a truncation graph. Any existing bounded-coefficient local graph enters the all-time truncation graph with its designated gain.*

Proof. Use the completed schedule constructed directly from the source. Each coordinate has finite canonical variation and quadratic energy measures. Boundedness gives L^1 and L^2 membership, respectively. Use the same H in all coordinates and glue. Refine for left limits, glue the càdlàg martingale and locally finite-variation parts, and obtain zero start from the completed graph. Apply the same construction to every $H^{(n)}$. For $n + 1 \geq b$ fixed-source uniqueness makes all gains indistinguishable, so test errors vanish on one common tail. Uniqueness also identifies any previously designated local gain with this constructed gain. $\square$

Proposition D.13 (Integral transfer under equivalent measures). *Suppose the same bounded price S has source data under equivalent probabilities μ, ν . Bounded predictable local integral graphs transfer with coefficient and gain unchanged. Consequently the general truncation graph transfers, and stopped Mémín realizations return to the original measure.*

Proof. First take global bounded-coefficient certificates under both measures. The sums of the two coordinate energy controls and of the two variation controls are finite measures with no mass at time zero. Common elementary density gives one sequence J_n approximating the coefficient in both L^2 energy and L^1 variation controls. The completed integral errors tend to zero uniformly over all tests. On each coordinate the elementary integral is the pathwise $J_n \cdot S$ stopped at that coordinate's stopping time.

Stop further at the minimum of the two coordinate stops. Finite stopping cannot increase a capped test envelope. Both integrals are thus limits of one stopped elementary sequence. Transfer convergence between measures and use zero-start uniqueness to identify these stopped gains. The sequence of minimum stops is exhaustive, so the stops can be removed. Apply this at each deterministic horizon for local certificates. Construct the bounded graph directly under the target source and identify its gain by this uniqueness argument. Transfer a general graph row by row through bounded truncations and then transfer convergence to its regular limit. For a stopped Mémín realization, absolute continuity preserves exhaustiveness of the stops. No unbounded gain's special decomposition is transferred in this argument. $\square$

Proposition D.14 (All-time realization by pasting stopping intervals). *If a zero-start regular X has a general truncation graph at every stage of a finite-valued exhaustive stopping sequence, then one predictable coefficient H realizes its all-time graph against the original source.*

Proof. Place bounded graphs on the source's directly constructed common schedule. The completed stopping rule gives graphs of arbitrary stopped gains. Stopping at a finite time

preserves convergence against elementary tests, so the rule passes to general truncation graphs. For $\sigma \leq \tau$, subtract the stopped rows to represent coefficient restriction to $(\sigma, \tau]$ and gain $X^\tau - X^\sigma$. Restriction commutes with coefficient truncation.

The common null set where monotonicity or divergence of the stops fails is measurable at time zero. Replace stops there by $n + 1$, obtaining an everywhere increasing sequence v_n . This does not change stopped gains up to indistinguishability. Restrict the first coefficient to $(0, v_0]$ and the subsequent ones to $(v_n, v_{n+1}]$. Their disjoint supports make truncation commute with finite sums. Additivity and convergence give the graph of each finite sum, whose gain telescopes to X^{v_n} . The coefficients stabilize pointwise on every $(0, v_n]$; their limit H is predictable and agrees there with the corresponding finite sum. Construct its bounded truncation integrals directly from the source, identify their stops with the finite-sum rows by uniqueness, and remove the localization error. This gives (H, X) without assuming coefficients agree on overlaps. $\square$

Unit-bounded transforms and Hahn components We prove items 4 and 5 of Theorem D.3. The localization coordinates below have true L^2 components on finite horizons.

Proposition D.15 (Completed bounded transforms and finite-horizon Hahn gains). *Let $Y = M + A$ be strongly adapted and right-continuous, with predictable global bounded-variation A and true martingale M satisfying $M_T \in L^2$. Suppose every unit elementary test obeys*

$$E \left[1 \wedge \sup_{t \leq T} |(J \cdot M)_t| \right] \leq b.$$

For predictable $|k| \leq 1$, its completed integral $I^k = (k1_{(0,T]}) \cdot M$ has the same test bound b . Moreover a positive predictable Hahn set B gives $R = 1_{B \cap (0,T]} \cdot Y = N + C$ with, almost surely for all $0 \leq a \leq c \leq T$,

$$0 \leq C_c - C_a, \quad A_c - A_a \leq C_c - C_a.$$

The component N has the same test bound and $E[1 \wedge \sup_{t \leq T} |N_t|] \leq b$.

Proof. The normalized variation and martingale energy controls are finite. Predictable ring density gives one elementary sequence with $|J_n| \leq 1$ converging to $k1_{(0,T]}$ in both controls. Energy continuity identifies its martingale integrals and gives elementary-Émery convergence to I^k . For another unit test J , finite-horizon associativity gives $J \cdot (J_n \cdot M) = (JJ_n) \cdot M$ with $|JJ_n| \leq 1$. Pass the bound b through the triangle inequality and arbitrarily small approximation error. Variation error is unnecessary for the martingale estimate. Obtain B from the normalized finite-variation bridge and restrict the horizon's realized unit integral. The completed/Stieltjes identification expresses $C_c - C_a$ as the signed mass of $B \cap (a, c]$; positivity of Hahn mass gives both increment bounds, including the right endpoint. The transform

estimate applies to N . Its zero start and the horizon unit test give the maximal-function bound. $\square$

For the local version remove stops as follows.

Use the integral graph's localizers σ_r and completion horizons U_r . Its normalized stopped martingale component is the completed integral of k ; restriction to $(0, U_r]$ does not change its energy class or unit bound. Apply the preceding approximation to each coordinate while keeping test horizon T fixed. Stopping at U_r before testing cannot increase the capped maximum relative to the test of M^{σ_r} . The stopped elementary gain identity bounds the latter by a test of M at the same T . No ordering of U_r and T is required. On $\{\sigma_r > T\}$ stopped and unstopped tests agree; otherwise the cap costs at most one. The full expectation is at most $b + P(\sigma_r \leq T)$; let $r \rightarrow \infty$. Centering does not change increments. A local finite-variation graph may first be stopped at $T + 1$.

Elementary density for a general graph follows by approximating each bounded truncation row and diagonalizing over integer horizons. Proposition C.12 makes its gain a good integrator.

D.5 Proof of closedness of the general integral graph

Proof of Theorem D.4. Choose one original-price approximation. The common null event on which the price bound fails belongs to the initial sigma-field by the usual conditions. Set the price to zero there; this gives an everywhere bound without changing gains up to indistinguishability. Apply bounded truncation and elementary density from Theorem D.3 to each Y_n . Diagonalize with geometrically small test-uniform errors on integer horizons to obtain elementary original-price gains P_n converging to the same X in Emery topology. By assumption X is zero-start, strongly adapted and càdlàg. Proposition C.12 makes it a good integrator, and Theorem C.8 gives a decomposition in the domain of the auxiliary topology. No integral representation of X is assumed.

Localize the same sequence. Apply Proposition C.24 to $P_n - X$ to obtain vanishing auxiliary r^1 cost. Its topology comparison uses completeness of the space of decomposable processes and a closed-graph theorem, independently of integral-range closedness. Take a faster subsequence with summable adjacent costs. Proposition C.25 chooses this subsequence, one equivalent measure Q , and one stopping sequence τ_r so that prelocal component costs and expected boundary jumps are jointly summable. It retains $\tau_r \leq r + 1$ and $\tau_r \rightarrow \infty$.

Pass the bounds to actual stopped integral components. Each $P_n^{\tau_r}$ is an already realized original-price integral. Lemma A.39 and equation (4) transfer the bounds obtained from arbitrary decompositions to the canonical components of this same gain. Subsection C.1.1 applies the bounds to every adjacent difference at the same r and constructs component

series. Their limits are locally uniform on the M side and in variation on the FV side; their sum is X^{τ_r} . No unrelated sequence or independently selected decomposition is substituted.

Realize both components with one coefficient. The joint stops of Subsection C.1.2 give terminal $L^2(Q)$ bounds on the approximating M components, uniform in their index. Together with summable adjacent FV variations, these are precisely the hypotheses of Lemma C.26. That lemma matches the coefficient limit under the FV variation control with the energy limit of convex averages of the same coefficients. One predictable coefficient therefore represents both stopped components, whose sum is the same X^{τ_r} . The FV control also determines the coefficient where the energy measure vanishes; no absolute continuity between the two measures is needed.

Return to the original measure and all time. Truncation inclusion and measure transfer from Theorem D.3 return each stopped general graph to the original measure. Apply the stopping-interval pasting proved in Section D.4 to this stopping sequence. Joining coefficients on disjoint intervals, equality of the stopped gains and $\tau_r \rightarrow \infty$ give the all-time original-price graph (H, X) . Pasting needs indistinguishable gains, not agreement of independently chosen coefficients on overlaps. The auxiliary special decomposition is not itself transferred between measures. $\square$

Gain domains and terminal sets An elementary-test realization consists of elementary predictable strategies with a deterministic bound on the sum of coefficient magnitudes in each row, and a strongly adapted càdlàg ucp limit of their gains, where the gains are Cauchy uniformly over elementary tests.

Proposition D.16 (Identification of gains and terminal claims). *For the same bounded source, the gain domains of the general truncation graph and elementary-test realizations agree up to indistinguishability. Their a -admissible terminal claim sets, defined using each candidate's own finite terminal limit, agree as well.*

Proof. Each elementary row has bounded coefficient, hence belongs to the general graph. Its Émery Cauchy property and ucp limit give Émery convergence to that same limit; apply Theorem D.4. Conversely approximate each bounded truncation row by elementary integrals of the original source. Removing a stop costs at most its probability of occurring before the horizon, independently of the test. Combine this with truncation convergence. At integer horizon $n + 1$ choose error at most $1/(n + 1)$. Horizon monotonicity gives convergence on every finite horizon, then the common test Cauchy bound and zero start give the required ucp limit. Indistinguishability transfers all-time lower bounds and the candidate's own terminal limit. Existence of a terminal limit is never inferred from finite-horizon convergence alone. $\square$

D.6 Connecting decomposition and market operations to the inputs

Proof of Proposition 2.3. By the elementary approximation in Theorem D.3, the gain Y is a good integrator. Transfer this property to Q and apply Theorem D.1 to the same gain with envelope $|q|$. Since $Y_0 = 0$ almost surely, the resulting zero-start components form a special decomposition of the same Y . $\square$

Proof of Proposition 2.4. Clauses [M1]–[M3] are the gain-domain and terminal-set identification in Section D.5. Taking $a = 1$ or the union over positive a retains each candidate's own terminal witness. For [M4], express the terminal value as the limit of the adapted gain at integer times to obtain a.e. measurability. Collect the lower bounds at those countably many times on one full-measure set and pass to the limit.

Regularity, the zero initial value and the intrinsic terminal in [M5] are data of the realization model; graph membership follows from the identification above. Replacing the terminal value by an a.e. equal representative leaves the same gain converging to it. For original coefficients H, K , switching uses $H + 1_E 1_{(\tau, \infty)}(K - H)$. It is predictable because $E \in \mathcal{F}_\tau$ and τ is a bounded stopping time. Stopping, restriction and addition in Corollary D.5 give the specified gain. Letting $t \rightarrow \infty$ gives its terminal formula from the two intrinsic terminal values and the finite stopped values.

For [M6], apply closed stopping from Theorem D.3 simultaneously to the gain and both components. Stopping a locally BV path at finite T makes it globally BV. For [M7], apply linearity of the general graph to a finite sum. Add the M/FV components with those same coefficients, preserving predictability, local BV, local martingale behavior and zero initial values. Thus the three prescribed weighted sums are realized simultaneously, without reselecting components. $\square$

The finite constructions below return realized integrals over auxiliary decompositions to the original price by Corollary D.5. That corollary treats even unbounded realized coefficients by truncation, preserving lower bounds and constancy after finite times. Only the general graph, not specialness, is transferred between equivalent measures.

D.7 Quantitative auxiliary-gain estimates

We give the numerical arguments used in Section 5. All components below are canonical, zero-start, and considered under one probability Q . The price to which the trading strategies return remains S . The constructions use only bounded predictable restriction, addition, positive scaling, and closed stopping. Corollary D.5 realizes every constructed admissible gain in the original market. The argument follows [1, Lemmas 4.7–4.10]; its formulation in terms of realized gains allows the auxiliary decomposition to vary with the index.

Lemma D.17 (An explicit jump envelope). *Let $Y = M + A$ be a regular special process with predictable FV part and $|Y_t| \leq q$ for all t , where $q \geq 0$ belongs to $L^2(Q)$. For each finite T there is a nonnegative $\xi_T \in L^2(Q)$ with*

$$|\Delta M_t| \leq \xi_T \quad (0 < t \leq T), \quad \|\xi_T\|_2 \leq 6\|q\|_2.$$

Proof. Let $Z_t = E_Q[q \mid \mathcal{F}_t]$ be its càdlàg martingale version and put $Z_T^* = \sup_{t \leq T} |Z_t|$. At a predictable stopping time $\sigma \leq T$, conditional expectation of the jump in the canonical decomposition gives

$$\Delta A_\sigma = E_Q[\Delta Y_\sigma \mid \mathcal{F}_{\sigma-}], \quad |\Delta A_\sigma| \leq 2Z_{\sigma-}.$$

To use the identity for local components, localize first and announce σ from below; conditional expectations pass to the jump using the integrable envelope of Y . Predictable jump enumeration covers all nonzero jumps of A by countably many such times. On their common full-measure set, $|\Delta A_t| \leq 2Z_T^*$ for every $0 < t \leq T$. Thus one can take

$$\xi_T = 2q + 2Z_T^*, \quad \|\xi_T\|_2 \leq 2\|q\|_2 + 4\|q\|_2 = 6\|q\|_2,$$

by Doob's L^2 inequality. This is the jump estimate of [1, Corollary 2.4], with the envelope and horizon explicitly specified. $\square$

Lemma D.18 (The finite-horizon vanishing-risk construction). *Let $Y = M + A$ be a realized original-market gain with $Y \geq -1$, $|Y| \leq q$, and $\|q\|_2 \leq B$. Suppose, for $0 < \alpha \leq 1/4$, integer $n > 0$, and finite $T > 0$,*

$$Q\left(\sup_{t \leq T} |M_t| > n^3\right) > 8\alpha.$$

Set $k = \lfloor \alpha n/4 \rfloor$, and assume $k > 0$, $6B \leq n^2$, $(B/n)^2 \leq \alpha$, and $m_n = \alpha^2 - (4B/(\alpha n^2))^2 \geq 0$. There is an original-market gain V_n , constant after T , such that

$$V_n(t) \geq -n^{-1/4} - \frac{2}{nk} \quad (\forall t),$$

$$Q\left(V_n(T) \geq \frac{\alpha m_n}{4} - n^{-1/4}\right) \geq \frac{m_n}{2} - \frac{16\sqrt{n}}{k}.$$

Proof. Take the first joint passage and the rescaled gain

$$\theta = T \wedge \inf\{t : |M_t| > n^3 \text{ or } |Y_t| > n\}, \quad L = n^{-2}Y^\theta = N + B^L.$$

The gain bound before passage and $Y \geq -1$ imply, including the jump at θ , that $\Delta L \geq -(n+1)/n^2 \geq -2/n$. Also $L^* \leq q/n^2$. The preceding lemma gives a jump envelope for N of L^2 norm at most $6B/n^2 \leq 1$. Before passage $|N| \leq n$, so N has an integrable all-time envelope bounded by n plus this jump envelope. It is consequently a true L^2 martingale. Chebyshev's inequality removes the gain-cap event, and right continuity at the first passage gives

$$Q(N_T^* > n/2) > 8\alpha - (B/n)^2 \geq 7\alpha.$$

The radius $n/2$ avoids any issue about strict inequality at the passage.

Starting at $\kappa_0 = 0$, successively stop at the first displacement of N strictly exceeding 1 after κ_{j-1} , or at T if there is none. Let $F_j = N_{\kappa_j} - N_{\kappa_{j-1}}$ for $1 \leq j \leq k$. The jump envelope yields $\|F_j\|_2 \leq 2$, and optional sampling gives $E_Q F_j = 0$. If the first k excursions have not been completed, the running maximum is bounded by the sum of their stopped excursion maxima. Doob bounds each such maximum in L^2 by 4. Hence

$$Q(N_T^* > n/2, \text{ fewer than } k \text{ completed excursions}) \leq \left(\frac{4k}{n/2}\right)^2 \leq \alpha.$$

In particular $Q(|F_j| \geq 1) > 6\alpha$ for every $j \leq k$. Therefore $EF_j^- = E|F_j|/2 > 3\alpha$. For $E_j = \{F_j^- > \alpha\}$, we have $E[F_j^- 1_{E_j}] > 2\alpha$, whereas Cauchy–Schwarz bounds this expectation by $2\sqrt{Q(E_j)}$. Thus $Q(E_j) > \alpha^2$. The corresponding gain increment has L^2 norm at most $2B/n^2$. Chebyshev at level $\alpha/2$ and $B^L = L - N$ give

$$Q(B_{\kappa_j}^L - B_{\kappa_{j-1}}^L \geq \alpha/2) \geq m_n.$$

Choose one predictable positive Hahn set P for dB^L and form the actual gain

$$R = (1_P 1_{(0, \kappa_k]}) \cdot L = N^R + C.$$

The process C is increasing and its increment over every excursion is at least the corresponding increment of B^L . For the count Z of the k events just obtained, $0 \leq Z \leq k$ and $EZ \geq km_n$. From $EZ \leq km_n/2 + kQ(Z \geq km_n/2)$ we obtain

$$Q(C_T \geq k\alpha m_n/4) \geq m_n/2.$$

No independence of excursion events is used. Predictable restriction contracts the L^2 norm of each martingale excursion. Orthogonality of disjoint stopped increments consequently gives $E|N_T^R|^2 \leq 4k$, so

$$Q\left(\sup_{t \leq T} |N_t^R| > d\right) \leq 16k/d^2.$$

Here all martingales are true L^2 martingales at the fixed horizon; optional sampling and the isometry justify the contraction and orthogonality.

Finally set

$$d = kn^{-1/4}, \quad \lambda = T \wedge \inf\{t : R_t < -d\}, \quad V_n = k^{-1}R^\lambda.$$

Restriction to a positive Hahn set selects either a gain jump or zero, so $\Delta R \geq -2/n$. The closed stop therefore satisfies $R^\lambda \geq -d - 2/n$, including its overshoot. Outside the martingale maximal event above, $R = N^R + C \geq -d$ up to T , and $R_T \geq C_T - d$. Intersect with the event for C_T and divide by k . Since $16k/d^2 = 16\sqrt{n}/k$, this proves both bounds. All operations defining L, R, V_n have the original-market realizations specified at the start of this subsection; the final stop makes the terminal value exactly $V_n(T)$. $\square$

The NFLVR contradiction using this finite-scale construction is proved in Proposition 5.1.

Lemma D.19 (Convex tails: admissibility and their explicit envelope). *Let $Y_i = M_i + A_i \geq -1$ be a realized family, each gain constant after a finite time, with common envelope $q \geq 0$ in $L^2(Q)$. Assume that $\{M_i^*\}_i$ is bounded in probability. Put $\tau_i(c) = \inf\{t : |M_i(t)| > c\}$ and $E_i(c) = \{\tau_i(c) < \infty\}$. For any finite convex weights w , the tail gain and its M component are*

$$Z^{w,c} = \sum_i w_i (Y_i - Y_i^{\tau_i(c)}), \quad L^{w,c} = \sum_i w_i (M_i - M_i^{\tau_i(c)}).$$

Their common gain envelope is

$$G^{w,c} = 2q \sum_i w_i 1_{E_i(c)}, \quad \sup_w \|G^{w,c}\|_2 \leq 2 \sup_i \|q 1_{E_i(c)}\|_2 \rightarrow 0.$$

On every finite horizon the family has a jump envelope $\xi_T^{w,c}$ satisfying

$$\sup_w \|\xi_T^{w,c}\|_2 \leq 6 \sup_w \|G^{w,c}\|_2 \rightarrow 0.$$

The cutoff for this bound is independent of T . Moreover, for each $\varepsilon > 0$ there is $N \geq 0$ such that, for every $0 < \delta \leq \varepsilon/4$, one can choose c_0 before the weights. If $c \geq c_0$ and the finite passage of $L^{w,c}$ at level ε has probability greater than ε , there are an original-market gain $V = M^V + A^V$ and an envelope g satisfying

$$V \geq -1, \quad |V| \leq g, \quad \|g\|_2 \leq 1, \quad Q\left((M^V)^* > \frac{\varepsilon}{2(N+1)\delta}\right) > \frac{\varepsilon}{2}.$$

This constructs one finite convex tail at one scale and does not assume NFLVR.

Proof. The assumed boundedness in probability of the maxima implies $\sup_i Q(E_i(c)) \rightarrow 0$ (use a slightly smaller passage level if necessary). Absolute continuity of the integral of q^2 gives $\sup_i \|q 1_{E_i(c)}\|_2 \rightarrow 0$. Each tail is zero before and at its passage, and thereafter its magnitude is at most $2q$. The envelope and its norm bound follow by the triangle inequality.

We prove the finite-scale construction. Fix $\varepsilon > 0$. Choose N with $Q(q \geq N) < \varepsilon/4$, and stop all gains at the first strict passage of their absolute values above N ; call this time ζ . Given $0 < \delta \leq \varepsilon/4$, choose the passage cutoff so that $Q(E_i(c)) \leq \delta^2$ for every i and $\|G^{w,c}\|_2 \leq \delta$ for all weights. These choices precede the offending weights. The active mass

$$F_t = \sum_i w_i 1_{\{\tau_i(c) < t\}}, \quad \gamma = \inf\{t : F_t > \delta\}$$

is increasing, adapted and left-continuous. Thus $F_\gamma \leq \delta$, whereas $EF_\infty \leq \delta^2$ implies $Q(\gamma < \infty) \leq \delta$. For $t \leq \zeta \wedge \gamma$, every active entry occurred strictly before ζ , so $Y_i(\tau_i(c)) \leq N$, while $Y_i(t) \geq -1$, including at the closed endpoint. Hence

$$Z_t^{w,c} \geq -(N+1)F_t \geq -(N+1)\delta.$$

Also stop at the passage of $L^{w,c}$ at level ε . Suppose its finite-passage event has probability greater than ε . Removing the events where the gain cap or active-mass stop occurs first leaves probability strictly greater than $\varepsilon - \varepsilon/4 - \delta \geq \varepsilon/2$. There the stopped martingale has absolute value at least ε , and therefore strictly greater than $\varepsilon/2$. Each gain in the finite sum is constant after a finite time, so the stopped gain also has a constant continuation after a finite time. Dividing by $(N + 1)\delta$ gives an original-market 1-admissible gain. Its envelope norm is at most $1/(N + 1) \leq 1$, and the displayed amplified probability bound follows. Finally apply Lemma D.17 with $q = G^{w,c}$. Explicitly, one can use

$$\xi_T^{w,c} = 2G^{w,c} + 2 \sup_{t \leq T} |E_Q[G^{w,c} \mid \mathcal{F}_t]|.$$

The norm estimate follows from Doob, independently of the horizon. $\square$

For [F1], take the terminal value $f = V_n(T)$ in Lemma D.18 and use $K_0 \subseteq K_0 - L_+^0$. Clause [F2] is precisely the conclusion of Lemma D.19.

D.8 Prefix energy and martingale approximation

Proof of [F3]–[F5]. For [F3], use only the envelope part of the proof of Lemma D.19, which does not use a gain lower bound. The convex-tail gain envelope $G^{w,c}$ satisfies $\sup_w \|G^{w,c}\|_2 \rightarrow 0$. Lemma D.17 gives a jump envelope $\xi_T^{w,c}$ for its M component with

$$\sup_w \|\xi_T^{w,c}\|_2 \leq 6 \sup_w \|G^{w,c}\|_2.$$

For $\eta > 0$, first choose a cutoff making this bound at most η . Stop $L^{w,c}$ at its level- η passage and at T . The pre-passage bound and jump envelope bound the all-time maximum of the resulting process R by $\eta + \xi_T^{w,c}$. This L^2 envelope removes the local martingale's localizers, so R is a true L^2 martingale with $\|R_T\|_2 \leq 2\eta$. The test estimate of Theorem D.3 gives $E_Q e_T^J(R, 0) \leq 4\eta$. Off the finite-passage event, the stopped and original tails agree; removing the stop costs at most $Q(B(w, c, \eta))$ in capped expectation. This proves [F3].

For [F4], apply the same argument to the original M at level c . Lemma D.17 gives $\|\xi_T\|_2 \leq 6\|q\|_2$, and $\sup_t |M_{t \wedge \tau(c) \wedge T}| \leq c + \xi_T$. Thus the stopped process is a true martingale with $\|P_T\|_2 \leq c + 6\|q\|_2$. Clause [F5] follows from preservation of right continuity and the zero initial value under closed stopping of càdlàg paths, without an envelope. $\square$

Proof of Proposition 2.6. For [A1], choose an approximating sequence with error $\varepsilon/3$ for a prescribed $\varepsilon > 0$. Make its Cauchy error at most $\varepsilon/3$ and use the triangle inequality for the two approximation errors and the middle difference. All bounds are uniform in the test, so the resulting cutoff is test-independent.

For [A2], stop $P_n - P_k$ at U . The L^2 test estimate of Theorem D.3, optional sampling and conditional-expectation contraction give

$$E_Q e_T^J(P_n, P_k) \leq 2 \|P_n(U) - P_k(U)\|_2$$

for every $T \leq U$ and J . Terminal L^2 Cauchy convergence supplies one cutoff making the right side at most ε . $\square$

D.9 Finite Hahn trades and closed-stop estimates

Proposition D.20 (Completed bounded transforms and finite-horizon Hahn gains). *Let $Y = M + A$ be strongly adapted and right-continuous, with predictable global bounded-variation A and true martingale M satisfying $M_T \in L^2$. Suppose every unit elementary test obeys*

$$E \left[1 \wedge \sup_{t \leq T} |(J \cdot M)_t| \right] \leq b.$$

For predictable $|k| \leq 1$, its completed integral $I^k = (k1_{(0,T]}) \cdot M$ has the same test bound b . Moreover a positive predictable Hahn set B gives $R = 1_{B \cap (0,T]} \cdot Y = N + C$ with, almost surely for all $0 \leq a \leq c \leq T$,

$$0 \leq C_c - C_a, \quad A_c - A_a \leq C_c - C_a.$$

The component N has the same test bound and $E[1 \wedge \sup_{t \leq T} |N_t|] \leq b$.

Proof. Item 4 of Theorem D.3 preserves the same test bound b under unit-bounded transforms. Restrict the Hahn set from item 5 to $[0, T]$ and use item 2 to decompose the same gain as $N + C$. The Stieltjes increments include the right endpoint and give both inequalities. The horizon unit test of zero-start N bounds its capped maximum by b . $\square$

Proposition D.21 (Local transforms and return to the original price). *For a regular local special source $Y = M + A$, assume the integral representation has true L^2 martingale components on an exhaustive sequence of stopped coordinates. A test bound b on M at a fixed T transfers to the martingale component N of a realized unit-bounded predictable transform. If $N_0 = 0$, its capped maximal expectation is also at most b . Local finite variation suffices; un-stopped global L^2 is not assumed. If Y is realized over bounded S , has normalized components and an L^2 gain envelope, the positive Hahn transform $V = N + C$ of Y^T satisfies*

$$0 \leq C_c - C_a, \quad A_{c \wedge T} - A_{a \wedge T} \leq C_c - C_a \quad (a \leq c)$$

and both martingale bounds b , and has a general graph over the original price.

Proof. Use the local transform estimate of Theorem D.3 at the fixed T . Removing localization costs only the probability of a stop before T , leaving the same bound b . The identified FV

integral retains the preceding Hahn increment bounds. For an original-market gain with an L^2 envelope, apply Corollary D.5 to this same Hahn gain. Only the general integral graph is transferred. $\square$

Application to two specified gains Let $R_n = \widetilde{M}_n + \widetilde{A}_n$ and $R_k = \widetilde{M}_k + \widetilde{A}_k$ be two specified special gains with common envelope $q \in L^2(Q)$, lower bound -1 , and bound b for every test of the M difference in both orders at fixed T . Their difference is dominated by $2q$ and carries exactly the specified component differences. Apply the L^2 stopped coordinates from Theorem D.1 and the preceding proposition to this difference. The resulting zero-start $V_{n,k} = N + C$ has an original-price general graph; C is increasing and dominates the stopped FV increments, while every test and the capped maximum of N have bound b . The indices merely distinguish the two gains; no sequence selection, NFLVR or maximality is used. Apply the following three propositions to this same trade.

Proposition D.22 (Closed finite stopping and admissibility). *Write $V_{n,k} = N + C$ and set, for $\delta \geq 0$,*

$$D_t = N_t - \max\{\widetilde{M}_n(t) - \widetilde{M}_k(t), 0\}, \quad \tau = \inf\{t \geq 0 : D_t < -\delta\}, \quad \sigma = \tau \wedge T.$$

Then $B = (R_k + V_{n,k})^\sigma \geq -(1 + \delta)$ at all times and its own terminal claim belongs to the original $K_{1+\delta}$.

Proof. Adapted right continuity makes the downside passage and its finite cut stopping times. Zero start and the Hahn increments give, for $t \leq T$, $C_t \geq 0$ and $C_t \geq \widetilde{A}_n(t) - \widetilde{A}_k(t)$. Before σ , $D_t \geq -\delta$, whence

$$(R_k + V_{n,k})_t \geq \max(R_n(t), R_k(t)) - \delta.$$

Take left limits at the stop. The actual Hahn restriction jumps by $\Delta(R_n - R_k)$ on its Hahn set and by zero on the complement, including at T . Thus the pasted gain's jump is either ΔR_n or ΔR_k . Add the selected input's jump to the left-limit inequality and use that input's bound -1 to get the closed-stop bound. At $\sigma = 0$ use zero start. Original-price addition and finite stopping realize this gain. It is constant after T , so its finite terminal value belongs to $K_{1+\delta}$. General graphs and lower bounds return to the original measure while component estimates stay under Q . $\square$

Proposition D.23 (Continuation on survival and failure probability). *Let $E = \{T < \tau\}$. For finite $U \geq T$ define*

$$Z_t = (R_k + V_{n,k})_{t \wedge \sigma} + 1_E(R_k(t \wedge U) - R_k(t \wedge T)).$$

This is an original-market general gain with $Z \geq -(1 + \delta)$, $Z_U \in K_{1+\delta}$, and on E ,

$$Z_U = R_k(U) + V_{n,k}(T).$$

If the capped maximal expectations of N and $\widetilde{M}_n - \widetilde{M}_k$ on $[0, T]$ are at most b , then for $0 < \delta \leq 1$,

$$Q(\tau \leq T) \leq 8b/\delta,$$

independently of U .

Proof. Since $E \in \mathcal{F}_T$, use predictable event-tail restriction after T , stop at U , and add the stopped gain. On E^c use the closed-stop lower bound. On E , $\sigma = T$ and after T the gain is $R_k(t \wedge U) + V_{n,k}(T)$. The inequalities $D_T \geq -\delta$ and $C_T \geq 0$ give $V_{n,k}(T) \geq -\delta$, preserving admissibility. The gain is constant after U . If both capped maxima are less than $\delta/4 < 1$, their uncapped path values are less than $\delta/4$, hence $D_t \geq -\delta/2$ for all $t \leq T$. Right continuity at T preserves a strict margin slightly beyond T ; therefore $\tau > T$. The failure event is contained in the union where one capped maximum is at least $\delta/4$. Markov's inequality gives $8b/\delta$. The original martingale difference's maximal bound follows from zero start and the horizon unit test in the assumed finite-scale estimate. $\square$

Proposition D.24 (Two Hahn orientations dominate variation). *Apply the construction to $A = \widetilde{A}_n - \widetilde{A}_k$ and $-A$, with resulting increasing components C^+, C^- . Almost surely,*

$$\text{Var}_{[0,T]} A \leq C_T^+ + C_T^-.$$

If $Q(\text{Var}_{[0,T]} A > \alpha) > \alpha$, one orientation satisfies

$$Q(C_T^\pm > \alpha/2) > \alpha/2,$$

retaining all the preceding estimates and original-market terminal membership for any finite $U \geq T$.

Proof. For $a \leq c \leq T$ the two increment inequalities give

$$A_c - A_a \leq C_c^+ - C_a^+, \quad -(A_c - A_a) \leq C_c^- - C_a^-.$$

Since both right-hand sides are nonnegative, $|A_c - A_a| \leq (C_c^+ - C_a^+) + (C_c^- - C_a^-)$. Sum over a partition and take the supremum. Right continuity identifies this path variation with the total variation of the stopped Stieltjes measure. The large-variation event is contained, up to a null set, in the union of the two events $\{C_T^\pm > \alpha/2\}$. Subadditivity chooses an orientation. The assumed bound b in both orders supplies both constructions simultaneously. $\square$

These constructions supply every conclusion of [H1] for the same two gains.

D.10 Terminal realization from component estimates

Lemma D.25 (From component estimates to Émery Cauchy estimates). *For finite T , put*

$$U_{n,k}^J = 1 \wedge \sup_{t \leq T} |(J \cdot (\widetilde{M}_n - \widetilde{M}_k))_t|, \quad V_{n,k} = \text{Var}_{[0,T]}(\widetilde{A}_n - \widetilde{A}_k).$$

Suppose that for every $\delta > 0$,

$$\begin{aligned} \sup_J Q_*(U_{n,k}^J > \delta) &\longrightarrow 0, \\ Q_*(V_{n,k} > \delta) &\longrightarrow 0 \end{aligned} \quad (n, k \longrightarrow \infty).$$

Then R_n is Émery Cauchy. The two indices tend to infinity independently: the cutoff for each probability error is fixed before n, k and the test J .

Proof. The variation estimate for Stieltjes integrals bounds the maximal test integral of the finite-variation difference by $V_{n,k}$. The capped test error $Z_{n,k}^J$ of the gain difference satisfies $0 \leq Z_{n,k}^J \leq 1$ and $Z_{n,k}^J \leq U_{n,k}^J + V_{n,k}$. For $0 < \delta < 1/2$,

$$E_{Q_*} Z_{n,k}^J \leq 2\delta + Q_*(U_{n,k}^J > \delta) + Q_*(V_{n,k} > \delta).$$

Choose δ and then the two probability bounds. The resulting cutoff is independent of J . Markov's inequality supplies the martingale probability estimate from the previously obtained expectation estimate. $\square$

Lemma D.26 (Émery convergence to the same uniform limit). *If R_n is Émery Cauchy and converges ucp to a regular process X , then it converges to X in Émery topology.*

Proof. For each fixed elementary test the finite-sum integral preserves ucp limits, and the capped error expectations tend to zero. Choose the Cauchy cutoff N for a prescribed error uniformly over all tests. For $n, k \geq N$ use the triangle inequality, then send $k \rightarrow \infty$ for fixed n, J . The remaining error between R_n and X has the same bound. Now take the supremum over J . $\square$

Proof of Proposition 2.8. Apply the preceding two lemmas to the same $Y_n = M_n + A_n$. The cutoffs for small probability errors are common to all tests, so $Y_n \rightarrow X$ in Emery topology under Q . Transfer convergence to the original measure by Theorem D.3 and realize the same X as an original-price general integral by Theorem D.4. The specified bound $X \geq -1$ and intrinsic terminal $X_\infty = h$ give $h \in K_1$. $\square$

E Mathematical steps and Lean declarations

The public import is `FTAPTheorem42`. Every declaration below occurs in the transitive dependencies of the main theorem's type or proof. Component estimates under Q_* and realization as general integrals under the original measure are separate steps.

Mathematical step	Lean declaration
Main theorem	<code>FTAPTheorem42.BoundedSourceIntegralMarket.theorem42_generalMarket</code>
General integral domain and terminal claims	<code>FTAPTheorem42.BoundedSourceIntegralMarket.truncatedTerminalClaimsBy_eq_generalMarket</code>
Closure of the original integral domain	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_truncatedIntegralGraph_of_emeryConverges</code>
Global realization under the original measure	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_realized_truncatedGraph</code>
Joint localization under an auxiliary measure	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_joint_actual_component_localization</code>
Elementary topology to auxiliary cost	<code>FTAPTheorem42.ElementaryMetricProcess.tendsto_r1_of_elementary</code>
Common localization and summable component costs	<code>FTAPTheorem42.J1Decomposition.exists_common_localizers_summable_component_costs</code>
Terminal improvement versus maximality	<code>FTAPTheorem42.not_persistent_terminalGain_improvement_of_vanishing_lowerError</code>
Cauchy estimates for both components of one sequence	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_selected_original_componentCauchy</code>
FV variation estimate from maximality	<code>FTAPTheorem42.BoundedSourceIntegralMarket.original_finiteVariation_cauchyInProbability</code>
Realization of a maximal closure element	<code>FTAPTheorem42.BoundedSourceIntegralMarket.maximal_mem_general_K1</code>
From terminal to process lower bounds	<code>FTAPTheorem42.BoundedSourceIntegralMarket.generalMarket_terminalLowerBoundControlsGain</code>
Assembly of Fatou and weak-star closedness	<code>FTAPTheorem42.GainProcessModel.theorem42ConcreteWeakStar_from_gainProcessModel_K1_maximalRealization</code>
Weak-star closedness of bounded slices	<code>FTAPTheorem42.isClosed_linftyWeakStar_claimSlice</code>
From a Fatou-closed cone to weak-star closedness	<code>FTAPTheorem42.linftyWeakStarClosed_of_fatouCone</code>
Forward-convex terminal compactness (Proposition 3.1)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.generalMarket_forwardConvexCandidates</code>

Mathematical step	Lean declaration
Uniform path limit from maximality (Lemma 4.5)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_uniform_terminal_limit_of_maximal</code>
Indexed maximal bound (Lemma 4.7)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_class_boundedInProbability</code>
Tail bound by normalization (Lemma 4.8)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_finiteTail_tendsToZero</code>
Uniform tail tests (Lemma 4.9)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_tail_tests</code>
M Cauchy estimates with common weights (Lemma 4.10)	<code>FTAPTheorem42.BoundedSourceIntegralMarket.exists_original_martingale_convexification</code>
Lower-bounded convex compactness (Lemma A1.1)	<code>FTAPTheorem42.hasForwardConvexAELimit_of_lowerBounded_convexSet_aestronglyMeasurable</code>
Existence of a maximal terminal claim (Lemma 4.3)	<code>FTAPTheorem42.exists_aeMaximal_ge_inMeasureSequentialClosure_of_mem</code>

Continuous-time gain identities use indistinguishability. Right continuity and a countable dense time set combine fixed-time null sets into one null set for all times. Passing between L^p classes and measurable representatives preserves lower bounds and terminal limits. The weak-star topology is induced by the integral pairing with L^1 .

Construction of the analytic inputs

Mathematical step	Lean declaration
Bounded-source decomposition	<code>FTAPTheorem42.HorizonFactorialGrid.exists_boundedSemimartingaleSource_globalSpecialDecompositionData</code>
Canonical component cost transfer	<code>FTAPTheorem42.ActualSIntegrableStrategy.centered_component_cost_le_witness_of_stopped</code>
Energy limit of the common coefficient	<code>FTAPTheorem42.LocalCompletedM2A.exists_intrinsicJointPassage_selectedIntegrand</code>
FV identification of the same coefficient	<code>FTAPTheorem42.LocalCompletedM2A.intrinsicJointPassageSelectedIntegrand_ae_eq_rangeVariation</code>
Finite-horizon vanishing-risk strategy	<code>FTAPTheorem42.LocalCompletedM2A.actualFirstLocalized_hahnPositive_stopDownside_of_originalHigh</code>

The boundary between the main proof and analytic implementation

Proof/Selection/ proves lower-bound control, terminal convex compactness, maximality-based pasting, and the uniform path limit. **Proof/Components/** corresponds to Lemmas 4.7–4.10: it derives the class bound from NFLVR, normalizes the tails, and chooses common Hilbert convex weights. **Proof/OrientedSequence.lean** assembles finite Hahn improvements

for the same sequence; `Closedness/` proves the maximality contradiction and FV Cauchy property corresponding to Lemma 4.11. Terminal convex compactness, maximal elements, and the final closure arguments use `PositiveTail/`, `Trading/`, and `Core/`.

The main proof does not directly reference `Stochastic/`; it receives the seven analytic propositions of Section 2 through `Interface/`. The following is the complete list of 22 declarations checked against the actual isolation build. All inputs are proved in the normal build. Only the isolated copy replaces them by assumptions and rebuilds the unchanged main proof. No theorem about selection, convexification, maximality or closedness is added as an assumption.

The complete isolated boundary

Identifiers refer to the clauses of Section 2. G0 and M0 are data; the remaining twenty entries are propositions.

Input	Lean declaration
[G0]	FTAPTheorem42.BoundedSourceIntegralMarket.GeneralIntegralGraph
[M0]	FTAPTheorem42.BoundedSourceIntegralMarket.generalMarket
[P1]	FTAPTheorem42.AnalyticInterface.regular_uniform_limit
[P2]	FTAPTheorem42.AnalyticInterface.regular_common_envelope
[P3]	FTAPTheorem42.AnalyticInterface.equivalent_envelope_measure
[D1]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_specialDecomposition
[M1]	FTAPTheorem42.BoundedSourceIntegralMarket.generalAdmissibleClaims_eq_generalMarket
[M2]	FTAPTheorem42.BoundedSourceIntegralMarket.generalTerminalClaims_eq_generalMarket_K0
[M3]	FTAPTheorem42.BoundedSourceIntegralMarket.generalAdmissibleClaims_on_e_eq_generalMarket_K1
[M4]	FTAPTheorem42.BoundedSourceIntegralMarket.generalMarket_terminal_properties
[M5]	FTAPTheorem42.BoundedSourceIntegralMarket.generalMarket_operations
[M6]	FTAPTheorem42.BoundedSourceIntegralMarket.originalSpecialGain_finiteStop
[M7]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_convexCombination
[F1]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_finiteRisk
[F2]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_finiteTail_normalization
[F3]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_finiteTail_test_estimate

Input	Lean declaration
[F4]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_prefix_energy
[F5]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_prefix_regula r
[A1]	FTAPTheorem42.AnalyticInterface.emeryCauchy_of_uniform_approximation s
[A2]	FTAPTheorem42.AnalyticInterface.martingaleTestsCauchy_of_terminalL2
[H1]	FTAPTheorem42.BoundedSourceIntegralMarket.originalGain_finiteHahn
[C1]	FTAPTheorem42.BoundedSourceIntegralMarket.truncatedTerminalClaims_on e_of_component_estimates

References

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- [4] M. Beiglböck, W. Schachermayer and B. Veliyev, *A direct proof of the Bichteler–Dellacherie theorem and connections to arbitrage*, authors' preprint, Definition 1.1 and Theorem 1.2.
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